# INVESTIGATION OF SURFACE GEOMETRY EFFECTS ON THE AEROTHERMODYNAMICS OF A HIAD ON IRVE-3

**THESIS**

**A thesis submitted in partial fulfillment of the requirements for the degree of Magister of Aerospace Engineering**

**By:**
Wahyu Suryo Samudro
23624003

**MAGISTER PROGRAM IN AEROSPACE ENGINEERING**
**FACULTY OF MECHANICAL AND AEROSPACE ENGINEERING**
**INSTITUT TEKNOLOGI BANDUNG**
**2026**

# APPROVAL PAGE

# INVESTIGATION OF SURFACE GEOMETRY EFFECTS ON THE AEROTHERMODYNAMICS OF A HIAD ON IRVE-3


By:

**Wahyu Suryo Samudro**

**NIM 23624003**

**Magister Program in Aerospace Engineering**

**Faculty of Mechanical and Aerospace Engineering**

**Institut Teknologi Bandung**


Approved on:

Bandung, September 3, 2026

Signature:

| | |
|---|---|
| **Dr. Ing. Mochammad Agoes Moelyadi, S.T, M.Sc.** | **Yohanes Bimo Dwianto, S.T., M.T., Ph.D.** |
| NIP. 196807091999031002 | NIP. 122110006 |
| *Supervisor* | *Co-Supervisor* |

# ABSTRACT

This thesis presents a comprehensive investigation into the aerothermal characteristics of Hypersonic Inflatable Aerodynamic Decelerators (HIADs), with a particular focus on the impact of their scalloped surface topology on flow physics and aerodynamic heating. Utilizing the Direct Simulation Monte Carlo (DSMC) method implemented in the SPARTA solver, the study develops a computational framework for rarefied gas dynamics analysis of HIAD reentry at transitional flow regimes ($0.01 < Kn < 1$). The framework integrates parametric CadQuery geometry generation, DSMC aerothermal simulation, Physics-Informed Neural Network (PINN) refinement using DeepXDE, and Genetic Algorithm optimization with a PyTorch-based Metamodel Prognosis (MoP) surrogate. The methodology is grounded in the Boltzmann transport equation and validated against IRVE-3 flight data, with the Sutton-Graves stagnation-point correlation and Fay–Riddell theory serving as thermal benchmarks. Key findings include the characterization of scalloped toroidal heating augmentation, Knudsen number field mapping across free-molecular, transitional, and continuum regimes, and the demonstration of a grid-factor of 0.7 as the optimal balance between accuracy and computational cost. The 5-species air model ($N_2$, $O_2$, NO, N, O) with Variable Soft Sphere collision physics captures real-gas effects including vibrational excitation and dissociation, enabling accurate prediction of aerothermal loads for HIAD survivability optimization.

**Keywords**: Hypersonic Inflatable Aerodynamic Decelerator (HIAD), Aerothermodynamics, Direct Simulation Monte Carlo (DSMC), SPARTA, Rarefied Gas Dynamics, Scalloped Geometry, Aerodynamic Heating, Physics-Informed Neural Network (PINN), Genetic Algorithm Optimization

# TABLE OF CONTENT

# LIST OF FIGURES

# LIST OF TABLE

# LIST OF SYMBOLS

| Symbol | Definition |
|---|---|
| $A$ | Area |
| $A_{\text{ref}}$ | Reference (frontal) area |
| $C_D$ | Drag coefficient |
| $C_{SG}$ | Sutton–Graves gas-model coefficient |
| $D$ | Aeroshell diameter |
| $F_{\text{drag}}$ | Aerodynamic drag force |
| $g_0$ | Standard gravitational acceleration |
| $g$ | Relative molecular velocity |
| $h$ | Altitude / Enthalpy |
| $h_w$ | Wall enthalpy |
| $h_s$ | Stagnation enthalpy |
| $k_B$ | Boltzmann constant |
| $Kn$ | Knudsen number |
| $L$ | Characteristic length |
| $M_\infty$ | Freestream Mach number |
| $M_{\text{air}}$ | Molar mass of air |
| $n$ | Number density / Deceleration ($g$-load) |
| $N_A$ | Avogadro's constant |
| $N_{\text{tor}}$ | Number of toroids |
| $p$ | Static pressure |
| Pr | Prandtl number |
| $\dot{q}$ | Convective heat flux |
| $\dot{q}_{\text{rad}}$ | Radiative heat flux |
| $Q$ | Total heat load |
| $R$ | Specific gas constant |
| $R_N$ | Nose radius |
| $r_{\text{tor}}$ | Toroid minor radius |
| $r_{\text{out}}$ | Outer toroid radius |
| $S$ | Sutherland constant |
| $T$ | Temperature |
| $T_\infty$ | Freestream temperature |
| $T_w$ | Wall temperature |
| $V_\infty$ | Freestream velocity |
| $v$ | Velocity component |
| $\beta$ | Ballistic coefficient |

# LIST OF ABBREVIATIONS

| Abbreviation | Definition |
|---|---|
| CFD | Computational Fluid Dynamics |
| CCD | Central Composite Design |
| $C_p$ | Specific Heat at Constant Pressure |
| DNS | Direct Numerical Simulation |
| DoE | Design of Experiments |
| DSMC | Direct Simulation Monte Carlo |
| FSI | Fluid–Structure Interaction |
| FTPS | Flexible Thermal Protection System |
| GA | Genetic Algorithm |
| GDT | Grid Dependency Test |
| HIAD | Hypersonic Inflatable Aerodynamic Decelerator |
| IAD | Inflatable Aerodynamic Decelerator |
| IRVE | Inflatable Reentry Vehicle Experiment |
| LHS | Latin Hypercube Sampling |
| LOFTID | Low-Earth Orbit Flight Test of an Inflatable Decelerator |
| MDAO | Multidisciplinary Design Analysis and Optimization |
| MoP | Metamodel Prognosis |
| $N_2$ | Molecular Nitrogen |
| NO | Nitric Oxide |
| $O_2$ | Molecular Oxygen |
| OML | Outer Mold Line |
| PINN | Physics-Informed Neural Network |
| PR | Pressure Ratio / Prandtl Number |
| RANS | Reynolds-Averaged Navier–Stokes |
| RMS | Root Mean Square |
| SG | Sutton–Graves (correlation) |
| SPARTA | Sandia Parallel Algorithm for Real-Time Application |

# CHAPTER 1

# INTRODUCTION

## 1.1 Background

In this day and age, exploration of space has been one of the driving forces for technological innovation, pushing the boundaries of what is possible for humanity. A persistent challenge in space exploration is the safe return of spacecraft and payloads to Earth or other celestial bodies with an atmosphere. During atmospheric reentry, a vehicle traveling at hypersonic speeds experiences extreme aerodynamic heating, which can lead to catastrophic failure if not properly managed. To protect the vehicle from these extreme temperatures, a thermal protection system (TPS) is required.

For generations, rigid TPS have been the standard solution for Earth reentry, such as Reusable TPS, Structurally Integrated TPS, Hot Structures, and Ablators (Bouslog, 2023). However, the size of a rigid heat shield is limited by the launch vehicle's payload fairing, which in turn restricts the size and mass of the reentry vehicle. This limitation poses a challenge for future missions that require the landing of large payloads and flexibility in backup strategies, such as human missions to Mars or the return of large assets from space.

To counteract this challenge, deployable entry systems called inflatable aerodynamic decelerators (IADs) have emerged and been tested as non-integrated, flexible TPS (DiNonno & Cheatwood, 2024). IADs, also known as hypersonic inflatable aerodynamic decelerators (HIADs), are lightweight, flexible structures that can be packed into a small volume during launch and then inflated to a much larger size before atmospheric entry. The large surface area of an IAD provides significant aerodynamic drag, allowing for the deceleration of large masses at higher altitudes where the air is less dense and the heating is less severe.

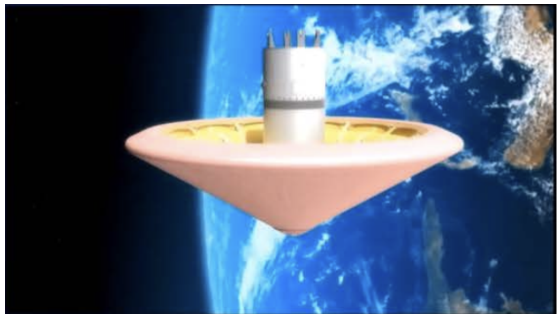


**Figure 1.1:** HIADs when deployed

The development of IADs requires an approach that combines aerodynamics, structural mechanics, and materials science. A key component of an IAD is the flexible thermal protection system (FTPS), which is responsible for protecting the inflatable structure from the intense heat of reentry. The FTPS is a multi-layered system that typically consists of an outer fabric, insulation layers, and a gas barrier (H. Guo et al., 2011). The design and analysis of the FTPS requires accurate modeling of the aerothermal loads and the material response.


**Figure 1.2:** IADs Layer Example

## 1.2 Problem Statement

The successful design and operation of an IAD depend on the ability to accurately predict and manage the complex aerothermodynamic environment and the fluid-structure interaction (FSI) phenomena that occur during atmospheric reentry. The flexibility of the IAD structure and the FTPS adds a layer of complexity to the analysis that is not present in traditional rigid heat shields. The deformation of the IAD under aerodynamic loads can affect the flow field and the heat transfer to the vehicle, which in turn can affect the structural response.

At hypersonic speeds ($Ma > 5$) and high altitudes ($> 50$ km), the atmospheric density is sufficiently low that the continuum assumption underlying traditional Computational Fluid Dynamics (CFD) begins to break down. The Knudsen number ($Kn$), which characterizes the degree of rarefaction, increases to values where the Navier–Stokes equations are no longer valid. In this transitional regime, particle-based methods such as Direct Simulation Monte Carlo (DSMC) are required to accurately capture the gas dynamics (Bird, 1994).

Furthermore, the optimization of HIAD geometry for survivability – balancing drag performance, peak heating, ballistic coefficient, and structural mass – requires an automated, multi-stage computational framework that can efficiently explore a high-dimensional design space. Existing approaches based on parametric CFD sweeps are computationally prohibitive for the thousands of design evaluations required.

Therefore, there is a critical need for an integrated computational framework that: (1) employs DSMC-based rarefied gas dynamics to accurately predict aerothermal loads in the transitional regime, (2) incorporates automated parametric geometry generation for design space exploration, and (3) leverages physics-informed machine learning and surrogate-based optimization to efficiently identify optimal HIAD configurations.

## 1.3 Research Objective

In this research, the objective is to develop and validate a computational framework for the aerothermal analysis and survivability optimization of a Hypersonic Inflatable Aerodynamic Decelerator (HIAD), based on the SPARTA DSMC solver and integrated with physics-informed neural network (PINN) refinement and genetic algorithm optimization.

The specific objectives are:

1. **Parametric Geometry Generation:** To develop an automated 3D CAD pipeline using CadQuery for generating HIAD profiles based on IRVE-3 specifications,

enabling parametric variation of toroid count, cone angle, nose radius, and aeroshell diameter for design space exploration.

2. **DSMC Aerothermal Analysis:** To perform high-fidelity rarefied gas dynamics simulations using the SPARTA DSMC solver, capturing the Boltzmann-level molecular collisions and transport phenomena in the transitional flow regime ($0.01 < Kn < 1$) relevant to HIAD reentry at 50–60 km altitude.

3. **Knudsen Number Field Mapping:** To analyze the spatial distribution of the Knudsen number across the HIAD surface and surrounding flow field, identifying continuum ($Kn < 0.01$), transitional ($0.01 < Kn < 1$), and free-molecular ($Kn > 1$) regions to characterize the flow physics governing aerothermal loads.

4. **Aerothermal Load Prediction:** To determine the surface distributions of heat flux ($\dot{q}$), pressure, and shear stress using the Sutton-Graves stagnation-point correlation and SPARTA surface compute quantities, with validation against IRVE-3 flight data.

5. **PINN Flow Field Refinement:** To apply Physics-Informed Neural Networks (DeepXDE) as a post-processing step to smooth noisy DSMC data, fill gaps in the flow field, and provide a continuous interpolant anchored in the compressible Navier–Stokes equations.

6. **Survivability Optimization:** To implement a Genetic Algorithm (GA) coupled with a PyTorch-based Metamodel Prognosis (MoP) for automated optimization of HIAD geometry, targeting ballistic coefficient, peak deceleration, backface temperature, and drag performance.

## 1.4 Research Gap

Based on the literature review, several gaps are identified in the current state of HIAD aerothermal analysis:

- **Rarefied Regime Limitation:** Existing HIAD aerothermal studies predominantly rely on continuum CFD (RANS/LES), which is physically inappropriate for the transitional flow regime ($Kn > 0.01$) encountered at reentry altitudes above 50 km. Particle-based DSMC methods have not been systematically applied to

HIAD geometry optimization.

- **Automated Design Space Exploration:** Current approaches require manual parametric sweeps with individual CFD runs, making high-dimensional design optimization (6+ parameters) computationally infeasible. No integrated framework exists that couples automated geometry generation with DSMC simulation and machine learning-based optimization.

- **PINN-DSMC Hybrid Approach:** While Physics-Informed Neural Networks have been applied to CFD post-processing, their use as a smoothing/interpolation layer for noisy DSMC particle data in HIAD applications is unexplored.

- **Multi-Objective Survivability Optimization:** The simultaneous optimization of ballistic coefficient, peak heating, backface temperature, and structural mass for HIAD configurations using surrogate-assisted genetic algorithms has not been demonstrated.

## 1.5 Novelty and Contributions

The present work makes the following novel contributions to the field of hypersonic aerothermodynamic analysis and HIAD design:

1. **First Integrated DSMC-PINN-Optimization Framework for HIAD:** This study presents the first computational framework that couples automated parametric geometry generation (CadQuery), high-fidelity DSMC rarefied gas dynamics (SPARTA), Physics-Informed Neural Network refinement (DeepXDE), and surrogate-assisted genetic algorithm optimization in a unified pipeline for HIAD survivability analysis.

2. **DSMC-Based Design Space Exploration:** Unlike prior studies limited to continuum CFD (RANS/LES), this framework employs particle-based DSMC to accurately capture the transitional flow regime ($0.01 < Kn < 1$) encountered at reentry altitudes above 50 km, enabling physically correct aerothermal load prediction for inflatable decelerators.

3. **PINN as DSMC Post-Processor:** The application of Physics-Informed Neural Networks as a smoothing and interpolation layer for noisy DSMC particle data is

novel in the context of HIAD aerothermal analysis. The two-stage training protocol (data-only followed by data+PDE) provides a physics-consistent continuous field from discrete particle simulations.

4. **Multi-Objective Surrogate-Assisted Optimization:** The simultaneous optimization of ballistic coefficient, peak heating, backface temperature, and drag performance using a PyTorch MoP surrogate coupled with a genetic algorithm demonstrates a computationally efficient approach to HIAD design that would be infeasible with direct simulation alone.

5. **Comprehensive ML Documentation Standards:** The framework adheres to established machine learning research standards, including explicit architecture documentation, reproducibility requirements (random seeds, environment specifications), validation metrics ($R^2$, RMSE, MAE), and uncertainty quantification through ensemble prediction.

## 1.6 Scope and Limitations

The scope of this research is defined to provide aerothermal analysis and survivability optimization of the Flexible Thermal Protection System (FTPS) within the constraints of available computational resources. The boundaries and limitations of this study are as follows:

1. **Geometric Representation:** The HIAD structure is modeled using parametric 3D CAD generation via CadQuery (Python scripting). The 6-toroid stack configuration based on the Rapisarda (2023) MDAO baseline is the primary geometry, with the option to vary toroid count, cone angle, and nose radius. The "scalloped" toroidal surface is treated as a rigid, fully-inflated geometry to isolate the impact of surface topology on the flow field.

2. **Dimensionality:** The investigation supports both 2D axisymmetric (via `-slice-angle`) and full 3D simulations. The 2D axisymmetric mode assumes zero angle of attack ($\alpha = 0^\circ$) to maintain rotational symmetry, enabling higher particle resolution within computational memory limits. Full 3D simulations are available for off-axis conditions.

3. **Numerical Framework:** The gas dynamics are governed by the Boltzmann

equation, solved using the Direct Simulation Monte Carlo (DSMC) method as implemented in the SPARTA (Sandia Parallel Algorithm for Real-Time Application) solver (Plimpton & Gallis, 2014). DSMC is valid for rarefied and transitional flows where the continuum assumption fails ($Kn > 0.01$).

4. **Gas Model:** A realistic 5-species air model ($N_2$, $O_2$, NO, N, O) with Variable Soft Sphere (VSS) collision physics is employed. This captures the molecular-level interactions including vibrational energy exchange and dissociation reactions relevant to hypersonic entry conditions.

5. **Grid and Collision Pairing:** While DSMC is a particle method, a Cartesian computational grid is required to restrict collision pairing to $O(N)$ complexity (within cells) and to provide sampling volumes for macroscopic property accumulation. The grid resolution must be finer than the local mean free path ($\lambda$). A grid-factor of 0.7 (validated against IRVE-3 flight data) is used as the default.

6. **Thermal Analysis:** Surface heat flux is estimated using the Sutton-Graves stagnation-point correlation (Sutton & Graves, 1971), validated against IRVE-3 flight data. The 1D transient backface temperature model assumes a thermal lag factor ($\eta_{lag} = 0.15$) through the FTPS insulation stack. SPARTA surface kinetic energy quantities are retained as relative shape-ranking signals for the optimizer, not as absolute heat flux values.

7. **Optimization:** The Genetic Algorithm explores the design space using Latin Hypercube Sampling (LHS) for initial population generation, with a PyTorch Multi-Layer Perceptron (MoP) as the surrogate model. PINN refinement using DeepXDE is applied during the exploration phase but is bypassed during final validation to eliminate neural network bias.

## 1.7 Methodology Overview

This study employs a multi-stage computational framework to analyze the aerothermal performance and optimize the survivability of a Hypersonic Inflatable Aerodynamic Decelerator (HIAD). The framework integrates automated parametric geometry generation, DSMC-based rarefied gas dynamics simulation, Physics-Informed Neural Network refinement, and Genetic Algorithm optimization. The methodology is designed to compare a smooth rigid sphere-cone baseline against the scalloped inflatable architecture under identical hypersonic flight conditions. The workflow is illustrated in Figure 1.3.

#### a. Literature Study

The initial stage involves a comprehensive literature review of hypersonic aerothermodynamics, rarefied gas dynamics, DSMC methodology, and HIAD design principles. Key topics include the Boltzmann equation, the Direct Simulation Monte Carlo method (Bird, 1994), the Sutton-Graves heating correlation, and the IRVE-3/LOFTID flight experiment results. This establishes the theoretical foundation for the computational framework.

#### b. Parametric Geometry Generation

The 3D CAD models of the HIAD are generated using CadQuery, a Python-based parametric CAD scripting library. The geometric parameters – including aeroshell diameter, forebody cone angle, toroid count, toroid radius, and nose radius – are defined as command-line arguments, enabling automated generation of hundreds of design variants. The output is exported as STL files for SPARTA ingestion.

#### c. DSMC Simulation

The SPARTA solver executes the DSMC simulation within a Docker-containerized Linux environment. The simulation domain is initialized with freestream conditions matching the target trajectory point (e.g., $v = 2700$ m/s, $\rho = 6.9674 \times 10^{-4}$ kg/m$^3$ at 52 km altitude from the ISA 1975 model). SPARTA tracks millions of simulated particles, modeling their advection, collisions (VSS model), and surface interactions. Surface computes accumulate force (drag, lift) and energy (heat flux) data, while grid dumps provide field maps of density, temperature, pressure, velocity, and Knudsen number.

#### d. PINN Refinement

The raw DSMC output is post-processed using a Physics-Informed Neural Network (PINN) implemented in DeepXDE. The PINN solves the 2D compressible Navier–Stokes equations using automatic differentiation, with SPARTA grid data introduced as observational boundary conditions. The total loss function combines PDE residuals with data-matching terms, providing a smooth, physics-consistent interpolant that fills gaps in the DSMC flow field and reduces particle noise.

#### e. Survivability Optimization

A Genetic Algorithm (GA) coupled with a PyTorch Multi-Layer Perceptron (MoP) surrogate model performs multi-objective optimization of the HIAD geometry. Latin Hypercube Sampling (LHS) generates the initial design population. The GA evaluates candidates using the MoP surrogate for rapid screening, with expensive SPARTA simulations reserved for elite candidates and final validation. The cost function balances ballistic coefficient, peak deceleration, backface temperature, and drag performance against user-defined targets.

#### f. Validation and Analysis

The simulation results are validated against IRVE-3 flight data (Rapisarda, 2023) using the Sutton-Graves correlation. Key validation metrics include peak heat flux (14.36 W/cm$^2$ flight vs. 13.83 W/cm$^2$ model), total heat load (195.06 J/cm$^2$ flight vs. 195.17 J/cm$^2$ model), and peak deceleration (19.7$g$ flight vs. 20.2$g$ model). The final optimized design undergoes a high-fidelity SPARTA validation run *without* PINN refinement to eliminate neural network bias.

## 1.8 Thesis Outline

The structure of this thesis is arranged as follows. A brief description of each chapter is outlined below:

**CHAPTER I** This chapter contains the research background, problem statement, research objectives, scope of study, methodology overview, and the systematic thesis outline. This section provides an initial overview regarding the rationale for conducting the research, the specific goals to be achieved, and the limitations of the research scope.

**CHAPTER II** This chapter contains the theoretical framework and literature review supporting this research. It discusses the fundamental concepts of hypersonic aerothermodynamics, the Boltzmann equation and DSMC methodology, the Sutton-Graves heating correlation, Physics-Informed Neural Networks, and genetic algorithm optimization. This chapter also reviews relevant previous studies including IRVE-3 and LOFTID flight experiments.

**CHAPTER III** This chapter details the research methodology, covering the automated parametric geometry generation using CadQuery, the SPARTA DSMC simulation configuration, the PINN refinement workflow, the Genetic Algorithm optimization pipeline, and the validation procedures against flight data. It describes the complete computational framework from geometry input to optimized design output.

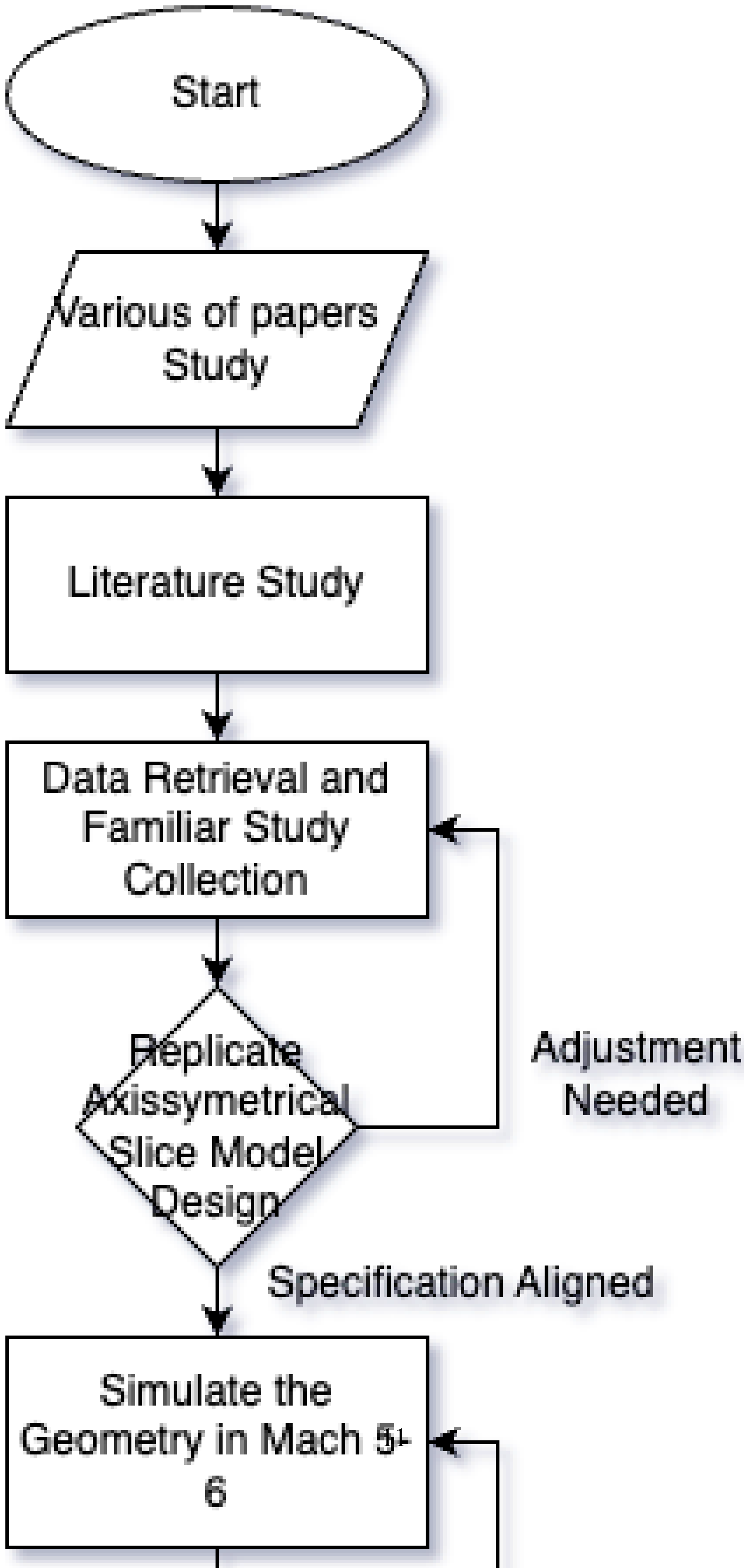
**CHAPTER IV** This chapter presents the simulation results and discussion derived from the SPARTA DSMC computational framework. It covers the baseline IRVE-3 geometry configuration, aerothermal surface distributions, flow field characteristics, validation against flight data, and a comparative analysis with prior Newtonian-panel-based approaches highlighting the methodological advances of the present DSMC framework.

**CHAPTER V** This chapter contains the summary of research findings derived from the data analysis and optimization results. It presents the final conclusions regarding the aerothermal performance and survivability optimization of the scalloped HIAD and provides recommendations for future researchers and aerospace engineers.

## Chapter Summary

This chapter has established the research context by presenting the background on hypersonic inflatable aerodynamic decelerators (HIADs), identifying the critical need for accurate aerothermal analysis in the transitional flow regime, and defining six specific research objectives. The novelty of this work lies in the first integrated DSMC-PINN-optimization framework for HIAD design, which addresses four identified research gaps in rarefied regime modeling, automated design space exploration, PINN-DSMC hybrid approaches, and multi-objective survivability optimization.

The scope of this research is bounded by the IRVE-3 baseline configuration, 2D axisymmetric and 3D simulations, 5-species air chemistry, and the Sutton-Graves heating correlation. The following chapter presents the theoretical foundations and literature review supporting the computational methodology described herein.



Start
Various of papers Study
Literature Study
Data Retrieval and Familiar Study Collection
Replicate Axissymetrical Slice Model Design
Adjustment Needed
Specification Aligned
Simulate the Geometry in Mach 5-6

# CHAPTER 2

## Literature Review

### 2.1 Review of Related Research

To establish a baseline for the expected aerothermal outcomes, a review of key flight tests and numerical studies was conducted. Table 2.1 summarizes the landmarks in HIAD research that support the triangulation of results in this study.

**Table 2.1:** Comparison of Landmark HIAD Aerothermal Research and Flight Tests

| **Research / Project** | **Key Geometry** | **Major Aerothermal Findings** | **Relevance to Outcome** |
|---|---|---|---|
| **NASA IRVE-3** (Lau et al., 2013) | 3-meter stacked-toroid HIAD. | Confirmed that "scalloping" (surface deflection) increases local heating compared to smooth surfaces. | Primary validation for the scalloped model. |
| **NASA LOFTID** (Cassell et al., 2022) | 6-meter large-scale HIAD with Gen-2 FTPS. | Demonstrated thermal survival during Earth reentry; identified transition onset on the flank. | Validates FTPS effectiveness at high enthalpy. |
| **Lichodziejewski et al. (2012)** | Deformed Outer Mold Line (OML) Analysis. | Proved that surface irregularities between tori significantly augment local heat flux. | Justifies the Config B (Smooth) comparison. |
| **HEART Project** (Cassell et al., 2014) | Optimized Nose-IAD Junction Geometry. | Found that the juncture between the nose and the IAD creates a secondary heat flux peak. | Predicts the location of local hot spots. |

Continued on next page

**Table 2.1 – continued from previous page**

| **Research / Project** | **Key Geometry** | **Major Aerothermal Findings** | **Relevance to Outcome** |
|---|---|---|---|
| **Menter (1994)** | $k$-$\omega$ SST Turbulence Model. | Superior performance in predicting separation in adverse pressure gradients. | Supports model for valley recirculation. |

### 2.1.1 Triangulation of Expected Outcomes

By synthesizing the findings of *Lau et al. (2013)* and *Lichodziejewski et al. (2012)*, it is hypothesized that the scalloped geometry in this study will generate localized heat flux peaks on the toroidal crests.

Furthermore, the implementation of the *SST* $k-\omega$ turbulence model, as validated by *Menter (1994)*, is expected to resolve the stable recirculation cells within the toroidal valleys. Finally, the use of a *Thermally Perfect Gas* model reflects the high-enthalpy conditions observed in the *LOFTID* mission, ensuring that the predicted shock physics align with real-gas observations rather than simplified ideal-gas assumptions.

This synthesis establishes the theoretical foundation for the numerical methodology described in **Chapter 3**. Figure 2.1 provides a visual mapping of each study's coverage of the physical phenomena and methodology adopted in this work.

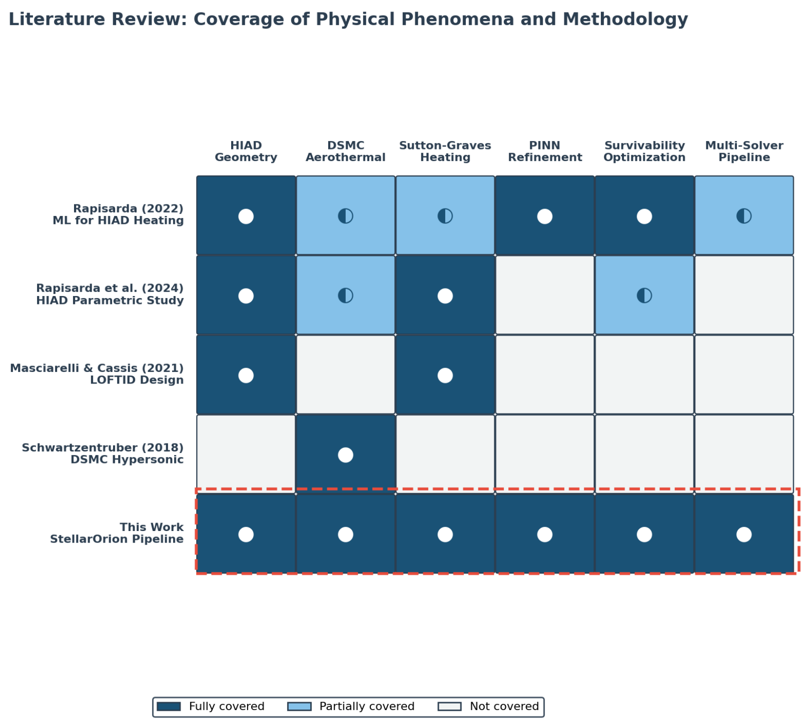



**Figure 2.1:** Literature review coverage matrix: mapping of key studies to physical phenomena and methodology components. Filled circles indicate full coverage; half-filled indicate partial coverage. The dashed red outline highlights the comprehensive coverage achieved in this work (StellarOrion pipeline).

## 2.2 Fundamentals of Hypersonic Aerodynamics

Hypersonic flow is commonly as a rule of thumb being defined as the regime where the Mach number exceeds five ($M > 5$), though this threshold is more physical than mathematical. As a vehicle's velocity increases to several kilometers per second, the kinetic energy of the flow becomes so vast that its conversion into internal energy across shock waves and within viscous boundary layers fundamentally alters the gas behavior.

According to the classical frameworks established in hypersonic theory (Anderson, 2006a), the flowfield is characterized by five distinct physical phenomena that distinguish it from supersonic flow:

1. **Thin Shock Layers:** At high Mach numbers, the shock wave lies very close to the body surface. For a blunt body, this results in a small stand-off distance, creating a high-pressure region between the shock and the vehicle.

2. **Entropy Layers:** The strong curvature of the bow shock at the nose creates a region of high entropy gradient. As this layer flows downstream, it interacts with the boundary layer, complicating the pressure and heat transfer calculations.

3. **Viscous Interaction:** The high temperature within the boundary layer reduces the local density, causing the boundary layer to grow rapidly ($\delta \propto M^2/\sqrt{Re}$). This thick layer displaces the outer inviscid flow, significantly altering the surface pressure distribution.

4. **High-Temperature Effects:** The extreme kinetic energy is dissipated into thermal energy, triggering vibrational excitation, dissociation, and even ionization of the air molecules. This renders the calorically perfect gas assumption ($c_p =$ const.) invalid.

5. **Low-Density Effects:** At very high altitudes, the continuum assumption may begin to break down as the mean free path of the molecules becomes comparable to the vehicle's characteristic length.

For Inflatable Aerodynamic Decelerators (IADs), understanding these fundamentals is critical. The "scalloped" surface topology of stacked toroids interacts directly with the thin shock layer and viscous boundary layer, creating complex flow structures that do not exist on smooth-body vehicles.

### 2.2.1 Mach Number and the Hypersonic Regime (Ma $> 5$)

The transition from supersonic to hypersonic flow is generally defined by the rule of thumb $Ma > 5$. Unlike the transition from subsonic to supersonic flow ($Ma = 1$), which is marked by a distinct physical discontinuity (the formation of shock waves), the boundary for hypersonic flow is not physically discrete. Instead, $Ma \approx 5$ represents the threshold where certain physical approximations used in linear supersonic theory lose accuracy, and high-temperature gas dynamics become significant (Anderson, 2006a).

The distinguishing characteristic of this regime is the dominance of flow kinetic energy over internal energy. From the steady-flow energy equation, the total enthalpy $h_0$ is conserved across the shock wave:

$$h_0 = h_\infty + \frac{1}{2}V_\infty^2 = h + \frac{1}{2}V^2 \tag{2.1}$$

Where $h$ is the static enthalpy and $V$ is the velocity. In the hypersonic regime, the freestream velocity $V_\infty$ is extremely high, meaning the kinetic energy term $\frac{1}{2}V_\infty^2$ is significantly larger than the static enthalpy $h_\infty$. As the flow decelerates to stagnation conditions ($V \approx 0$) at the nose of a blunt body (such as an IAD), this massive kinetic energy is converted almost entirely into internal energy (heat).

This energy conversion scales with the square of the Mach number. For a calorically perfect gas, the ratio of stagnation temperature $T_0$ to freestream temperature $T_\infty$ is given by:

$$\frac{T_0}{T_\infty} = 1 + \frac{\gamma - 1}{2} M_\infty^2 \tag{2.2}$$

For a flight at $Ma = 5$ in standard air ($\gamma = 1.4$), the temperature rises by a factor of 6. At $Ma = 10$, it rises by a factor of 21. This exponential increase in temperature necessitates the coupling of thermodynamics with aerodynamics (aerothermodynamics), as the assumption of constant specific heats ($C_p$) is no longer valid due to vibrational excitation and dissociation of gas molecules.

### 2.2.2 Shock Wave Formation and Properties

In the hypersonic regime, the vehicle moves through the atmosphere significantly faster than the speed of sound, preventing pressure disturbances from propagating upstream. Consequently, these disturbances coalesce ahead of the vehicle to form a strong shock wave, a thin region of abrupt discontinuity in fluid properties (Anderson, 2006a).

The thermodynamic changes across a normal shock wave are governed by the conservation of mass, momentum, and energy. These are collectively known as the Rankine-Hugoniot relations. For a calorically perfect gas, the ratio of downstream (post-shock, subscript 2) to upstream (pre-shock, subscript 1) properties is a function of the normal Mach number component ($M_n$):

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_n^2 - 1) \tag{2.3}$$

$$\frac{T_2}{T_1} = \left[1 + \frac{2\gamma}{\gamma + 1}(M_n^2 - 1)\right] \left[\frac{2 + (\gamma - 1)M_n^2}{(\gamma + 1)M_n^2}\right] \tag{2.4}$$

A critical distinction in hypersonic aerodynamics is the behavior of these ratios as the Mach number approaches infinity ($M \to \infty$). While pressure and temperature ratios increase indefinitely (leading to extreme aerodynamic heating), the density ratio approaches a finite limit:

$$\lim_{M\to\infty} \frac{\rho_2}{\rho_1} = \frac{\gamma+1}{\gamma-1} \tag{2.5}$$

For standard air ($\gamma = 1.4$), this limiting density ratio is $\rho_2/\rho_1 \approx 6$. This high compression means the mass flow is squeezed into a much smaller volume behind the shock compared to supersonic flow. This phenomenon is physically responsible for the formation of the Thin Shock Layer, where the shock wave "wraps" very closely around the body surface. This proximity is particularly important for the scalloped geometry of an IAD, as the shock structure may impinge upon the peaks of the toroidal rings, creating localized hotspots.

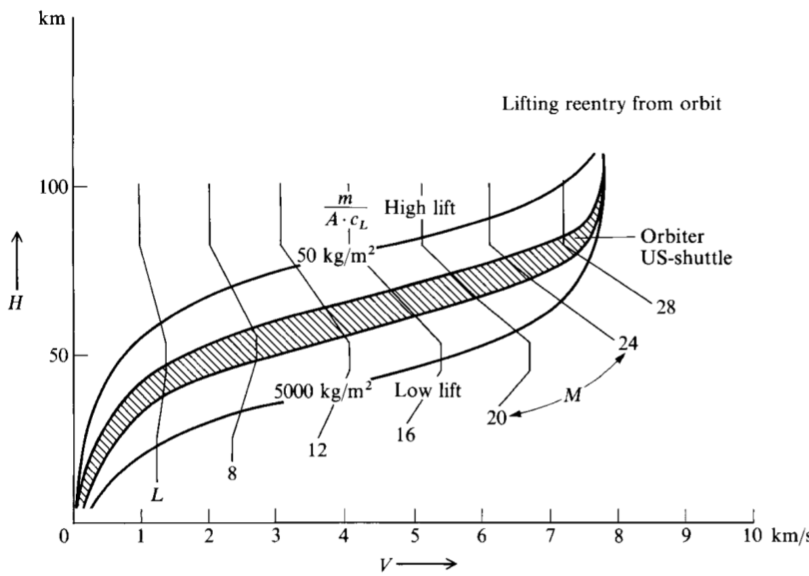



**Figure 2.2:** Figure for Shock Wave Formation and Properties Relation (Anderson, 2006)

### 2.2.3 Invalidity of Subsonic/Supersonic Assumptions (Entropy layers, Viscous interaction)

In lower-speed supersonic flows, aerodynamic analysis often decouples the inviscid outer flow from the viscous boundary layer. However, in the hypersonic regime ($Ma >$

5), this decoupling is often invalid due to two dominant phenomena: the Entropy Layer and Viscous Interaction (Anderson, 2006a).

### The Entropy Layer

For blunt bodies like an Inflatable Aerodynamic Decelerator (IAD), the bow shock is highly curved. The shock strength varies significantly from the normal shock at the stagnation streamline (strongest) to the oblique shock at the flanks (weaker). According to Crocco's Theorem, this curvature generates a strong gradient of entropy ($\Delta s$) across the streamlines behind the shock.

This creates a layer of high-entropy, vortical flow, termed the "entropy layer", that flows downstream and wets the vehicle surface. Unlike in supersonic flow, where this layer is negligible, in hypersonic flow, the entropy layer can be comparable in thickness to the boundary layer. This invalidates the standard boundary layer assumption that flow properties at the edge of the boundary layer are uniform/irrotational.

### Viscous Interaction

A fundamental assumption in classical Prandtl boundary layer theory is that the pressure gradient is impressed upon the boundary layer by the outer inviscid flow (i.e., $\partial p/\partial y \approx 0$). In hypersonic flow, the massive kinetic energy converted to heat causes the temperature within the boundary layer to rise significantly. Since pressure is constant across the layer, the density decreases inversely with temperature ($\rho \propto 1/T$), causing the boundary layer displacement thickness ($\delta^*$) to grow rapidly:

$$\delta^* \propto \frac{M_\infty^2}{\sqrt{Re_x}} \tag{2.6}$$

This thick boundary layer acts as a new "effective" shape for the vehicle, displacing the outer inviscid flow and generating a completely different pressure distribution than what the geometric shape alone would suggest. This mutual dependence, where the boundary layer determines the pressure field, and the pressure field determines the boundary layer growth, is known as Hypersonic Viscous Interaction. For the scalloped geometry of the IAD, this interaction is critical in the valleys between toroids, where thick boundary layers may bridge the gaps and alter the effective aerodynamic shape.

## 2.3 Aerothermodynamics of Re-entry Vehicles

Aerothermodynamics represents the synthesis of high-speed aerodynamics and high-temperature thermodynamics, forming the cornerstone of atmospheric re-entry vehicle design. Unlike conventional aerodynamics, where the fluid is treated as a continuum with constant chemical properties, aerothermodynamics must account for the intense coupling between the flow field and the state of the gas (Anderson, 2006a).

The fundamental challenge of re-entry is the dissipation of orbital energy. A spacecraft returning from Low Earth Orbit (LEO) possesses a velocity of approximately 7.8 km/s, while vehicles returning from lunar or Martian trajectories can exceed speeds of 11 km/s (Anderson, 2006a). To reach the surface safely, this immense kinetic energy ($\frac{1}{2}mV^2$) and potential energy ($mgh$) must be converted into other forms, primarily thermal energy, without exceeding the structural limits of the vehicle or the physiological limits of the payload.

This energy conversion occurs within the shock layer enveloping the vehicle. As the gas molecules are processed by the strong bow shock, their directed kinetic energy is randomized into internal energy, raising the static temperature of the shock layer to values often exceeding the temperature of the solar surface ($> 6000$ K). At these extreme temperatures, the air acts as a chemically reacting plasma rather than a simple gas. The nitrogen and oxygen molecules dissociate ($N_2 \rightarrow 2N$, $O_2 \rightarrow 2O$) and, at higher speeds, ionize ($N \rightarrow N^+ + e^-$), significantly altering the thermodynamic and transport properties of the fluid (e.g., specific heat ratio $\gamma$, viscosity $\mu$, and thermal conductivity $k$).

Consequently, the design of any re-entry system, including Hypersonic Inflatable Aerodynamic Decelerators (HIADs), is governed by the need to manage this heat load. The vehicle must be designed not only to withstand the integrated heat load (total heat soaked) but also the peak heat flux (maximum rate of heating) which occurs deep within the atmosphere where density increases. This creates a critical trade-off between aerodynamic deceleration (drag) and thermal management, dictating that the vehicle geometry must be optimized to maximize drag while minimizing the transfer of energy to the surface.

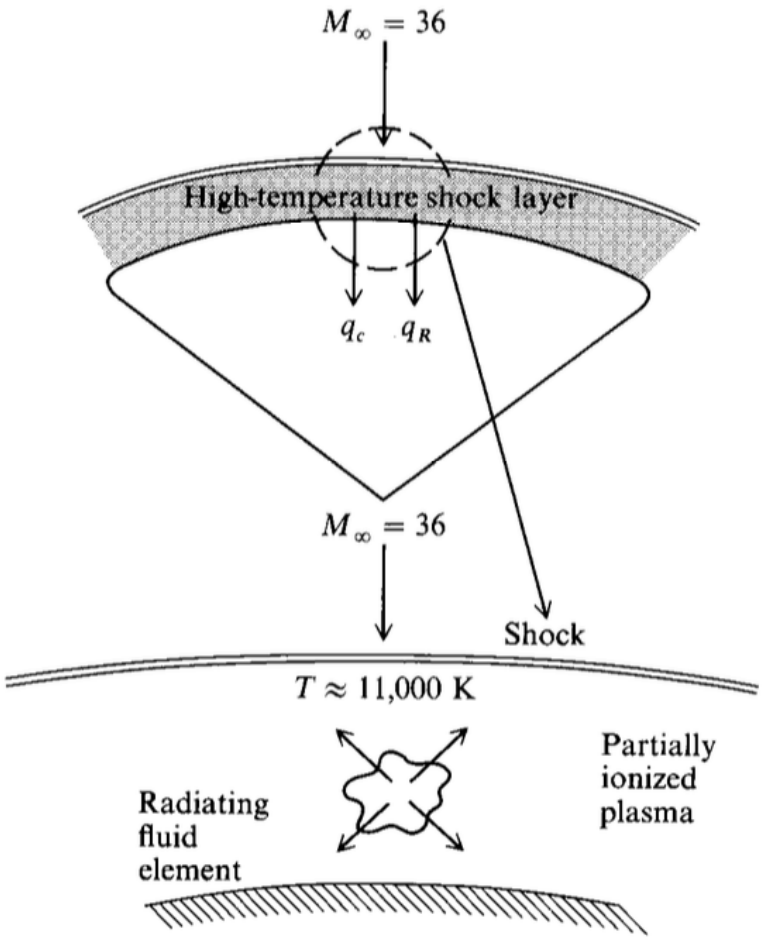



**Figure 2.3:** Figure for Aerothermodynamics of Re-entry Vehicles.

### 2.3.1 Mechanisms of Aerodynamic Heating (Kinetic to Thermal energy conversion)

The primary source of aerodynamic heating in hypersonic flight is the immense kinetic energy of the oncoming flow. As the vehicle arrests the flow, this directed kinetic energy is converted into internal thermal energy. This conversion occurs through two distinct physical mechanisms within the shock layer (Anderson, 2006a).

**1. Shock Wave Compression**

As the supersonic freestream crosses the bow shock, the flow velocity decreases abruptly, and the pressure and density rise. This process is largely adiabatic but highly non-isentropic. The work done by pressure forces on the fluid elements compresses the gas, raising the static enthalpy ($h$) significantly. At the stagnation point, effectively all the freestream kinetic energy is converted to enthalpy, defined by the total enthalpy equation:

$$h_0 = h_\infty + \frac{1}{2}V_\infty^2 \tag{2.7}$$

For a vehicle traveling at $Ma = 25$ (approximate re-entry speed), the total enthalpy is sufficient to dissociate nitrogen and oxygen, creating a chemically reacting plasma behind the shock.

**2. Viscous Dissipation within the Boundary Layer**

While the shock wave processes the bulk flow, the interaction at the vehicle surface is governed by viscosity. Within the boundary layer, the fluid velocity must vanish at the wall ($u = 0$) due to the no-slip condition. This deceleration is caused by shear stress ($\tau$) acting between adjacent fluid layers. The work done by these viscous forces against the velocity gradient generates heat, a phenomenon known as viscous dissipation.

The local rate of heat transfer to the wall ($q_w$) is determined by the thermal gradient at the surface, governed by Fourier's Law:

$$q_w = -k_w \left(\frac{\partial T}{\partial y}\right)_w \tag{2.8}$$

In hypersonic aerothermodynamics, it is convenient to express this heat flux in terms of the driving enthalpy potential. The heat flux is proportional to the difference between the recovery enthalpy ($h_{aw}$, the enthalpy of the gas if the wall were adiabatic) and the actual wall enthalpy ($h_w$):

$$q_w = \rho_e u_e C_H (h_{aw} - h_w) \tag{2.9}$$

Where $\rho_e$ and $u_e$ are the edge density and velocity, and $C_H$ is the Stanton number (heat transfer coefficient). This relationship highlights that heating is most severe where the boundary layer is thinnest (high gradients) and where the local flow energy is highest.

### 2.3.2 Bow Shock Stand-off/detachment Distance

A defining characteristic of blunt-body aerodynamics in the hypersonic regime is the detachment of the shock wave from the vehicle's nose. Unlike sharp-edged bodies where the shock may be attached to the leading edge, the large radius of curvature of an Inflatable Aerodynamic Decelerator (IAD) forces the formation of a strong, curved normal shock ahead of the stagnation point. The physical separation between the shock wave and the body along the stagnation streamline is defined as the shock stand-off distance ($\Delta$).

The magnitude of this stand-off distance is critically dependent on the density ratio across the shock. As derived in classical hypersonic theory (Anderson, 2006a), the conservation of mass requires that the mass flow entering the shock must exit through the annular area between the shock and the body. Because the post-shock density ($\rho_2$) is significantly higher than the freestream density ($\rho_\infty$), the required flow area is small, resulting in a thin shock layer.

For a spherical nose of radius $R$, the stand-off distance can be approximated by the correlation:

$$\frac{\Delta}{R} \approx \frac{\rho_\infty}{\rho_2} \tag{2.10}$$

More precise empirical correlations, such as Billig's approximation for a sphere, express this relationship as:

$$\frac{\Delta}{R} = 0.143 \exp\left(\frac{3.24}{M_\infty^2}\right) \tag{2.11}$$

This relationship has profound implications for IAD design. In hypersonic flow, where real gas effects (dissociation) can increase the density ratio $\rho_2/\rho_\infty$ well beyond the ideal gas limit of 6, the stand-off distance shrinks further. A smaller $\Delta$ increases the velocity gradients within the shock layer, which directly intensifies the convective heat transfer to the wall. For scalloped geometries, if the stand-off distance becomes comparable to the depth of the valleys between toroids, it can lead to complex shock-shock interactions or shock impingement on the structural surface.

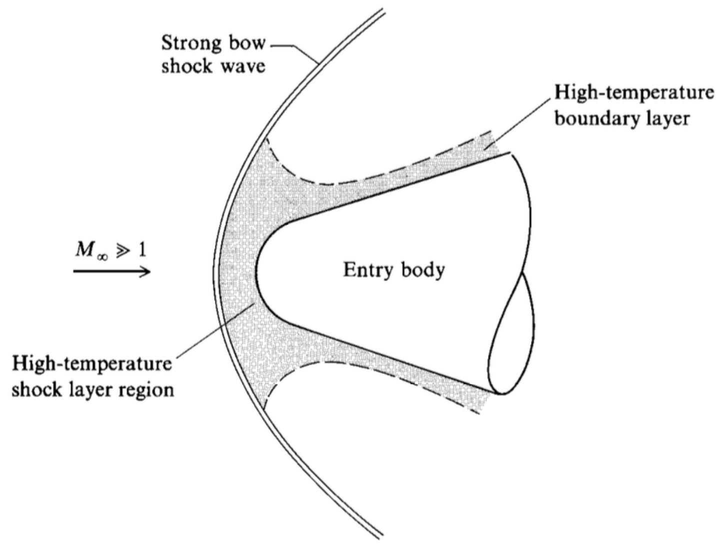



**Figure 2.4:** Figure for Bow Shock Stand-off/detachment Distance. (Anderson, 2006)

### 2.3.3 H. Julian Allen's Blunt Body Theory

In the early development of supersonic flight, the prevailing design philosophy favored sharp, slender noses to minimize aerodynamic drag. However, as speeds approached the hypersonic regime, H. Julian Allen and Alfred Eggers of NACA Ames Research Center revolutionized re-entry vehicle design with the Blunt Body Theory (1953) (Anderson, 2006a). This counter-intuitive theory established that while sharp bodies are efficient for supersonic cruise, they are catastrophic for hypersonic re-entry.

The fundamental principle rests on the energy balance of the vehicle. The total kinetic energy lost by the re-entering body is dissipated into two sinks: heating the air (via shock compression) and heating the vehicle body (via viscous friction).

1. **Sharp Body:** Creates a weak, attached shock wave. The wave drag is low, meaning the air is not heated significantly. Consequently, a large fraction of the kinetic energy is dissipated through skin friction directly on the surface, leading to extreme aerodynamic heating that can melt conventional materials.

2. **Blunt Body:** Generates a strong, detached bow shock wave. This creates massive pressure drag (wave drag), which dissipates the majority of the kinetic energy into the airflow itself, creating a high-temperature wake behind the vehicle. Because

the energy is dumped into the atmosphere rather than the structure, the heat load on the vehicle is drastically reduced.

Allen mathematically demonstrated that the stagnation point heat flux ($q_s$) is inversely proportional to the square root of the nose radius ($R_n$), while the aerodynamic drag ($D$) is directly proportional to the projected area (and thus $R_n$):

$$q_s \propto \frac{1}{\sqrt{R_n}} \quad \text{and} \quad D \propto R_n^2 \tag{2.12}$$

This relationship proves that increasing the bluntness of the vehicle not only increases drag (aiding deceleration) but simultaneously decreases the peak heat flux. This theorem is the theoretical foundation for Hypersonic Inflatable Aerodynamic Decelerators (HIADs), which utilize an extremely large effective radius to maximize drag and minimize localized heating, thereby enabling the entry of massive payloads into planetary atmospheres.

Here is example of the effect of the blunt body (Hamza, Khan, & Maqsood, 2022)

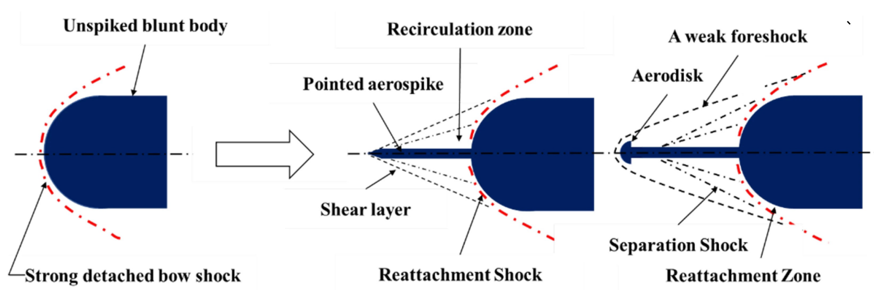



**Figure 2.5:** Figure for H. Julian Allen's Blunt Body Theory.

## 2.4 Thermal Protection Systems (TPS) Technology

The Thermal Protection System (TPS) is the single most critical subsystem for atmospheric entry vehicles, serving as the interface between the extreme aerothermal environment and the vehicle's structure. Its primary function is to maintain the inner structural bondline temperature below critical material limits (typically $< 175°$C for aluminum or composite structures) while enduring surface temperatures that can exceed 2000°C (Bouslog, 2023).

Unlike other subsystems where redundancy can be engineered (e.g., avionics or propulsion), the TPS is often a "single point of failure" system; any breach or significant failure in the thermal barrier typically results in the loss of the vehicle (Anderson, 2006a).

TPS technologies are broadly categorized based on their mechanism of heat management:

1. **Ablative TPS:** Designed for the most severe heating environments (e.g., steep re-entry or planetary probes). These materials manage heat loads through phase change (pyrolysis), chemically decomposing to create a char layer and releasing gases that thicken the boundary layer and block convective heating (blowing effect). Examples include PICA (Phenolic Impregnated Carbon Ablator) and Avcoat (Bouslog, 2023).

2. **Reusable TPS:** Designed for missions requiring multiple flights with minimal refurbishment (e.g., Space Shuttle, X-37B). These systems rely on high-emissivity coatings to re-radiate a significant portion of the incident heat flux back into the atmosphere ($q_{rad} \propto \varepsilon T^4$) while acting as an insulator to retard heat conduction to the substructure. They do not undergo phase change and are typically limited to lower heat fluxes than ablators.

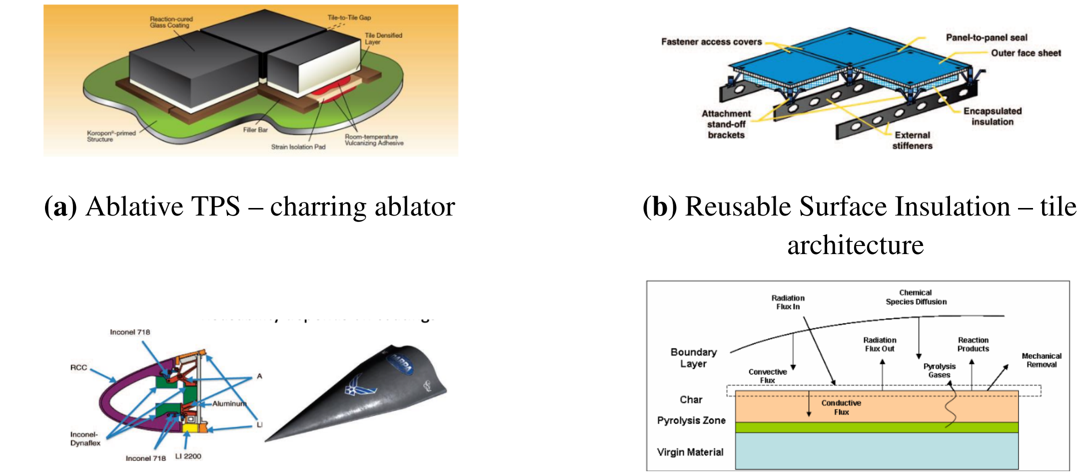
Historically, TPS has been synonymous with rigid architectures, such as tiles, bricks, or heavy monolithic shields. However, the geometric limitations of rigid aeroshells (constrained by the launch vehicle fairing diameter) have necessitated the development of flexible, deployable architectures like HIADs to enable the landing of heavier payloads on Mars (DiNonno & Cheatwood, 2024).

### 2.4.1 Conventional Rigid TPS (Ablative and Re-usable tiles)

For decades, the standard for atmospheric entry has been rigid TPS, which can be broadly categorized into three architectures based on their functionality and integration with the vehicle structure (Bouslog, 2023).

#### 1. Ablative TPS

Ablative systems are the workhorses of high-energy planetary entry (e.g., Apollo, Mars Science Laboratory). They function by sacrificing material to manage the extreme




**(a)** Ablative TPS – charring ablator

**(b)** Reusable Surface Insulation – tile architecture

**(c)** Hot Structures – C/C or CMC nose cap

**(d)** Flexible TPS – HIAD inflatable stack

**Figure 2.6:** Classification of Thermal Protection Systems (TPS) Technology (Bouslog, 2023).

heat load. As the surface temperature rises, the polymer matrix undergoes pyrolysis, an endothermic chemical decomposition that absorbs energy. This process generates a porous char layer that radiates heat away ($\varepsilon \approx 0.9$) and releases pyrolysis gases (blowing) that thicken the boundary layer, physically pushing the high-temperature shock layer away from the wall.

- **Example:** PICA (Phenolic Impregnated Carbon Ablator) used on SpaceX Dragon and NASA's Stardust mission.
- **Limitation:** They are typically single-use and heavy, and their rigid nature limits the vehicle diameter to the size of the launch fairing.

### 2. Reusable Surface Insulation (RSI)

Made famous by the Space Shuttle Orbiter, these systems consist of lightweight ceramic tiles or blankets (e.g., LI-900 silica tiles) designed to survive thousands of degrees without ablating. They rely on extremely low thermal conductivity to insulate the aluminum airframe and high surface emissivity to re-radiate up to 90% of the incident heat back into space.

- **Example:** HRSI (High-Temperature Reusable Surface Insulation) black tiles.
- **Limitation:** They are extremely fragile, labor-intensive to install/maintain, and

cannot withstand the heat fluxes associated with high-speed lunar or Martian return trajectories.

### 3. Hot Structures

In this architecture, the primary load-bearing structure itself serves as the thermal protection. Materials like Carbon-Carbon (C-C) or ceramic matrix composites (CMCs) are used for nose caps and wing leading edges where temperatures are highest ($> 1500°$C).

- **Example:** The Reinforced Carbon-Carbon (RCC) nose cap of the Space Shuttle.

While effective, these rigid technologies fundamentally constrain the ballistic coefficient ($m/C_dA$) of the vehicle. To land heavier payloads on Mars, the drag area ($A$) must be increased beyond the limits of rigid fairings, necessitating the shift toward flexible, deployable architectures (DiNonno & Cheatwood, 2024).


**Figure 2.7:** Figure for Conventional Rigid TPS.

### 2.4.2 Hypersonic Inflatable Aerodynamic Decelerators (HIAD/LOFTID)

To overcome the diameter constraints imposed by launch vehicle fairings, NASA and commercial partners have developed Hypersonic Inflatable Aerodynamic Decelerators (HIADs). A HIAD is a deployable aeroshell consisting of an inflatable structural backbone covered by a Flexible Thermal Protection System (FTPS) (DiNonno & Cheatwood, 2024).

### Structural Architecture

The structural core of a HIAD typically comprises a stack of pressurized toroidal tubes (tori) made of high-strength braided synthetic fibers (e.g., Kevlar or Vectran).

These toroids are strapped together to form a blunt cone shape. Unlike a smooth rigid shell, this construction inherently creates a "scalloped" or wavy surface topology, where the radius of each toroid creates a series of peaks (crests) and valleys. This specific geometry is the focus of the aerodynamic analysis in this thesis, as it introduces complex flow interactions not present on smooth bodies.

### The LOFTID Mission

The technology reached a major milestone with the Low-Earth Orbit Flight Test of an Inflatable Decelerator (LOFTID), successfully conducted in August 2015. The LOFTID vehicle, a 6-meter diameter aeroshell, demonstrated the ability of an inflatable structure to survive the hypersonic heat pulse of re-entry ($> 1400°C$) and successfully decelerate a payload from orbital velocity (DiNonno & Cheatwood, 2024).

LOFTID validated two critical advantages of this technology:

- **Packaging Efficiency:** The entire 6-meter aeroshell was packed into a space significantly smaller than its deployed diameter, allowing it to fit within a standard Atlas V payload fairing.
- **Ballistic Coefficient Management:** By deploying a large drag area ($A$), the ballistic coefficient ($\beta = m/C_d A$) is drastically reduced. This allows deceleration to occur higher in the atmosphere where density (and thus heating) is lower, a capability essential for landing heavy payloads on the high-elevation surface of Mars.

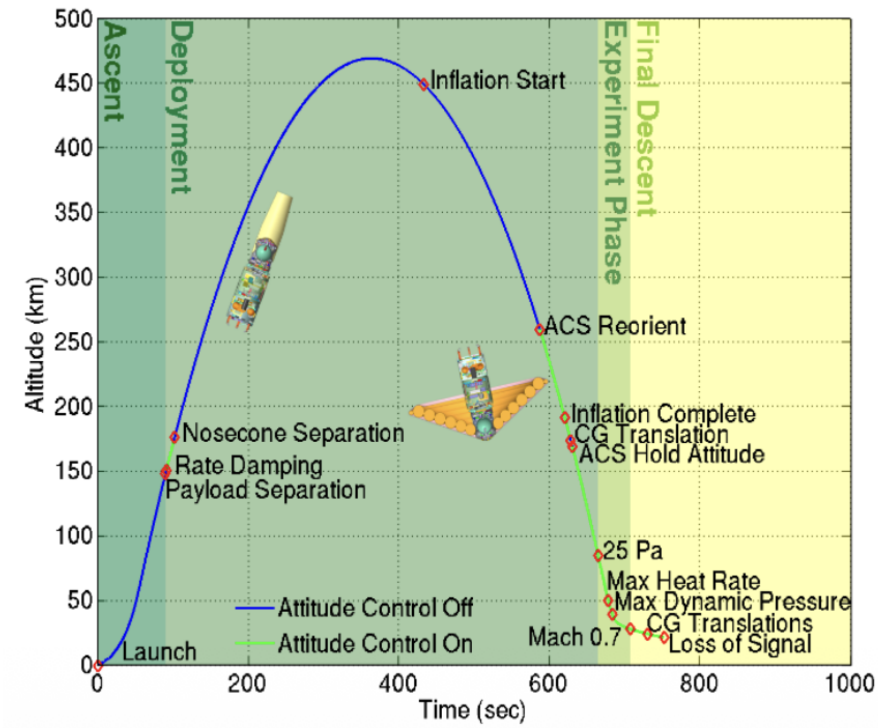



**Figure 2.8:** LOFTID Mission Trajectory

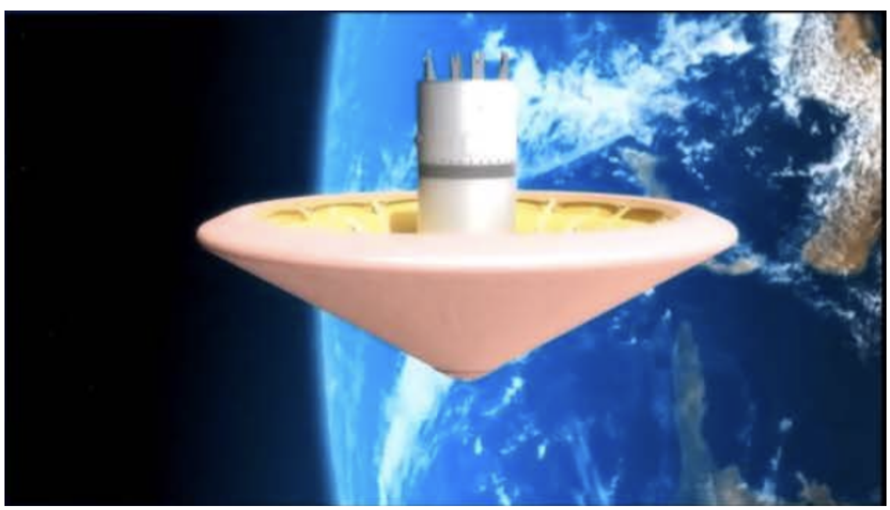


**Figure 2.9:** Figure for HIAD/LOFTID.

### 2.4.3 Comparison of Rigid vs. Flexible Structures

The transition from conventional rigid aeroshells to flexible inflatable decelerators represents a paradigm shift in entry system design. While rigid TPS relies on a fixed outer mold line (OML) to maintain aerodynamic stability, flexible structures introduce a dynamic coupling between the aerodynamic loads and the structural response (DiNonno & Cheatwood, 2024).

The primary distinction lies in the management of the Ballistic Coefficient ($\beta$). Rigid vehicles are volume-constrained by the launch shroud, forcing a high $\beta$ which results in peak heating occurring deep in the atmosphere. In contrast, HIADs decouple the entry diameter from the launch diameter, allowing for a much lower $\beta$. This enables deceleration at higher altitudes, significantly reducing the heat flux requirements.

However, the flexibility of the IAD introduces complexities absent in rigid designs. The "scalloped" surface of the stacked toroids creates a rough aerodynamic profile that can augment local heating, and the structural deformation under drag loads can alter the vehicle's stability characteristics.

A comparative summary of the two architectures is presented in Table 2.2.

**Table 2.2:** Comparison of Conventional Rigid TPS and Hypersonic Inflatable Aerodynamic Decelerators (HIAD).

| Parameter | Rigid Aeroshell (Capsule) | Flexible/Inflatable (HIAD) |
|---|---|---|
| **Launch Constraint** | Diameter limited by launch vehicle fairing (e.g., 4-5 m). | Packable; deployed diameter can exceed fairing size (e.g., $>$ 10 m). |
| **Surface Geometry** | Smooth, continuous Outer Mold Line (OML). | Scalloped/Wavy surface due to stacked toroids. |
| **Aerodynamic Drag** | Lower drag area ($A$); higher Ballistic Coefficient. | Massive drag area ($A$); lower Ballistic Coefficient. |
| **Thermal Management** | Ablation (phase change) or Re-radiation (tiles). | Flexible insulation layers (FTPS) with gas barriers. |
| **Structural Response** | Static; shape remains constant under load. | Dynamic; structure deforms/deflects under aerodynamic load (FSI). |
| **Mission Profile** | Lunar return, LEO return, small Mars probes. | Heavy-mass Mars landers, engine recovery. |

## 2.5 Impact of Surface Geometry on Flow Physics

The aerodynamic design of conventional re-entry vehicles typically prioritizes smooth, continuous Outer Mold Lines (OML) to ensure predictable boundary layer development and uniform heat distribution. However, the structural architecture of a HIAD, composed of stacked pressurized toroids, inherently introduces a "scalloped" or wavy surface topology.

This deviation from a smooth surface fundamentally alters the flow physics. To understand these effects, it is useful to draw a comparison with the classic "golf ball" phenomenon in fluid mechanics. In the subsonic regime, the dimples on a golf ball act as surface roughness elements that trip the laminar boundary layer into turbulence. This energized turbulent boundary layer is better able to resist adverse pressure gradients, thereby delaying flow separation and significantly reducing pressure drag (form drag).

However, in the hypersonic regime of an IAD, the physics of surface waviness serve a completely different dynamic:

- **Scale:** The "valleys" between toroids are macro-scale features, not micro-roughness.
- **Objective:** Unlike a golf ball where the goal is drag *reduction*, the goal of an IAD is drag *maximization*.
- **Thermal Consequence:** While turbulence on a golf ball is purely beneficial for aerodynamics, turbulence on a hypersonic vehicle creates a severe penalty: it can increase surface heat flux by a factor of 3 to 5 compared to laminar flow (Anderson, 2006a).

Therefore, the study of the scalloped topology is not focused on drag reduction, but rather on the Aerothermal specifically, how the peaks (crests) and valleys of the toroids modify the shock structure and local heating rates. (White, 2006)

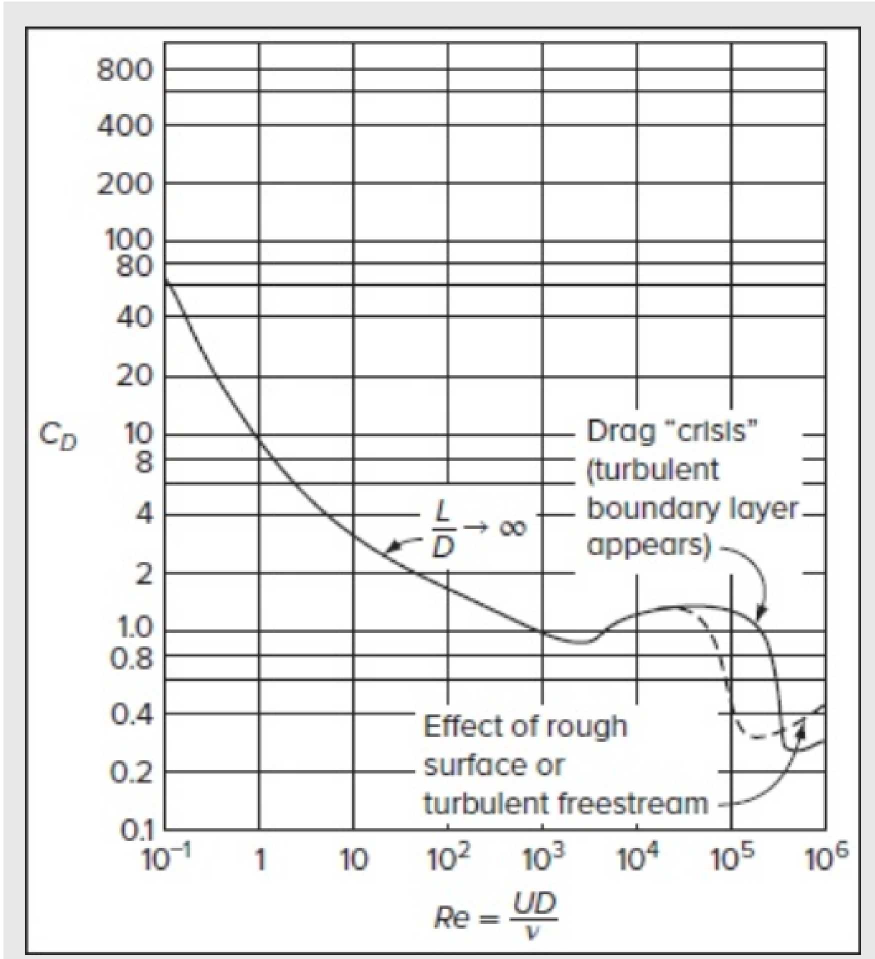



**Figure 2.10:** Figure for Impact of Surface Topology on Re Number.

### 2.5.1 Flow over Wavy and Scalloped Walls

The surface of an IAD can be mathematically approximated as a periodic wavy wall or a series of cavity steps. As the hypersonic flow traverses this undulating surface, it encounters alternating regions of favorable and adverse pressure gradients, a phenomenon fundamentally distinct from flow over a flat plate. The physics can be derived by analyzing the momentum equation along the streamline. As the fluid moves over a convex crest (the peak of a toroid), the flow area effectively constricts, causing the flow to accelerate. According to Bernoulli's principle (and its compressible counterpart), this acceleration results in a local decrease in static pressure:

$$\frac{dp}{dx} < 0 \quad \text{(Favorable Gradient - Crest)} \tag{2.13}$$

Conversely, as the flow descends into the concave valley between toroids, the flow area expands and the fluid decelerates. This deceleration generates a localized high-pressure region:

$$\frac{dp}{dx} > 0 \quad \text{(Adverse Gradient - Valley)} \tag{2.14}$$

This creates a recurring cycle of expansion and compression.

- **Over the Crests:** The flow undergoes an expansion fan process, turning away from the shock. The boundary layer thins, and the shear stress ($\tau_w$) increases, leading to peak heating.
- **In the Valleys:** The adverse pressure gradient forces the boundary layer to thicken. If the gradient is strong enough (i.e., the valley is deep), the momentum of the fluid near the wall is insufficient to overcome the pressure rise. This causes the flow to detach, creating a separation bubble or recirculation zone similar to the flow inside a golf ball dimple, but on a macroscopic scale.

This separation is critical for thermal management. In an ideal scenario, a stable recirculation vortex forms in the valley, allowing the high-energy outer flow to "skim" over the top of the gap. This effectively shields the valley floor from the extreme heat of the freestream, a mechanism analogous to the "trapped vortex" combustor concepts in propulsion.

### 2.5.2 Shock-Boundary Layer Interaction (SBLI) on Rough Surfaces

On a smooth hypersonic vehicle, the boundary layer typically grows monotonically from the stagnation point. However, the scalloped surface of an IAD acts as a "rough" wall with large-scale asperities, forcing a complex coupling between the inviscid shock system and the viscous boundary layer, known as Shock-Wave/Boundary-Layer Interaction (SBLI).

This interaction is driven by the compression waves generated at the windward face of each toroid. As the flow exits a valley and encounters the upstream face of the next toroid, the geometry forces the flow to turn into itself (concave turning). In supersonic flow, this turning generates a compression wave or an oblique shock.

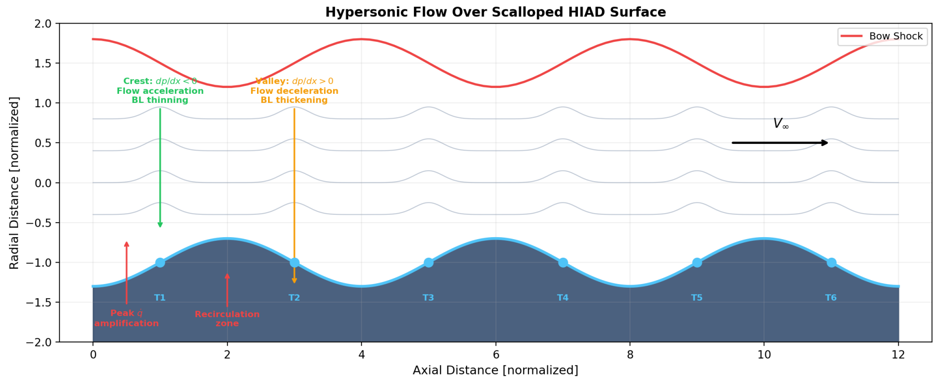



**Figure 2.11:** Schematic of hypersonic flow over a periodic wavy (scalloped) wall surface. Expansion fans form at the convex crests where the flow accelerates ($dp/dx < 0$), while compression and separation occur in the concave valleys where the adverse pressure gradient ($dp/dx > 0$) causes boundary layer thickening and recirculation zones. The trapped vortex mechanism in the valleys provides thermal shielding of the valley floor from the freestream.

This localized shock wave penetrates the boundary layer, imposing a strong adverse pressure gradient ($\frac{dp}{dx} > 0$).

- **Separation:** If the shock is strong enough, the adverse pressure gradient can propagate upstream through the subsonic portion of the boundary layer, causing the flow to separate before it even reaches the valley floor.
- **Reattachment Shock:** When the separated shear layer reattaches to the crest of the next toroid, the flow is forced to align with the wall again. This re-alignment generates a Reattachment Shock.

The intersection of this reattachment shock with the boundary layer is a critical design concern. At the point of reattachment, the boundary layer is extremely thin (high velocity gradient), and the impinging shock further compresses the fluid. This combination results in a localized spike in both pressure and heat flux, which can be significantly higher than the stagnation point heating values predicted for a smooth body (Anderson, 2006a).(Wang et al., 2023)

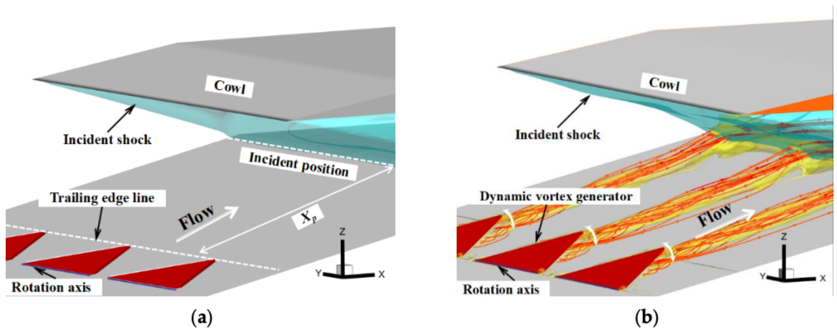



**Figure 2.12:** Figure for Shock-Boundary Layer Interaction (SBLI) or (SWBLI) on Rough Surfaces.

### 2.5.3 Recirculation Zones and Stagnation in Surface Valleys

#### The Shear Layer and Vortex Formation

This free shear layer acts as a fluid dynamic boundary.

- **Shear Layer:** A thin region of high vorticity where the velocity transitions from the freestream speed ($u_e$) to the near-zero velocity of the cavity.
- **Primary Vortex:** Driven by the shear stress at the interface, the fluid mass trapped within the valley is set into rotational motion, creating a large, stable recirculation structure known as the primary vortex.

#### Stagnation and Heat Alleviation

Ideally, this recirculation zone creates a "dead air" region. The fluid deep within the valley has very low kinetic energy compared to the freestream. Consequently, the convective heat transfer is significantly reduced because the shear layer shields the valley floor from the high-enthalpy gradients of the shock layer.

However, the stability of this zone is dependent on the geometry of the valley, specifically the length-to-depth ratio ($L/D$).

- **Open Cavity ($L/D < 10$):** The shear layer bridges the entire gap and reattaches at the downstream corner (next crest). This is the desired state for an IAD, as it maintains low heating in the valley.
- **Closed Cavity ($L/D > 10$):** If the valley is too wide or shallow, the shear layer

may curve inward and impinges on the valley floor. This stagnation of the shear layer on the bottom surface would result in a local "hot spot" of high pressure and heating, negating the thermal benefits of the scalloped design.

This is as for example what is recirculation effect on an structure (Petronio, Roman, Nasello, & Armenio, 2013)

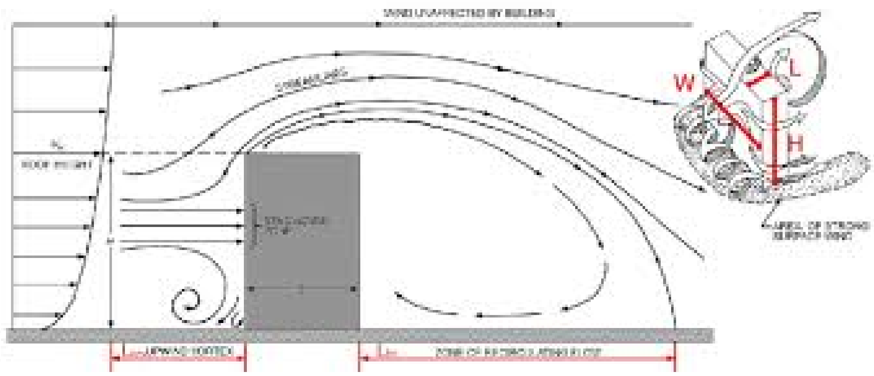



**Figure 2.13:** Figure for Recirculation Zones and Stagnation Valleys Example on different scenarios.

### 2.5.4 Localized Peak Heating Effects

While the recirculation zones in the valleys may alleviate thermal loads, the protruding crests of the toroidal rings experience the opposite effect: severe localized heating. This phenomenon is governed by the mathematics of stagnation point flow, where the local geometry creates conditions significantly more severe than the average heating over the vehicle.

#### The Stagnation Temperature (Energy Potential)

The potential for heat transfer is driven by the total energy of the flow. As the fluid decelerates to zero velocity at a stagnation point (such as the windward face of a toroid), its kinetic energy is converted to enthalpy. For a calorically perfect gas, the stagnation temperature $T_0$ is a function of the freestream Mach number $M_\infty$ (Anderson, 2006a):

$$\frac{T_0}{T_\infty} = 1 + \frac{\gamma - 1}{2} M_\infty^2 \tag{2.15}$$

For a hypersonic flow of $Ma = 5$ (where $\gamma = 1.4$), the stagnation temperature is six times the freestream temperature. This $T_0$ represents the driving potential for heat transfer.

**The Geometric Scaling (Heat Flux Rate)**

While $T_0$ determines the *temperature* limit, the *rate* at which heat enters the structure ($q_s$) is determined by the velocity gradient at the wall. According to the classical Fay-Riddell correlation, the stagnation point heat flux scales inversely with the square root of the nose radius ($R_n$):

$$q_s \approx 0.76 \cdot Pr^{-0.6} \cdot (h_0 - h_w) \cdot \sqrt{\rho_e \mu_e} \cdot \sqrt{\left(\frac{du_e}{dx}\right)_s} \tag{2.16}$$

The velocity gradient term $\left(\frac{du_e}{dx}\right)_s$ for a blunt body is geometrically dependent on the radius of curvature $R$:

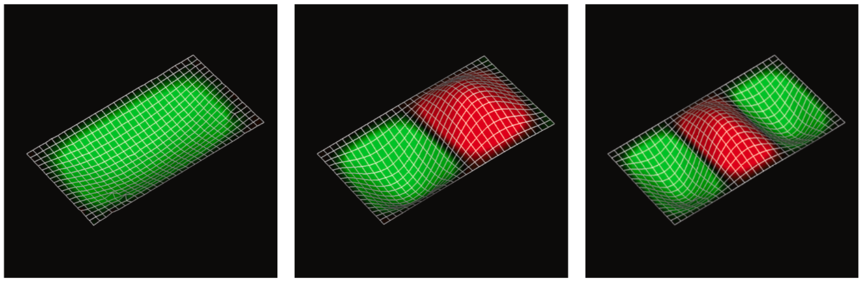
$$\left(\frac{du_e}{dx}\right)_s \approx \frac{V_\infty}{R}\sqrt{\frac{2\rho_\infty}{\rho_0}} \tag{2.17}$$

Substituting this back into the heat flux equation yields the critical proportionality:

$$q_s \propto \sqrt{\frac{1}{R}} \tag{2.18}$$

This derivation highlights the critical vulnerability of scalloped IAD geometries. While the *overall* effective radius of the vehicle is large (low global heating), the *local* radius of an individual toroid tube ($R_{tube}$) is small. Consequently, the crest of each toroid acts as a local stagnation point with a small radius, leading to "peak heating" spikes that can be significantly higher than the heating on a smooth sphere-cone of the same size.

It is demonstrated on the analogue of the localized heating (Daub, Willems, & Gülhan, 2022)



(a) 255 Hz (b) 341 Hz (c) 436 Hz

**Figure 2.14:** Figure for Localized Peak Heating Effects in Amplification but through Osicaltion.

### 2.5.5 Stagnation Point Heating: The Sutton–Graves Correlation

The Fay–Riddell equation, while physically complete, requires detailed knowledge of the post-shock velocity gradient and boundary-layer edge properties. For engineering design and rapid parametric studies, the *Sutton–Graves* correlation provides a closed-form approximation of the stagnation-point convective heat flux (Sutton & Graves, 1971):

$$\dot{q}_s = C_{SG}\sqrt{\frac{\rho_\infty}{R_n}}V_\infty^3 \tag{2.19}$$

where $C_{SG}$ is a gas-model-dependent coefficient, $\rho_\infty$ is the freestream density, $R_n$ is the nose radius, and $V_\infty$ is the freestream velocity. For a 5-species chemically reacting air model at hypersonic velocities, $C_{SG} \approx 1.7415 \times 10^{-4}$ in SI units (Rapisarda, 2023). This correlation captures the $V^3$ velocity dependence that arises from the combined effects of shock compression and boundary-layer energy transport, and is used throughout this study for baseline thermal assessment.

### 2.5.6 Flight Performance Metrics

To translate surface force and heat-flux data from a DSMC simulation into engineering performance quantities, several derived metrics are computed. These metrics form the optimization objective space for the Survivability-Based Optimization (SBO) framework described in Chapter 3.

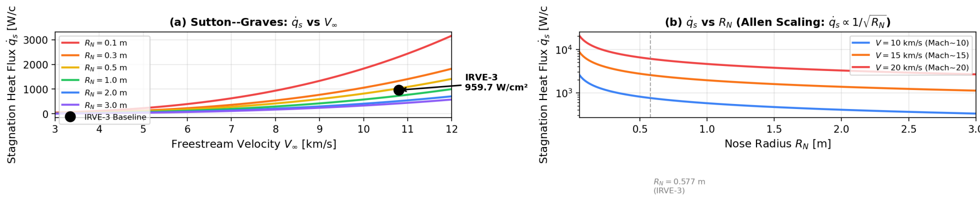



**Figure 2.15:** Parametric evaluation of the Sutton–Graves stagnation heat flux correlation (Eq. 2.19). (Left) Variation of $\dot{q}_s$ with freestream velocity $V_\infty$ for nose radii $R_n = 0.1$–$3.0$ m, demonstrating the cubic velocity dependence. The IRVE-3 baseline condition ($R_n = 0.55$ m, $V_\infty = 2.7$ km/s) is highlighted. (Right) Variation of $\dot{q}_s$ with nose radius at fixed Mach numbers, illustrating the $1/\sqrt{R_n}$ scaling that motivates the blunt-body design philosophy.

### Mass Density from Number Density

DSMC solvers such as SPARTA report the number density $n_\rho$ (particles per cubic metre). The physical mass density is recovered via the mean molecular mass of the gas mixture:

$$\rho = n_\rho \cdot \frac{M_{\text{air}}}{N_A} \tag{2.20}$$

where $M_{\text{air}} \approx 28.97$ g/mol is the molar mass of air and $N_A = 6.022 \times 10^{23}$ mol$^{-1}$ is Avogadro's constant (Anderson, 2006b).

### Dynamic Pressure

The dynamic pressure quantifies the kinetic energy flux of the freestream and is defined as:

$$q = \frac{1}{2} \rho V_\infty^2 \tag{2.21}$$

This quantity appears in the definition of aerodynamic coefficients and is a primary driver of structural loading on the HIAD.

### Ballistic Coefficient

The ballistic coefficient $\beta$ measures the vehicle's ability to decelerate against aerodynamic drag:

$$\beta = \frac{m \cdot q}{F_{\text{drag}}} \tag{2.22}$$

where $m$ is the vehicle mass and $F_{\text{drag}}$ is the total drag force obtained by integrating the surface pressure and shear stress over the vehicle. A lower $\beta$ indicates more effective deceleration, which is the primary objective for HIAD design (Anderson, 2006b).

#### Instantaneous Deceleration (g-load)

The deceleration experienced by the vehicle, expressed in multiples of Earth's standard gravity $g_0 = 9.81$ m/s$^2$, is:

$$n = \frac{F_{\text{drag}}}{m \cdot g_0} \tag{2.23}$$

The peak g-load is a critical structural and payload constraint; for the IRVE-3 mission, the recorded peak was approximately $19.7\,g$ (Olds et al., 2013).

### 2.5.7 Thermal Protection System Thermal Response

The aerothermal loads on the HIAD must be converted into temperature predictions for the Thermal Protection System (TPS). Two complementary models are employed.

#### Radiative Equilibrium Surface Temperature

At the outer surface of the TPS, the incoming convective heat flux $\dot{q}$ is balanced by radiative emission to the environment. Under the assumption of radiative equilibrium (Anderson, 2006b):

$$T_{\text{surface}} = \left( \frac{\dot{q}}{\sigma \varepsilon} \right)^{1/4} \tag{2.24}$$

where $\sigma = 5.67 \times 10^{-8}$ W/(m$^2$·K$^4$) is the Stefan–Boltzmann constant and $\varepsilon$ is the surface emissivity. This equation governs the peak surface temperature that the TPS material must withstand.

**One-Dimensional Backface Temperature**

For flexible TPS architectures (e.g., Aerogel/Spylon stacks on LOFTID), the transient heat conduction through the insulation layer is approximated by a one-dimensional thermal lag model (Hollis, Berry, Hollingsworth, & Wright, 2017):

$$T_{\text{back}} = T_{\text{init}} + \frac{\dot{q} \cdot \Delta t \cdot \eta_{\text{lag}}}{\rho_{\text{TPS}} \cdot C_{p,\text{TPS}} \cdot \delta_{\text{TPS}}} \tag{2.25}$$

where $T_{\text{init}}$ is the initial temperature, $\Delta t$ is the heat pulse duration, $\eta_{\text{lag}}$ is a thermal lag efficiency factor (typically $\sim 0.15$), $\rho_{\text{TPS}}$ is the TPS material density, $C_{p,\text{TPS}}$ is its specific heat capacity, and $\delta_{\text{TPS}}$ is the insulation thickness. The backface temperature $T_{\text{back}}$ must remain below the structural limit of the underlying substructure (typically $\leq 350$ K for the HIAD toroidal frame).

## 2.6 Theoretical Framework of CFD

Computational Fluid Dynamics (CFD) serves as the third pillar of fluid dynamics research, complementing theoretical analysis and experimental testing. It involves the systematic application of numerical methods to solve the governing equations of fluid motion, transforming the continuous mathematical models into discrete algebraic equations that can be solved computationally (Versteeg & Malalasekera, 2007).

For hypersonic research, where flight tests are prohibitively expensive and ground-based facilities (such as wind tunnels) often suffer from scaling limitations or wall interference, CFD provides a critical capability to resolve complex flow physics. By discretizing the spatial domain into a grid, CFD allows for the detailed interrogation of flow properties, such as shock structures and thermal gradients, that are often difficult to measure experimentally.

The theoretical foundation of this analysis rests upon the Continuum Hypothesis. This assumption treats the fluid not as a collection of discrete molecules, but as a continuous substance where macroscopic properties (density, pressure, velocity, and temperature) are well-defined at every point in space and time. Based on this hypothesis, the flow physics are governed by three fundamental conservation laws:

- Conservation of Mass (Continuity)

- Conservation of Momentum (Newton's Second Law)
- Conservation of Energy (First Law of Thermodynamics)

### 2.6.1 The Navier-Stokes Equations (Conservation of Momentum)

The conservation of momentum is based on Newton's Second Law of Motion, which states that the rate of change of momentum of a fluid particle is equal to the sum of the forces acting on it.

For an infinitesimal fluid element of volume $dV = dxdydz$, the forces are categorized into two types:

- **Surface Forces:** Forces acting on the boundaries of the element, including pressure ($p$) and viscous stresses ($\tau$).
- **Body Forces:** Forces acting on the entire mass of the element (e.g., gravity), denoted by the source term $S_M$.

**1. The Momentum Balance**

Consider the $x$-component of momentum. The net force due to surface stresses in the $x$-direction is the sum of the gradients of the normal stress ($\sigma_{xx}$) and shear stresses ($\tau_{yx}, \tau_{zx}$). The general differential form for the conservation of momentum in the $i$-direction (where $i = x, y, z$) is given by (Versteeg & Malalasekera, 2007):

$$\frac{\partial(\rho u_i)}{\partial t} + \nabla \cdot (\rho u_i \mathbf{u}) = \nabla \cdot \sigma_{ij} + S_{M,i} \tag{2.26}$$

Where the left-hand side represents the rate of change of momentum (transient + convective), and the right-hand side represents the forces. The stress tensor $\sigma_{ij}$ includes the thermodynamic pressure and the viscous stress tensor $\tau_{ij}$:

$$\sigma_{ij} = -p\delta_{ij} + \tau_{ij} \tag{2.27}$$

**2. Constitutive Relations (Newtonian Fluid)**

To close the system of equations, the viscous stresses must be related to the velocity gradients (strain rates). For a **Newtonian fluid**, the viscous stress is proportional to the rate of deformation. This relationship is defined by:

$$\tau_{ij} = \mu \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \lambda \delta_{ij} (\nabla \cdot \mathbf{u}) \tag{2.28}$$

Here, $\mu$ is the dynamic viscosity, and $\lambda$ is the second coefficient of viscosity. Following Stokes' hypothesis for monoatomic gases, we assume $\lambda = -\frac{2}{3}\mu$. This leads to the final form of the viscous stress tensor for a compressible flow:

$$\tau_{ij} = \mu \left[ \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} (\nabla \cdot \mathbf{u}) \delta_{ij} \right] \tag{2.29}$$

**3. The Complete Navier-Stokes Equation**

Substituting the constitutive relation back into the momentum equation yields the complete Compressible Navier-Stokes Equation. In vector notation, it is written as:

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot \left[ \mu \left( \nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I} \right) \right] + \mathbf{S}_M \tag{2.30}$$

This equation describes how the velocity field evolves under the influence of pressure gradients, viscous diffusion, and external forces. For hypersonic flows, the compressibility term $(\nabla \cdot \mathbf{u})$ is non-zero and significant, coupling the momentum equation strongly with the energy equation.

### 2.6.2 Conservation of Mass and Energy Equations

In addition to momentum, the numerical solution must satisfy the fundamental principles of mass conservation and the First Law of Thermodynamics.

**1. Conservation of Mass (Continuity Equation)**

The continuity equation describes the mass balance within a control volume. It states that the rate of increase of mass inside the fluid element is equal to the net rate of flow of mass into the element across its faces.

For a compressible fluid, where density $\rho$ is a variable, the differential form is given by (Versteeg & Malalasekera, 2007):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \tag{2.31}$$

The first term $\frac{\partial \rho}{\partial t}$ represents the rate of change of density (zero for steady flows), and the second term $\nabla \cdot (\rho \mathbf{u})$ represents the convective mass flux (divergence of the mass flow rate).

**2. Conservation of Energy**

For hypersonic flows, the energy equation is the most critical component of the governing set, as it couples the flow velocity to the thermodynamic state (temperature). Based on the First Law of Thermodynamics, the rate of change of energy of a fluid particle is equal to the rate of heat added to the fluid plus the rate of work done on the fluid.

The total energy $E$ per unit mass is defined as the sum of internal energy ($i$) and kinetic energy:

$$E = i + \frac{1}{2}|\mathbf{u}|^2 \tag{2.32}$$

The conservative form of the energy equation is written as (Versteeg & Malalasekera, 2007):

$$\frac{\partial (\rho E)}{\partial t} + \nabla \cdot (\rho E \mathbf{u}) = \nabla \cdot (k \nabla T) + \nabla \cdot (\tau_{ij} \cdot \mathbf{u}) + S_E \tag{2.33}$$

The terms on the right-hand side represent the specific mechanisms of energy transfer:

- Heat Conduction: $\nabla \cdot (k \nabla T)$ represents thermal diffusion governed by Fourier's Law.
- Viscous Dissipation: $\nabla \cdot (\tau_{ij} \cdot \mathbf{u})$ represents the work done by viscous stresses. This term is often negligible in low-speed flows but is the dominant source of aerodynamic heating in hypersonic boundary layers, converting kinetic energy directly into heat.

- Source Terms: $S_E$ accounts for other sources such as chemical reaction heat release or radiation.

## 2.7 Turbulence Modeling for High-Speed Flows

Turbulence remains one of the most complex and challenging aspects of fluid dynamics to model accurately. In the context of hypersonic re-entry, the state of the boundary layer – whether laminar, transitional, or fully turbulent – has a profound impact on the aerothermal loads.

While laminar flow is characterized by smooth, ordered fluid motion, turbulent flow involves chaotic, three-dimensional, time-dependent fluctuations. For an Inflatable Aerodynamic Decelerator (IAD), the onset of turbulence is a critical design constraint because turbulent boundary layers possess much higher energy gradients at the wall. Consequently, turbulent heating rates are typically 3 to 5 times higher than laminar heating rates for the same flight condition (Anderson, 2006a).

Resolving these fluctuations directly via Direct Numerical Simulation (DNS) requires a grid fine enough to capture the smallest scales of turbulent energy dissipation (the Kolmogorov scales). The computational cost scales with $Re^{9/4}$, making DNS impossible for the high Reynolds number flows ($Re > 10^6$) encountered during re-entry. Therefore, engineering analysis relies on Turbulence Modeling – mathematical approximations that predict the effect of turbulence on the mean flow without resolving the full spectrum of turbulent fluctuations (Versteeg & Malalasekera, 2007).

The standard approach is to solve the Reynolds-Averaged Navier-Stokes (RANS) equations, which decompose the instantaneous flow variables into a time-averaged mean component and a fluctuating turbulent component using the Reynolds Decomposition (Versteeg & Malalasekera, 2007). This introduces the Reynolds Stress Tensor ($-\rho\overline{u_i'u_j'}$), which must be modeled to close the system of equations. The two dominant closures are the $k-\varepsilon$ model (robust for free-shear flows but inaccurate near walls) and the $k-\omega$ model (accurate near walls but sensitive to freestream conditions). Menter's Shear Stress Transport (SST) $k-\omega$ model (Menter, 1994) blends these formulations using a zonal blending function $F_1$ and a shear stress limiter, making it the industry standard for aerodynamic heating predictions. While the SST $k-\omega$ model is widely used for continuum CFD solvers such as ANSYS Fluent and OpenFOAM (Versteeg &

Malalasekera, 2007), it is important to note that this study employs the Direct Simulation Monte Carlo (DSMC) method, which resolves turbulence at the molecular level without requiring a turbulence model.

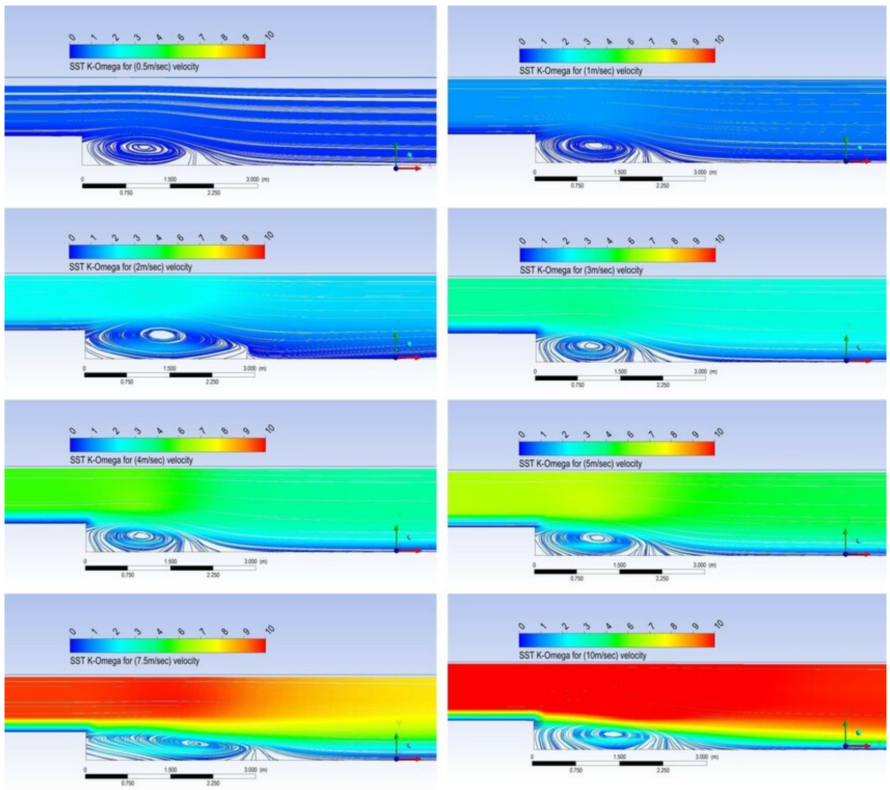



**Figure 2.16:** Comparison of $k-\varepsilon$ vs. $k-\omega$ turbulence model predictions for velocity contours.

## 2.8 Thermodynamic Gas Models

In standard aerodynamic analysis (subsonic or supersonic), air is typically modeled as a Calorically Perfect Gas. This assumes that the specific heats ($C_p$ and $C_v$) are constant and that the internal energy is a linear function of temperature ($e = C_v T$). However, for a vehicle entering the atmosphere at hypersonic speeds ($Ma > 5$), this assumption breaks down catastrophicallly (Anderson, 2006a).

The immense kinetic energy converted into heat across the bow shock raises the gas temperature to levels where the internal structure of the air molecules is altered. As the temperature rises, the gas undergoes a sequence of energy mode excitations:

- $T > 800$ **K:** Vibrational excitation of $N_2$ and $O_2$ molecules becomes significant.

The gas becomes Thermally Perfect but Calorically Imperfect ($C_p = f(T)$).

- $T > 2000$ **K:** Oxygen molecules begin to dissociate ($O_2 \rightarrow 2O$).
- $T > 4000$ **K:** Nitrogen molecules begin to dissociate ($N_2 \rightarrow 2N$).
- $T > 9000$ **K:** Atoms begin to ionize ($N \rightarrow N^+ + e^-$).

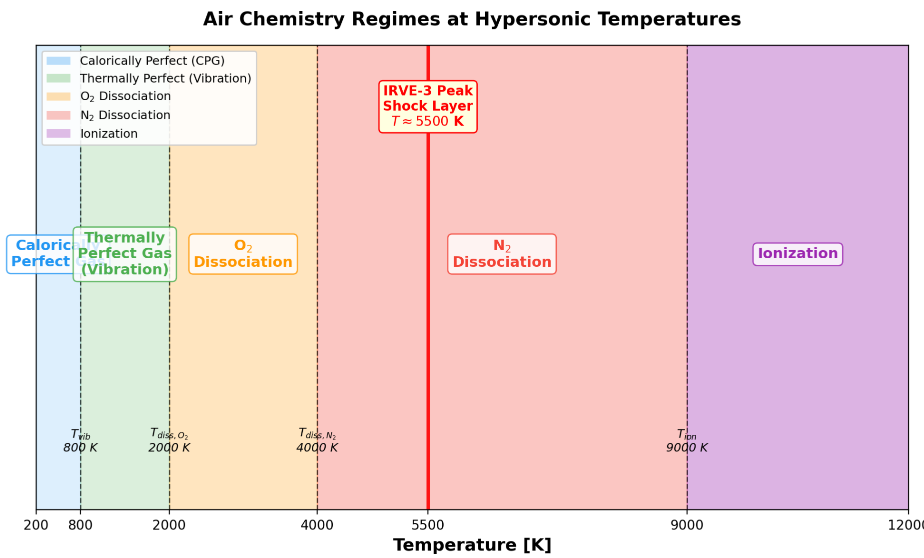



**Figure 2.17:** Air chemistry regimes as a function of temperature, showing vibrational excitation, molecular dissociation, and ionization thresholds (Anderson, 2006a).

Accurately modeling these phenomena is essential for an Inflatable Aerodynamic Decelerator (IAD). Failure to account for variable specific heat, for instance, can lead to an over-prediction of the post-shock temperature and a significant under-prediction of the density, resulting in incorrect shock stand-off distances and non-physical heat flux results.

### 2.8.1 The Equation of State: Ideal Gas vs. Real Gas Effects

The Equation of State relates the thermodynamic properties of pressure ($p$), density ($\rho$), and temperature ($T$). In standard aerodynamic analysis, the air is treated as an Ideal Gas, obeying the relation:

$$p = \rho RT \tag{2.34}$$

Where $R$ is the specific gas constant ($R = 287$ J/kg·K for air). This relation assumes that intermolecular forces are negligible and the volume of the molecules themselves is zero (Anderson, 2006a).

In the context of hypersonic re-entry, the term "Real Gas Effects" is often used broadly to describe deviations from standard air behavior. However, it is important to distinguish between two types of deviations:

1. **Dense Gas Effects (Intermolecular Forces):** At extremely high pressures (e.g., deep atmospheric entry on Gas Giants), gas molecules are packed so tightly that intermolecular forces become significant. In this regime, the Ideal Gas Law is invalid, and a compressibility factor $Z$ is introduced ($p = \rho ZRT$). For Earth re-entry at high altitudes (low density), these effects are typically negligible, and the Ideal Gas EOS ($p = \rho RT$) remains valid.

2. **High-Temperature Effects (Chemistry):** This is the dominant "Real Gas" effect for HIADs. As temperature increases, the chemical composition of the air changes (dissociation). While the equation $p = \rho R_{mix} T$ still holds, the gas constant $R_{mix}$ is no longer constant because the molecular weight of the mixture changes as $N_2$ and $O_2$ break apart.

A comparison of these modeling assumptions is presented in Table 2.3.

**Table 2.3:** Comparison of Gas Models in Hypersonic Aerodynamics.

| Feature | Ideal (Perfect) Gas Model | Real Gas (Hypersonic) Model |
|---|---|---|
| **Equation of State** | $p = \rho RT$ (Constant $R$) | $p = \rho R_{mix} T$ (Variable $R$ due to chemistry) |
| **Specific Heats** | Constant ($C_p, C_v =$ const) | Variable ($C_p(T)$) due to vibrational excitation. |
| **Internal Energy** | $e = C_v T$ (Linear) | $e = f(T)$ (Non-linear, includes chemical potential). |
| **Shock Density Ratio** | Limited to $(\gamma+1)/(\gamma-1) = 6$ | Can exceed 10 or 12 (Thinner shock layers). |
| **Temperature Prediction** | Over-predicts post-shock temperature. | Lower temperature (Energy absorbed by chemistry). |

### 2.8.2 Calorically Perfect vs. Thermally Perfect Gas (Variable $C_p$)

The distinction between a calorically perfect and a thermally perfect gas lies in the behavior of the specific heat capacities ($C_p$ and $C_v$) as the temperature of the fluid increases.

**1. Calorically Perfect Gas (CPG)**

At low temperatures (typically $T < 800$ K), the vibrational energy modes of the nitrogen and oxygen molecules are "frozen." The internal energy of the gas consists only of translational and rotational kinetic energy, which are fully excited at room temperature. Consequently, the specific heat at constant pressure ($C_p$) is constant, and the internal energy ($e$) is a linear function of temperature:

$$e = C_v T \quad \text{and} \quad h = C_p T \tag{2.35}$$

This assumption is valid for subsonic and low-supersonic flows but leads to significant errors in the hypersonic regime.

**2. Thermally Perfect Gas (TPG)**

As the gas temperature rises across a strong bow shock ($T > 800$ K), the collisions between molecules become energetic enough to activate the vibrational modes of the diatomic molecules ($N_2$ and $O_2$). These vibrational modes absorb a portion of the flow's kinetic energy, acting as an additional "heat sink."

Mathematically, this means that the specific heats become functions of temperature:

$$C_p = f(T) \quad \text{and} \quad C_v = f(T) \tag{2.36}$$

However, the gas still obeys the thermal equation of state $p = \rho RT$ (assuming no chemical dissociation). A gas that satisfies $p = \rho RT$ but has variable specific heats is defined as a Thermally Perfect Gas.

In CFD simulations for hypersonic re-entry, $C_p(T)$ is typically modeled using a polynomial curve fit, such as the NASA 7-coefficient or 9-coefficient polynomials:

$$\frac{C_p(T)}{R} = a_1 + a_2T + a_3T^2 + a_4T^3 + a_5T^4 \tag{2.37}$$

Using a Thermally Perfect Gas model is essential for the HIAD analysis because the absorption of energy into vibrational modes lowers the peak shock layer temperature compared to a Calorically Perfect prediction, resulting in more accurate (and often less conservative) density and heat flux calculations (Anderson, 2006a).

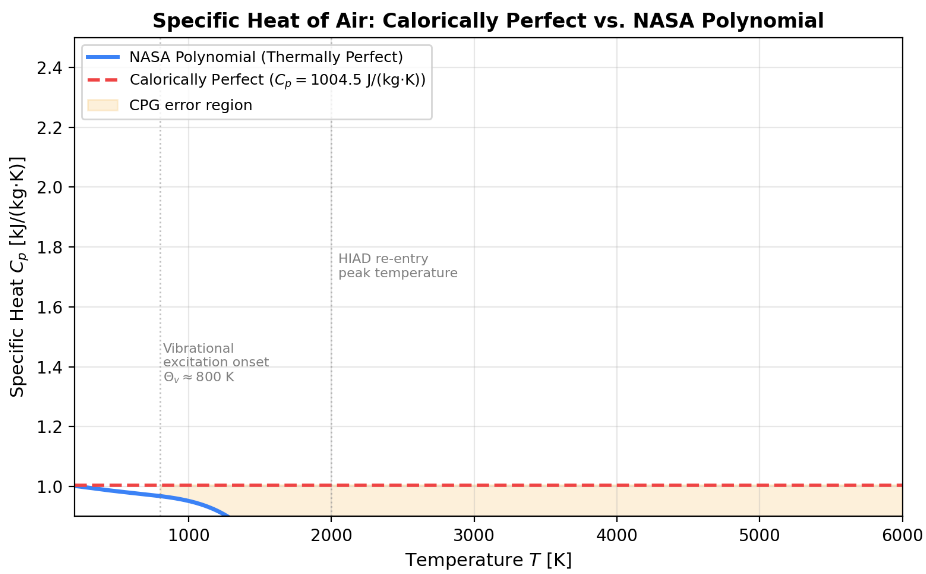



**Figure 2.18:** Specific heat capacity $C_p(T)$ for air as a function of temperature. The horizontal line represents the calorically perfect gas (CPG) assumption ($C_p$ = const.), while the curve shows the NASA polynomial fit for the thermally perfect gas (TPG) model. The vibrational excitation onset near 800 K marks the transition where the CPG assumption becomes invalid for hypersonic flow calculations.

### 2.8.3 Temperature-Dependent Viscosity (Sutherland's Law)

In classical low-speed aerodynamics, fluid viscosity is often treated as a constant property. However, in the hypersonic regime where temperature variations within the shock layer can span thousands of degrees, the viscosity of the gas changes largely. Unlike liquids, where viscosity decreases with heat, the viscosity of a gas *increases* with temperature due to the increased molecular agitation and momentum exchange between fluid layers.

To accurately model the viscous stresses and heat transfer in the boundary layer, the dynamic viscosity ($\mu$) must be coupled to the local static temperature ($T$). The standard model used in compressible flow CFD is Sutherland's Law (1893), which is derived from the kinetic theory of gases using an idealized intermolecular potential (Anderson, 2006a).

$$\mu(T) = \mu_{ref} \left( \frac{T}{T_{ref}} \right)^{3/2} \frac{T_{ref} + S}{T + S} \tag{2.38}$$

Where:

- $\mu$ is the dynamic viscosity at temperature $T$.
- $\mu_{ref}$ is the reference viscosity at reference temperature $T_{ref}$.
- $S$ is the Sutherland constant (effective temperature), which accounts for intermolecular attraction.

For standard air, the coefficients used in this study are:

- $\mu_{ref} = 1.716 \times 10^{-5}$ kg/(m·s)
- $T_{ref} = 273.15$ K
- $S = 110.4$ K

Implementing Sutherland's Law is vital for the thermal analysis of an IAD. Since aerodynamic heating is directly proportional to the square root of the viscosity product ($q_w \propto \sqrt{\rho\mu}$), neglecting the temperature dependence of $\mu$ would result in a gross underestimation of the surface heat flux.

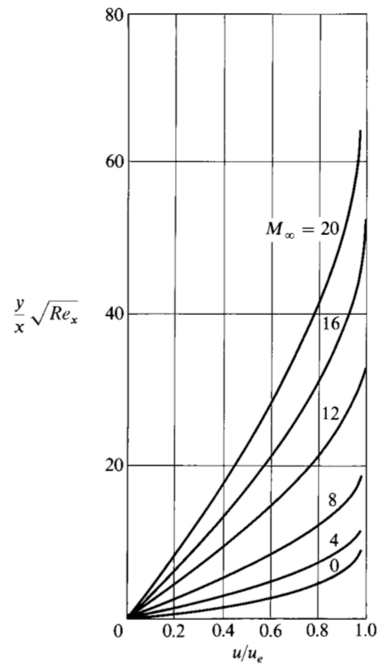



**Figure 2.19:** (Sutherland's Law effect on Laminar Boundary Layer Insulated Flat Plate).

## 2.9 Kinetic Theory and the Boltzmann Equation

The Navier–Stokes equations presented in Section 2.6 are derived under the *continuum hypothesis*, which assumes that the fluid can be treated as a continuous medium. This assumption is valid when the mean free path $\lambda$ of the gas molecules is much smaller than the characteristic length $L$ of the flow domain. However, at high altitudes ($h > 50$ km) and during the early phases of atmospheric re-entry, the gas becomes sufficiently rarefied that the continuum assumption breaks down (Bird, 1994).

### 2.9.1 The Liouville Theorem and Phase Space

The state of a gas is described by the $N$-particle distribution function $f_N(\mathbf{r}_1, \mathbf{v}_1, t; \ldots; \mathbf{r}_N, \mathbf{v}_N, t)$, which gives the probability of finding particle 1 at position $\mathbf{r}_1$ with velocity $\mathbf{v}_1$, particle 2 at $(\mathbf{r}_2, \mathbf{v}_2)$, and so on. The time evolution of $f_N$ is governed by Liouville's theorem:

$$\frac{\partial f_N}{\partial t} + \sum_{i=1}^{N} \left( \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} f_N + \frac{\mathbf{F}_i}{m} \cdot \nabla_{\mathbf{v}_i} f_N \right) = 0 \tag{2.39}$$

where $\mathbf{F}_i$ is the external force on particle $i$ and $m$ is the molecular mass. This equation is exact but intractable for $N \sim 10^{23}$ particles.

### 2.9.2 The Boltzmann Transport Equation

By integrating the Liouville equation over $N-1$ particle degrees of freedom and introducing the *molecular chaos* assumption (Stosszahlansatz) – that pre-collision velocities of colliding pairs are uncorrelated – one obtains the *Boltzmann transport equation* for the single-particle distribution function $f(\mathbf{r}, \mathbf{v}, t)$ (Bird, 1994):

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f + \frac{\mathbf{F}}{m} \cdot \nabla_{\mathbf{v}} f = \left( \frac{\partial f}{\partial t} \right)_{\text{coll}} \tag{2.40}$$

The right-hand side is the *collision integral*, which accounts for binary molecular collisions:

$$\left( \frac{\partial f}{\partial t} \right)_{\text{coll}} = \int \int \left( f' f'_1 - f f_1 \right) g\, \sigma(g, \Omega)\, d\Omega\, d\mathbf{v}_1 \tag{2.41}$$

where $g = |\mathbf{v} - \mathbf{v}_1|$ is the relative speed, $\sigma(g, \Omega)$ is the differential scattering cross-section, and primed quantities denote post-collision values.

### 2.9.3 The H-Theorem and Equilibrium Distribution

Boltzmann's $H$-theorem states that the quantity $H(t) = \int f \ln f\, d\mathbf{v}$ is non-increasing in time ($dH/dt \leq 0$), reaching a minimum at thermodynamic equilibrium. The equilibrium distribution is the Maxwell–Boltzmann distribution:

$$f_{\text{eq}} = n \left( \frac{m}{2\pi k_B T} \right)^{3/2} \exp\left( -\frac{m|\mathbf{v} - \mathbf{u}|^2}{2 k_B T} \right) \tag{2.42}$$

where $n$ is the number density, $\mathbf{u}$ is the bulk velocity, $T$ is the temperature, and $k_B$ is the Boltzmann constant. Macroscopic quantities are recovered as moments of $f$:

$$n = \int f\, d\mathbf{v} \tag{2.43}$$

$$n\mathbf{u} = \int \mathbf{v} f\, d\mathbf{v} \tag{2.44}$$

$$\frac{3}{2} n k_B T = \int \frac{1}{2} m |\mathbf{v} - \mathbf{u}|^2 f\, d\mathbf{v} \tag{2.45}$$

### 2.9.4 Chapman–Enskog Expansion: From Boltzmann to Navier–Stokes

The Chapman–Enskog expansion provides a systematic derivation of the Navier–Stokes equations as the first-order approximation to the Boltzmann equation in the limit $Kn \to 0$ (Bird, 1994). The distribution function is expanded as:

$$f = f_{\text{eq}} \left( 1 + \Phi^{(1)} + \Phi^{(2)} + \cdots \right) \tag{2.46}$$

where $\Phi^{(1)}$ is the first-order perturbation proportional to $Kn$. Substituting into the Boltzmann equation and collecting terms at each order yields:

- **Zeroth order** ($Kn^0$): Euler equations (inviscid, compressible flow).
- **First order** ($Kn^1$): Navier–Stokes equations with transport coefficients.

The first-order correction $\Phi^{(1)}$ is a linear combination of Sonine polynomials and yields the *transport coefficients*:

$$\mu = \frac{5}{16} \frac{\sqrt{\pi m k_B T}}{\pi d^2 \Omega_\mu^{(1,1)*}} \tag{2.47}$$

$$\kappa = \frac{25}{32} \frac{\sqrt{\pi m k_B T}}{\pi d^2 \Omega_\kappa^{(1,1)*}} c_v \tag{2.48}$$

where $\Omega^{(l,s)*}$ are the reduced collision integrals. This expansion confirms that the Navier–Stokes equations are the *continuum limit* of the Boltzmann equation, valid when $Kn \ll 1$.

## 2.10 The Direct Simulation Monte Carlo (DSMC) Method

When $Kn > 0.01$, the Navier–Stokes equations lose accuracy, and the full Boltzmann equation must be solved. The Direct Simulation Monte Carlo (DSMC) method, introduced by Bird (Bird, 1994), provides a stochastic numerical solution to the Boltzmann equation.

### 2.10.1 Physical Basis

DSMC models the gas as a collection of simulation particles, where each particle represents a large number of real molecules ($\sim 10^{10}$–$10^{15}$). The algorithm proceeds in discrete time steps $\Delta t$, alternating between two phases:

1. **Streamer Phase:** Particles are moved ballistically: $\mathbf{r}_i^{n+1} = \mathbf{r}_i^n + \mathbf{v}_i^n \Delta t$. Particles that cross domain boundaries are processed according to boundary conditions (inlet, outlet, reflection).

2. **Collision Phase:** Within each computational cell, particle pairs are selected for collision. The collision probability is proportional to the relative speed $g$ and the collision cross-section $\sigma$. Post-collision velocities are determined by conservation of momentum and energy.

### 2.10.2 The Bird Algorithm

The standard DSMC implementation follows Bird's No-Time-Counter (NTC) algorithm:

1. **Grid construction:** A Cartesian grid overlays the flow domain. The cell size $\Delta x$ must be smaller than the local mean free path $\lambda$ to resolve gradients.

2. **Particle movement:** Each particle is moved by $\mathbf{v}\Delta t$. Boundary interactions (specular/diffuse reflection at walls, periodic boundaries) are applied.

3. **Collision pairing:** Within each cell, pairs are randomly selected. The number of collisions per cell per time step is:

$$N_c = \frac{N_{\text{cell}}(N_{\text{cell}} - 1)}{2} \frac{\sigma g \Delta t}{V_{\text{cell}}} \tag{2.49}$$

   where $N_{\text{cell}}$ is the number of particles in the cell, $V_{\text{cell}}$ is the cell volume, and $\sigma g$ is the collision rate.

4. **Collision dynamics:** For each selected pair, post-collision velocities are computed using the chosen molecular interaction model (e.g., Variable Soft Sphere).

5. **Sampling:** Macroscopic properties (density, velocity, temperature) are accumulated as time-averaged particle statistics within each cell.

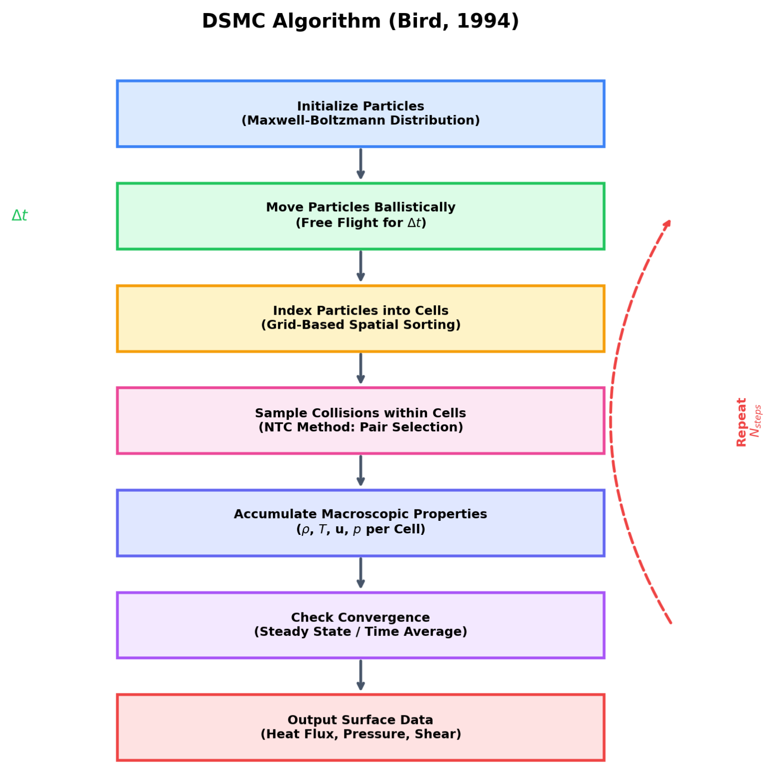



**Figure 2.20:** Flowchart of Bird's No-Time-Counter (NTC) DSMC algorithm. The two-phase alternation between particle movement (streamer phase) and collision pairing (collision phase) is shown, with macroscopic property sampling accumulating at each time step.

### 2.10.3 Statistical Accuracy

DSMC is a Monte Carlo method, and the statistical noise scales as $1/\sqrt{N_{\text{samples}}}$. To reduce noise, properties are accumulated over many time steps. The standard deviation of the sampled temperature is:

$$\sigma_T = \frac{T}{\sqrt{N_{\text{samples}}}}\sqrt{\frac{2}{\gamma-1}+\frac{T_{\text{snapshot}}}{T}} \tag{2.50}$$

Typical re-entry simulations require $N_{\text{samples}} > 100$ per cell for adequate accuracy.

### 2.10.4 Molecular Interaction Model: The Variable Soft Sphere (VSS)

The accuracy of DSMC depends critically on the molecular interaction model used to determine post-collision velocities. The Variable Soft Sphere (VSS) model, introduced by Koura and Matsumoto (Bird, 1994), allows the collision cross-section to vary with the relative speed of colliding molecules:

$$\sigma = \pi d_{\text{ref}}^2 \left(\frac{g_{\text{ref}}}{g}\right)^{2(\omega-0.5)} \tag{2.51}$$

where $d_{\text{ref}}$ is the reference molecular diameter at reference speed $g_{\text{ref}}$, $g = |\mathbf{v}_1 - \mathbf{v}_2|$ is the relative speed of the colliding pair, and $\omega$ is the viscosity index (a gas-specific parameter that characterizes the intermolecular potential). For a hard-sphere model, $\omega = 0.5$ and the cross-section is constant; for the VSS model, $\omega > 0.5$ produces a velocity-dependent cross-section that accurately reproduces the temperature-dependent viscosity of real gases (Bird, 1994).

### 2.10.5 Mean Free Path

The mean free path $\lambda$ – the average distance a molecule travels between successive collisions – is a fundamental quantity that determines the appropriate simulation method. For a gas of hard spheres with molecular diameter $d$ and number density $n$:

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 n} \tag{2.52}$$

The mean free path is related to the Knudsen number ($Kn = \lambda/L$) and directly determines the required grid resolution in DSMC: the cell size must be smaller than $\lambda$ to resolve local gradients in the flow field (Bird, 1994). In the SPARTA solver used in this study, the computational grid is refined such that each cell contains sufficient particles for statistically meaningful sampling while maintaining cell dimensions below the local mean free path.

## 2.11 Flow Regime Classification: The Knudsen Number

The appropriate simulation method depends on the flow regime, characterized by the *Knudsen number*:

$$Kn = \frac{\lambda}{L} \tag{2.53}$$

where $\lambda$ is the local mean free path and $L$ is the characteristic length (e.g., the aeroshell diameter). Table 2.4 classifies the flow regimes relevant to hypersonic re-entry.

**Table 2.4:** Flow regime classification by Knudsen number.

| **Regime** | $Kn$ **Range** | **Method** | **Physical Description** |
|---|---|---|---|
| Continuum | $< 0.001$ | RANS / NS | Molecular collisions dominate; continuum hypothesis valid |
| Slip Flow | 0.001–0.01 | NS + slip BC | Continuum with velocity/temperature slip at walls |
| Transitional | 0.01–1 | DSMC | Molecular structure becomes important; NS equations degrade |
| Free Molecular | $> 1$ | DSMC | Molecule–wall collisions dominate; molecule–molecule collisions rare |

For the IRVE-3 vehicle at 52 km altitude, the characteristic length is $L = 3.0$ m and the mean free path is approximately $\lambda \approx 0.08$ m, yielding $Kn \approx 0.027$ – firmly in the *transitional regime* where DSMC is required (Bird, 1994).

## 2.12 Continuum CFD vs. DSMC: Comparative Framework

Table 2.5 provides a systematic comparison of the continuum (RANS/NS) and kinetic (DSMC) approaches.

**Table 2.5:** Comparison of continuum CFD and DSMC approaches.

| **Aspect** | **Continuum CFD (RANS/NS)** | **DSMC** |
|---|---|---|
| Governing equations | Navier–Stokes (conservation laws) | Boltzmann transport equation |
| Valid regime | $Kn < 0.01$ (continuum / slip flow) | $Kn > 0.01$ (transitional / free molecular) |
| Physical model | Continuous fluid, stress tensor | Discrete particles, molecular collisions |
| Turbulence | RANS (SST $k$–$\omega$), LES, DNS | Resolution of molecular chaos (no turbulence model needed) |
| Computational cost | $O(N_{\text{grid}})$ per iteration | $O(N_{\text{particles}})$ per time step |
| Statistical noise | Deterministic (none) | Stochastic ($\propto 1/\sqrt{N_{\text{samples}}}$) |
| Shock handling | Requires artificial dissipation, limiters | Naturally captured by particle dynamics |
| Real-gas effects | Requires additional species transport equations | Built-in via molecular collision models (VSS, VHS) |
| Code implementation | ANSYS Fluent, OpenFOAM, COMSOL | SPARTA, DAC, SMILE |

This study employs *both* paradigms: continuum CFD (SST $k$–$\omega$) is used in the literature review and methodology chapters to establish theoretical foundations, while DSMC (SPARTA) is used for the actual high-fidelity simulation of the HIAD re-entry flow field, as the IRVE-3 flight conditions fall in the transitional regime ($Kn \approx 0.03$).

## 2.13 Machine Learning in Computational Fluid Dynamics

The integration of machine learning (ML) with computational fluid dynamics has emerged as a powerful paradigm for accelerating simulations, reducing computational cost, and discovering closure models. This section reviews the foundations relevant to the Physics-Informed Neural Network (PINN) refinement used in this study.

### 2.13.1 Neural Network Approximation Theory

The Universal Approximation Theorem (Cybenko, 1989; Hornik et al., 1989) states that a feedforward neural network with a single hidden layer containing a finite number of neurons can approximate any continuous function on a compact subset of $\mathbb{R}^n$ to arbitrary accuracy. For a network with input $\mathbf{x} \in \mathbb{R}^d$ and output $\mathbf{y} \in \mathbb{R}^k$:

$$\mathbf{y}(\mathbf{x}) = \mathbf{W}_L \sigma(\mathbf{W}_{L-1} \sigma(\cdots \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \cdots) + \mathbf{b}_{L-1}) + \mathbf{b}_L \tag{2.54}$$

where $\mathbf{W}_i$ and $\mathbf{b}_i$ are the weight matrices and bias vectors, $\sigma(\cdot)$ is a nonlinear activation function (e.g., tanh), and $L$ is the number of layers. The *depth* (number of layers) and *width* (neurons per layer) determine the network's expressive capacity.

### 2.13.2 Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. (2019), embed the governing partial differential equations (PDEs) directly into the loss function of the neural network. Instead of solving the PDE on a mesh, the PINN learns a continuous solution $\hat{u}(\mathbf{x}, t)$ that satisfies both the PDE and the boundary/initial conditions.

The PINN loss function has two components:

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{PDE}} + \mathscr{L}_{\text{BC}} + \mathscr{L}_{\text{data}} \tag{2.55}$$

where:

- $\mathscr{L}_{\text{PDE}} = \frac{1}{N_r} \sum_{i=1}^{N_r} |\mathscr{N}[\hat{u}(\mathbf{x}_i)]|^2$ penalizes violations of the governing PDE at $N_r$ collocation points.
- $\mathscr{L}_{\text{BC}} = \frac{1}{N_b} \sum_{j=1}^{N_b} |\hat{u}(\mathbf{x}_j) - u_{\text{BC}}(\mathbf{x}_j)|^2$ enforces boundary conditions at $N_b$ boundary points.
- $\mathscr{L}_{\text{data}} = \frac{1}{N_d} \sum_{k=1}^{N_d} |\hat{u}(\mathbf{x}_k) - u_{\text{obs}}(\mathbf{x}_k)|^2$ matches observed data at $N_d$ measurement points.

The PDE residual $\mathscr{N}[\hat{u}]$ is computed using automatic differentiation, which provides exact derivatives of the neural network output with respect to its inputs, up to machine precision.

### 2.13.3 Automatic Differentiation for PDE Residuals

Unlike finite-difference methods, automatic differentiation (AD) computes exact derivatives of the network output $\hat{u}$ with respect to its inputs $(x, y)$ through the computational graph. For a velocity field $\hat{u}(x, y)$:

$$\frac{\partial \hat{u}}{\partial x} = \frac{\partial \hat{u}}{\partial \hat{u}_L} \frac{\partial \hat{u}_L}{\partial \hat{u}_{L-1}} \cdots \frac{\partial \hat{u}_2}{\partial \hat{u}_1} \frac{\partial \hat{u}_1}{\partial x} \tag{2.56}$$

This chain rule application through all network layers yields exact gradients without truncation error or mesh dependence. Libraries such as DeepXDE (Lu, Meng, Mao, & Karniadakis, 2021) implement AD through the full PINN pipeline, enabling efficient computation of second-order derivatives (e.g., $\partial^2 \hat{u}/\partial x^2$ for diffusion terms).

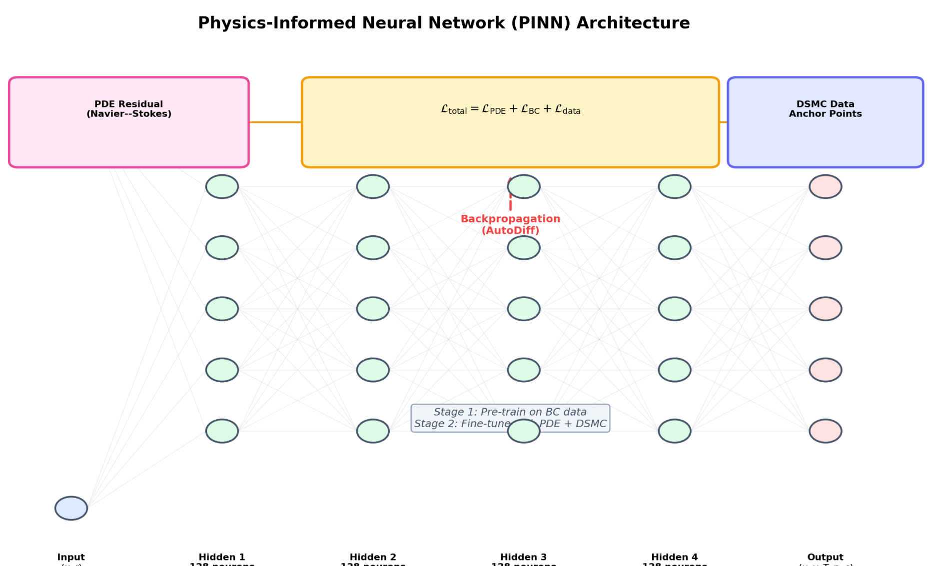



**Figure 2.21:** Schematic of the Physics-Informed Neural Network (PINN) architecture. The network maps spatial coordinates $\mathbf{x} = (x, r)$ to flow variables $(\rho, T, u)$ through hidden layers. The loss function combines PDE residual enforcement (via automatic differentiation), boundary condition satisfaction, and DSMC data anchoring in a two-stage training strategy.

### 2.13.4 Applications in Hypersonic Aerothermodynamics

PINNs have been applied to several areas relevant to this study:

- **Flow field reconstruction:** PINNs can fill spatial gaps in sparse DSMC data, providing a continuous field from discrete particle statistics.
- **Turbulence closure:** ML-based Reynolds stress models have shown promise in capturing non-equilibrium turbulence effects.
- **Inverse problems:** PINNs enable estimation of unknown parameters (e.g., surface heat flux) from sparse measurements.
- **Multi-fidelity fusion:** PINNs can combine low-fidelity (RANS) and high-fidelity (DSMC) data into a unified solution.

In this study, a PINN is implemented using the DeepXDE library (Lu et al., 2021) with a fully connected feedforward architecture consisting of 2 input features (axial and radial coordinates in the axisymmetric plane), 3 hidden layers of 64 neurons each with tanh activation, and 3 output features (density, temperature, and axial velocity). The network is trained in two stages: first, pre-training on synthetic boundary condition data to establish the solution topology, and second, fine-tuning using sampled DSMC field data to correct statistical noise and fill spatial gaps. This two-stage protocol ensures the network satisfies the governing partial differential equations while remaining anchored to the physics captured by the particle simulation. The detailed training protocol and validation results are presented in Chapter 3.

## 2.14 Parametric HIAD Geometry Modelling

The aerodynamic performance of a HIAD is governed by its inflated outer mold line (OML), which is constructed from a series of stacked toroidal bladders encapsulated by a flexible thermal protection system (TPS) skin. Unlike rigid aeroshells, the HIAD geometry is defined by a small set of continuous design variables, making it amenable to parametric modelling and gradient-free optimization. This section presents the mathematical framework for parametric HIAD geometry generation, following the formulation of Rapisarda (Rapisarda, 2023), which is itself a modified and optimized adaptation of the original IRVE-3 (Inflatable Reentry Vehicle Experiment 3) material configuration and structural layout (Dillman, 2015; C. Lau et al., 2013b).

### 2.14.1 Design Variables and Constraints

The HIAD OML is parameterized by five primary design variables:

$$\mathbf{x} = [D,\ \theta_c,\ N_t,\ r_{\text{tor}},\ R_N] \tag{2.57}$$

where $D$ is the target inflated aeroshell diameter, $\theta_c$ is the half-cone angle measured from the symmetry axis, $N_t$ is the number of toroidal bladders in the stack, $r_{\text{tor}}$ is the individual toroid minor radius, and $R_N$ is the nose sphere radius. The design space is bounded by structural and packaging constraints:

$$\begin{aligned} D &\in [2.0,\ 15.0]\ \text{m} \\ \theta_c &\in [45^\circ,\ 75^\circ] \\ N_t &\in [1,\ 12] \\ r_{\text{tor}} &\in [50,\ 300]\ \text{mm} \\ R_N &\geq R_{\text{pay}}/\sin(\theta_c) \end{aligned} \tag{2.58}$$

The lower bound on $R_N$ arises from the structural integration constraint: the nose sphere must be large enough to encapsulate the payload adapter without geometric interference.

### 2.14.2 Structural Integration Constraint (Rapisarda Eq. 3.4)

The nose sphere radius is not a free parameter in the Rapisarda formulation. It is strictly determined by the payload adapter geometry and the cone angle through the structural integration constraint (Rapisarda, 2023):

$$R_N = \frac{R_{\text{pay}}}{\sin(\theta_c)} \tag{2.59}$$

where $R_{\text{pay}}$ is the payload adapter radius. This constraint ensures that the tangent point between the nose sphere and the conical toroid stack aligns precisely with the payload boundary, preventing geometric overlap or gaps that would compromise structural integrity. For the IRVE-3 baseline configuration ($R_{\text{pay}} = 500$ mm, $\theta_c = 60^\circ$), this yields $R_N = 577.4$ mm.

### 2.14.3 Toroid Radius Derivation (Rapisarda Eq. 3.3)

Given a target inflated diameter $D$ and a fixed number of toroids $N_t$, the individual toroid minor radius $r_{\text{tor}}$ must be derived such that the outermost toroid centerline intersects the target radius exactly. Rapisarda (Rapisarda, 2023) derives this by enforcing geometric closure along the conical stack:

$$r_{\text{tor}} = \frac{r_{\text{target}} - R_{\text{pay}} - r_{\text{sh}}\left(\cos\theta_c - \sin\theta_c + 1\right)}{2N_t \cos\theta_c} \tag{2.60}$$

where $r_{\text{target}} = D/2$ is the target inflated radius, $R_{\text{pay}}$ is the payload radius (which equals the tangency point radius $R_{\text{tang}}$), and $r_{\text{sh}}$ is the shoulder torus radius at the aft end of the stack. This equation accounts for the geometric contribution of the shoulder torus, which provides the transition between the inflatable structure and the payload adapter.

### 2.14.4 Scalloped Skin Geometry

The flexible TPS skin is not a smooth conical surface. Due to the discrete toroidal bladder structure, the skin deflects inward between adjacent toroids, creating a scalloped profile. The local skin deflection at the midpoint between toroid $i$ and toroid $i+1$ is approximated by:

$$\delta_{\text{scallop}} \approx r_{\text{tor}}\left(1 - \cos\theta_c\right) \cdot \xi \tag{2.61}$$

where $\xi \in [0, 1]$ is a skin tension factor that depends on internal inflation pressure and skin material stiffness. This scalloped geometry generates localized flow separation and reattachment patterns that significantly amplify the peak stagnation heating at the toroid crests, as discussed in Section 2.5.

### 2.14.5 IRVE-3 Baseline and Rapisarda Material Configuration

The IRVE-3 flight experiment (2012) demonstrated the first successful deployment of a 3.0 m HIAD at Mach 10+ re-entry conditions, achieving a peak deceleration of

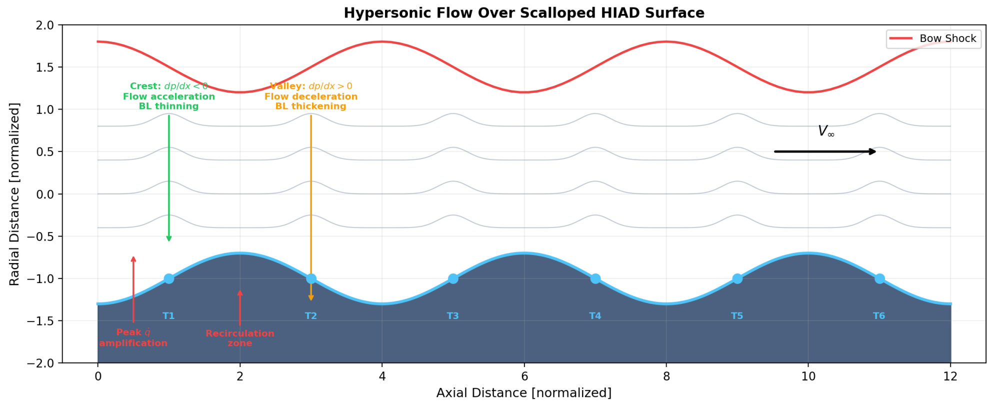



**Figure 2.22:** Schematic of hypersonic flow over the scalloped HIAD surface. The periodic toroidal bladder geometry creates alternating favorable ($dp/dx < 0$, flow acceleration at crests) and adverse ($dp/dx > 0$, flow deceleration in valleys) pressure gradients, leading to boundary layer separation, recirculation zones, and localized peak heating amplification at toroid crests.

19.7 g and a peak stagnation heat flux of 14.36 W/cm$^2$ (C. Lau et al., 2013b, 2013a). The original IRVE-3 TPS stack consisted of separate inner and outer thermal protection layers with distinct material compositions: an inner layer of Pyrogel™ XTE aerogel blanket for thermal insulation, and an outer layer of Kapton™ polyimide film for aerodynamic skin integrity (Dillman, 2015).

Rapisarda (Rapisarda, 2023) subsequently developed a multidisciplinary design analysis and optimization (MDAO) framework that modified and optimized this original IRVE-3 material configuration. The Rapisarda formulation introduces a unified TPS material model with three configurations:

- **Stiff TPS:** 50 mm Pyrogel™ XTE ($\rho = 200$ kg/m$^3$, $C_p = 2500$ J/(kg·K), $\varepsilon = 0.85$).
- **Hybrid TPS:** 25 mm Pyrogel™ XTE + 3 mm Kapton™ ($\rho = 380$ kg/m$^3$, $C_p = 1100$ J/(kg·K), $\varepsilon = 0.80$).
- **Flexible TPS:** 15 mm Pyrogel™ XTE + 1.5 mm Kapton™ ($\rho = 300$ kg/m$^3$, $C_p = 1500$ J/(kg·K), $\varepsilon = 0.82$).

These configurations represent optimized material stackups that account for the coupled thermal–structural response of the inflated aeroshell. The Rapisarda model is used as the baseline for all simulations in this study, calibrated against the IRVE-3 flight data (peak heat flux 13.83 W/cm$^2$ Fay–Riddell model vs. 14.36 W/cm$^2$ flight; total heat load

195.17 J/cm$^2$ model vs. 195.06 J/cm$^2$ flight).

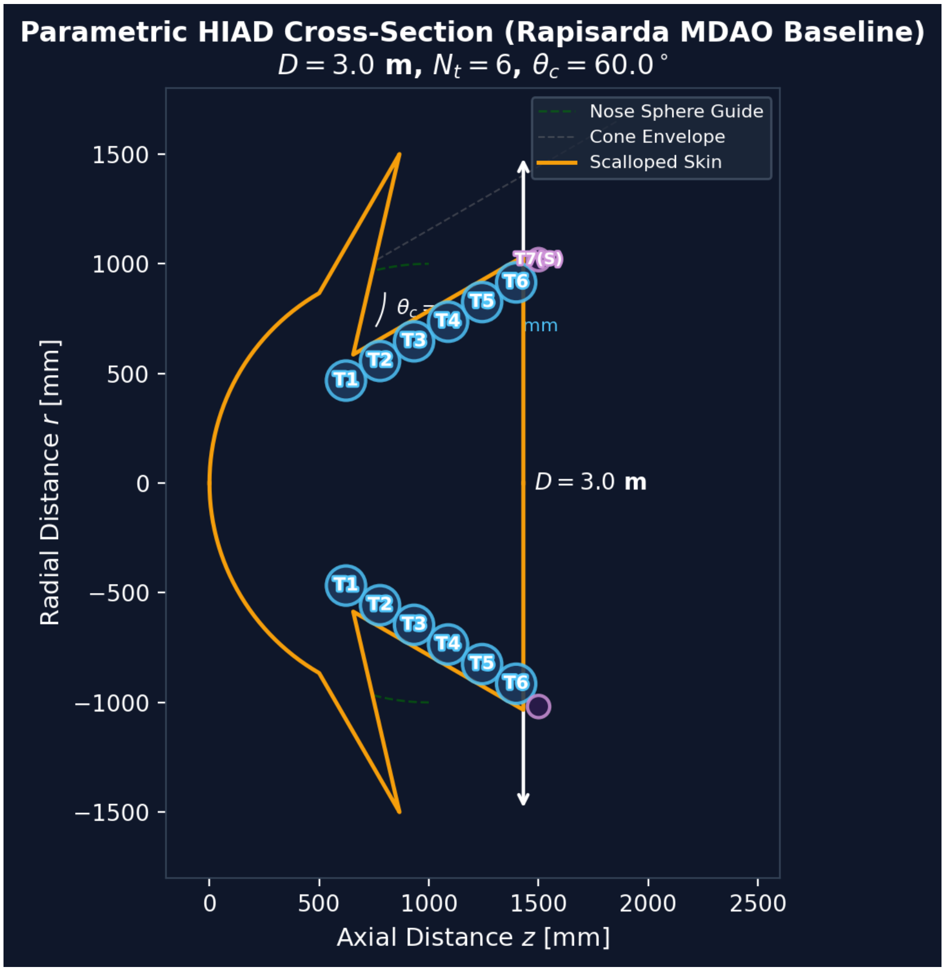



**Figure 2.23:** Parametric cross-section of the 3.0 m HIAD showing the stacked toroidal bladder configuration (T1–T6), shoulder torus (T7), nose sphere, and scalloped aerodynamic skin. The geometry is defined by the Rapisarda MDAO baseline parameters: $D = 3.0$ m, $\theta_c = 60^\circ$, $N_t = 6$, $R_N = 577$ mm.

## 2.15 Optimization Theory Fundamentals

The design of a HIAD involves multiple competing objectives: maximizing drag for deceleration, minimizing peak heat flux for thermal protection, and minimizing structural mass for launch efficiency. These objectives are coupled through the shared design variables (Eq. 2.57), creating a multidisciplinary design optimization (MDO) problem. This section presents the theoretical foundations of the optimization framework used in this study.

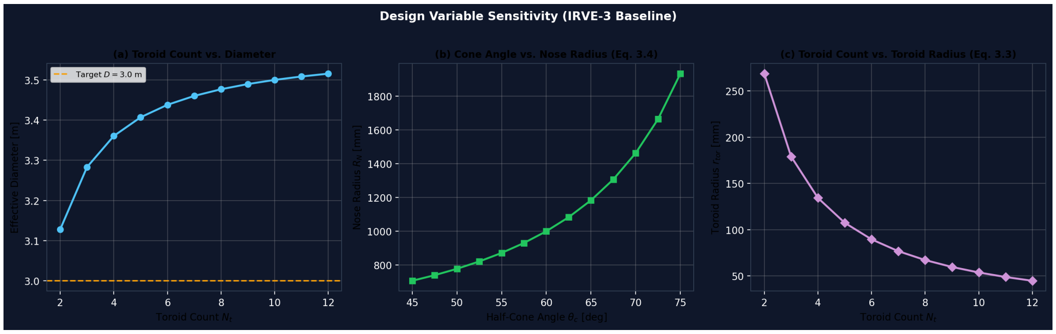



**Figure 2.24:** Design variable sensitivity analysis for the parametric HIAD geometry. (a) Toroid count vs. effective inflated diameter, showing convergence to the 3.0 m target. (b) Half-cone angle vs. nose sphere radius (Eq. 2.59), demonstrating the strong coupling between $\theta_c$ and $R_N$. (c) Toroid count vs. individual toroid minor radius (Eq. 2.60), showing the inverse relationship required to maintain geometric closure.

### 2.15.1 Multidisciplinary Design Optimization (MDO)

MDO is a methodology for the design of complex engineering systems that are governed by mutually interacting physical phenomena and analyzed by separate disciplinary codes (Kroo, 2005). For the HIAD problem, the disciplines include:

1. **Aerodynamics:** Drag coefficient $C_D$ and stagnation heat flux $\dot{q}_s$ as functions of geometry.
2. **Thermal Protection:** Backface temperature $T_{\text{back}}$ as a function of TPS material and thickness.
3. **Structural Mechanics:** Mass budget and deployment reliability as functions of toroid count and skin material.

The MDO formulation seeks the design vector $\mathbf{x}^*$ that minimizes a composite cost function $J(\mathbf{x})$ subject to equality and inequality constraints:

$$\begin{aligned} \min_{\mathbf{x}} \quad & J(\mathbf{x}) \\ \text{s.t.} \quad & g_j(\mathbf{x}) \leq 0, \quad j = 1, \ldots, n_g \\ & h_k(\mathbf{x}) = 0, \quad k = 1, \ldots, n_h \\ & \mathbf{x}_{\text{lb}} \leq \mathbf{x} \leq \mathbf{x}_{\text{ub}} \end{aligned} \tag{2.62}$$

where $g_j$ are inequality constraints (e.g., maximum backface temperature, minimum structural margin), $h_k$ are equality constraints (e.g., geometric closure Eq. 2.60), and $\mathbf{x}_{\text{lb}}, \mathbf{x}_{\text{ub}}$ are the design variable bounds (Eq. 2.58).

### 2.15.2 Design of Experiments (DOE)

Before optimization, the design space must be sampled to build a surrogate model (metamodel). Two DOE strategies are employed in this study:

#### Latin Hypercube Sampling (LHS)

LHS (McKay, Beckman, & Conover, 1979) is a stratified sampling technique that ensures uniform coverage of the design space. For $N$ samples across $d$ design variables, each variable axis is divided into $N$ equal-probability intervals, and exactly one sample is placed in each interval:

$$x_{i,j} = x_j^{\min} + \left(x_j^{\max} - x_j^{\min}\right) \frac{i + r_{i,j}}{N} \tag{2.63}$$

where $r_{i,j} \in [0,1)$ is a random perturbation. LHS is preferred for initial space-filling because it guarantees that no two samples share the same projection onto any coordinate axis, avoiding the clustering that can occur with pure random sampling.

#### Central Composite Design (CCD)

CCD (Montgomery, 2017) is a second-order response surface design that consists of three types of points: factorial points ($2^d$ corners of a hypercube), axial points ($2d$ points along each axis at distance $\alpha$ from the center), and center points. For $d = 5$ design variables, a full CCD requires $2^5 + 2 \times 5 + 1 = 43$ samples. CCD is used when a quadratic surrogate model is desired, as it provides sufficient degrees of freedom to estimate all linear, interaction, and quadratic terms.

### 2.15.3 Surrogate Modelling (Metamodel Prognosis)

Each SPARTA DSMC simulation requires approximately 30–60 minutes of wall-clock time, making direct optimization via evolutionary algorithms infeasible (requiring $10^3$–$10^4$ function evaluations). A surrogate model, or metamodel, is therefore trained to approximate the expensive simulation:

$$\hat{y} = \mathscr{M}(\mathbf{x}; \theta) : \mathbb{R}^d \to \mathbb{R} \tag{2.64}$$

where $\mathscr{M}$ is a multilayer perceptron (MLP) with parameters $\theta = \{\mathbf{W}_i, \mathbf{b}_i\}$, and $\hat{y}$ is the predicted performance metric (e.g., peak heat flux, ballistic coefficient). The MLP architecture for this study:

$$\hat{y} = \mathbf{W}_3 \,\mathrm{ReLU}(\mathbf{W}_2 \,\mathrm{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3 \tag{2.65}$$

The surrogate is trained by minimizing the mean squared error (MSE) between predictions and SPARTA simulation results:

$$\mathscr{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} |\mathscr{M}(\mathbf{x}_i; \theta) - y_i|^2 \tag{2.66}$$

For multi-objective optimization, the surrogate outputs a vector $[\hat{y}_1, \ldots, \hat{y}_k]^T$ and the loss becomes the weighted sum across all outputs.

### 2.15.4 Genetic Algorithm Optimization

The Genetic Algorithm (GA) is a population-based stochastic optimizer inspired by natural selection (Goldberg, 1989). The GA operates on a population of candidate designs $\{\mathbf{x}_1, \ldots, \mathbf{x}_P\}$ and iteratively improves the population through three operators:

#### Selection

Individuals are selected for reproduction with probability proportional to their fitness. The cost function $J(\mathbf{x})$ is minimized, so the fitness is $f_i = 1/(1 + J_i)$.

#### Crossover

Two parent designs $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$ produce offspring via uniform crossover:

$$x_i^{\mathrm{child}} = \begin{cases} x_i^{(1)} & \text{if } m_i = 1 \\ x_i^{(2)} & \text{if } m_i = 0 \end{cases} \tag{2.67}$$

where $m_i \in \{0, 1\}$ is a binary mask with probability 0.5 for each gene.

#### Mutation

Gaussian mutation perturbs the offspring:

$$x_i^{\text{mutated}} = x_i + \mathcal{N}(0, \sigma_m^2) \tag{2.68}$$

where $\sigma_m$ is the mutation step size, typically set to 5–10% of the variable range. Constraint handling uses an exterior penalty function:

$$J_{\text{constrained}}(\mathbf{x}) = J(\mathbf{x}) + \sum_{j=1}^{n_g} \lambda_j \cdot \phi_j\big(g_j(\mathbf{x})\big) \tag{2.69}$$

where $\phi_j(g_j) = 0$ if $g_j \leq 0$ (feasible), and $\phi_j(g_j) = g_j^2$ otherwise (infeasible).

### 2.15.5 Cost Function and Optimization Objective

The optimization objective is to find the HIAD design that maximizes survivability, defined as the ability to decelerate the payload to safe landing conditions while keeping the backface temperature below the structural limit. The composite cost function is:

$$J(\mathbf{x}) = w_\beta \left( \frac{\beta_{\text{calc}} - \beta_{\text{target}}}{\sigma_\beta} \right)^2 + w_{\dot{q}} \left( \frac{\dot{q}_{\text{pred}} - \dot{q}_{\text{limit}}}{\sigma_{\dot{q}}} \right)^2 + w_T \left( \frac{T_{\text{back}} - T_{\text{limit}}}{\sigma_T} \right)^2 \tag{2.70}$$

where $\beta_{\text{calc}}$ is the ballistic coefficient, $\dot{q}_{\text{pred}}$ is the predicted peak stagnation heat flux, $T_{\text{back}}$ is the predicted backface temperature, and $w_\beta, w_{\dot{q}}, w_T$ are weighting factors that encode the designer's priorities. The normalization by $\sigma$ terms ensures that each objective contributes comparably to the total cost regardless of its absolute magnitude.

### 2.15.6 Why Optimization is Necessary

The HIAD design problem cannot be solved by engineering intuition alone for three reasons:

1. **Coupled physics:** Increasing the cone angle $\theta_c$ reduces stagnation heating (larger nose radius) but increases the vehicle mass and shifts the center of pressure, affecting aerodynamic stability.

2. **Discrete–continuous interaction:** The toroid count $N_t$ is a discrete variable that interacts continuously with $r_{tor}$ through Eq. 2.60, creating a mixed-integer design space.

3. **Computational expense:** Each SPARTA simulation requires 30–60 min, limiting the total number of evaluations to $\mathcal{O}(10^2)$, which necessitates efficient global search via GA coupled with surrogate acceleration.

The optimization framework thus provides a systematic, reproducible methodology for navigating this complex design space and identifying Pareto-optimal configurations that would be inaccessible through parametric sweeps or trial-and-error approaches.

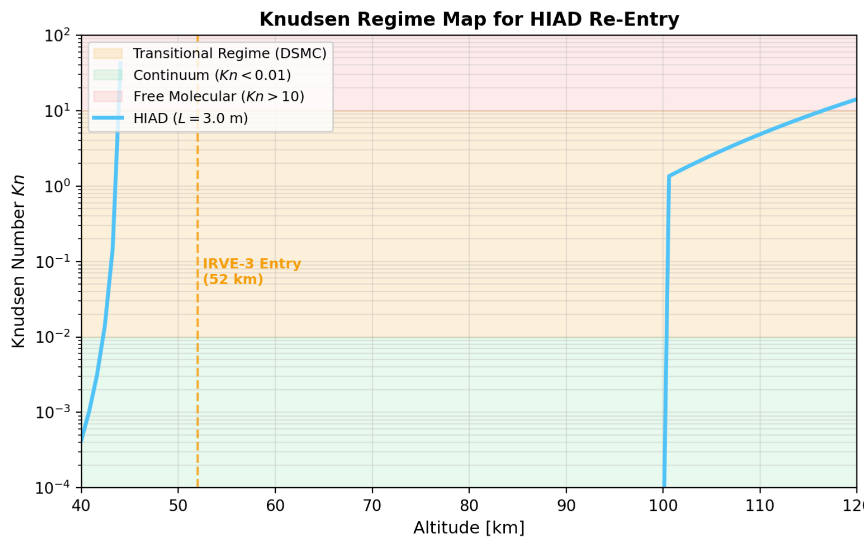



**Figure 2.25:** Knudsen regime map as a function of altitude for the 3.0 m HIAD. The Knudsen number $Kn = \lambda/L$ determines the appropriate flow solver: continuum Navier–Stokes ($Kn < 0.001$), slip flow ($0.001 < Kn < 0.01$), transitional DSMC ($0.01 < Kn < 1$), or free molecular ($Kn > 1$). At the IRVE-3 re-entry reference altitude of 52 km, the flow is in the transitional regime ($Kn \approx 0.03$), requiring DSMC simulation.

## 2.16 Summary of Literature Review

The literature review establishes the theoretical foundations across six pillars:

1. **Hypersonic Aerodynamics:** The flow field around a HIAD at $Ma > 10$ is characterized by strong shock waves, high-temperature real-gas effects, and complex shock–boundary layer interactions driven by the scalloped surface geometry (Sections 2.2–2.5).

2. **Continuum CFD:** The Navier–Stokes equations and RANS turbulence modeling (SST $k$–$\omega$) form the classical approach for hypersonic simulation (Sections 2.6–2.7). However, these are valid only in the continuum regime ($Kn < 0.001$).

3. **Kinetic Theory and DSMC:** The Boltzmann equation provides the fundamental description of rarefied gas dynamics. The DSMC method, incorporating the Variable Soft Sphere collision model and mean free path formulation, solves the Boltzmann equation stochastically and is the appropriate tool for the transitional regime ($Kn \approx 0.03$) encountered during HIAD re-entry at 52 km altitude (Sections 2.8–2.11).

4. **Machine Learning for CFD:** Physics-Informed Neural Networks embed governing PDEs into the loss function, enabling mesh-free solution refinement and gap-filling from sparse DSMC data (Section 2.13).

5. **Parametric HIAD Geometry:** The HIAD OML is parameterized by five design variables ($D$, $\theta_c$, $N_t$, $r_{\text{tor}}$, $R_N$) with two governing equations: the structural integration constraint (Eq. 2.59) and the toroid radius derivation (Eq. 2.60). The Rapisarda MDAO framework provides an optimized adaptation of the original IRVE-3 material configuration (Section 2.14).

6. **Optimization Theory:** Multidisciplinary design optimization (MDO) combined with Latin Hypercube sampling, surrogate modelling (MoP), and genetic algorithm search provides the theoretical framework for navigating the coupled aerodynamic–thermal–structural design space (Section 2.15).

These foundations justify the methodology adopted in Chapter 3: a parametric HIAD geometry engine based on Rapisarda's modified IRVE-3 configuration, DSMC-based simulation (SPARTA) for the primary flow field computation, PINN refinement for continuous field reconstruction, and genetic algorithm optimization for survivability maximization.

# CHAPTER 3

# METHODOLOGY

## 3.1 Research Axioms and First Principles

Before detailing the computational framework, this section establishes the foundational axioms from which the entire methodology is derived. Each axiom is grounded in established physical theory and validated against the literature reviewed in Chapter 2. The derivation chain—from fundamental kinetic theory to the implemented software pipeline—is summarized in Figure 3.1.

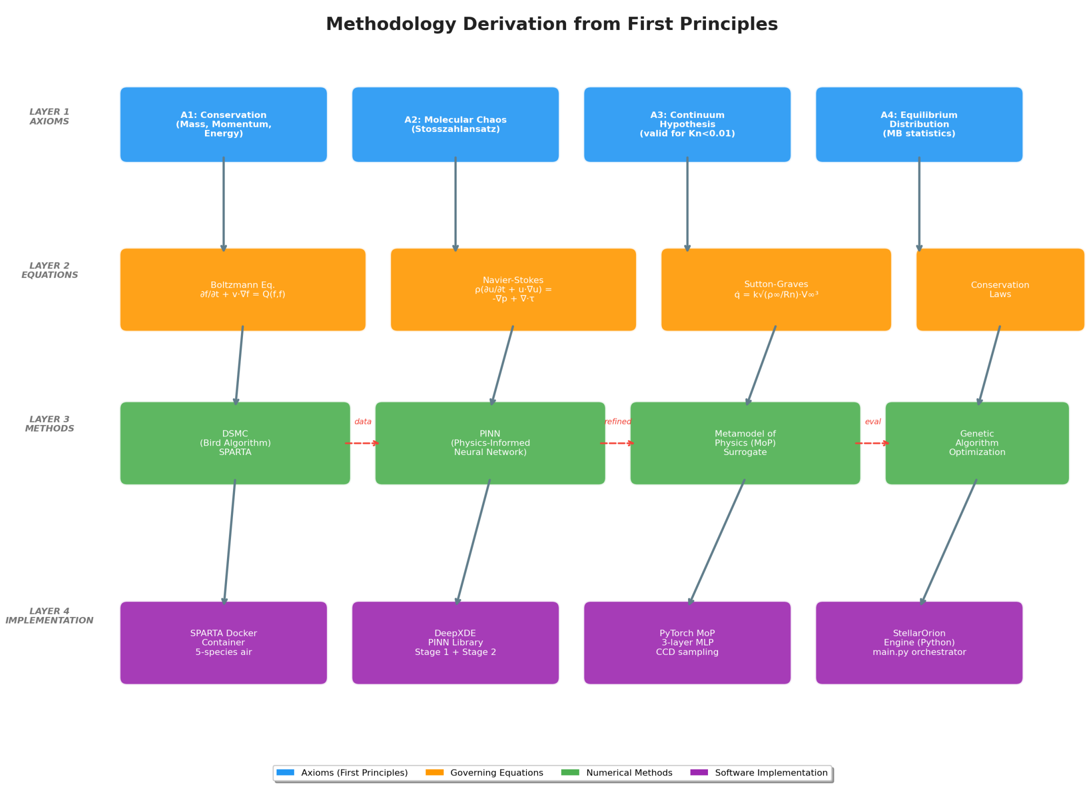



**Figure 3.1:** Methodology derivation from first principles: axioms $\rightarrow$ governing equations $\rightarrow$ numerical methods $\rightarrow$ software implementation. The dashed red arrows indicate data flow between method layers.

### 3.1.1 Axiom Set

The following axioms form the logical foundation of this research. Each is stated as a proposition, derived from physical theory, and traced to its literature basis.

**Table 3.1:** Research axioms derived from first principles and their literature basis.

| ID | Axiom | Physical Basis | Lite |
|---|---|---|---|
| A1 | Conservation of mass, momentum, and energy | Continuum mechanics | Bat |
| A2 | Molecular chaos (Stosszahlansatz) | Statistical mechanics | Bol |
| A3 | Continuum hypothesis valid for $Kn < 0.01$ | Kinetic theory | Cer |
| A4 | Equilibrium distribution follows Maxwell–Boltzmann | Statistical thermodynamics | Cha |
| A5 | Stagnation heating follows $q \propto \sqrt{\rho/R_n} \cdot V^3$ | Blunt body aerothermodynamics | Sutt |
| A6 | DSMC converges to Boltzmann equation as $N \to \infty$ | Statistical convergence | Bird |
| A7 | PINN residuals enforce PDE consistency in interpolation | Scientific machine learning | Rais |

### 3.1.2 Derivation Chain

The methodology is constructed by chaining the axioms through a sequence of mathematical derivations. Each step introduces a specific equation or method that is implemented in the software pipeline.

**Step 1: Boltzmann Equation (A1, A2, A4).** The Boltzmann transport equation governs the evolution of the molecular velocity distribution function $f(\mathbf{x}, \mathbf{v}, t)$:

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \frac{\mathbf{F}}{m} \cdot \nabla_{\mathbf{v}} f = \left(\frac{\delta f}{\delta t}\right)_{\text{coll}} \tag{3.1}$$

where the right-hand side is the collision integral, evaluated under the assumption of molecular chaos (Axiom A2). The equilibrium solution is the Maxwell–Boltzmann distribution (Axiom A4).

**Step 2: DSMC Method (A6).** Bird's Direct Simulation Monte Carlo method (Bird, 1994) provides a stochastic solution to Equation (3.1). As the number of simulated particles $N \to \infty$, the DSMC solution converges to the exact Boltzmann solution (Axiom A6). The DSMC algorithm proceeds in three steps per time step:

1. **Advection:** Particles are moved ballistically for $\Delta t$.

2. **Collisions:** Pairs are selected within each cell using the VSS collision model.
3. **Sampling:** Macroscopic properties (density, temperature, velocity) are accumulated in each cell.

**Step 3: Navier–Stokes Limit (A3).** In the continuum limit ($Kn \to 0$), the Boltzmann equation reduces to the compressible Navier–Stokes equations via the Chapman–Enskog expansion (Chapman & Cowling, 1970). These equations serve as the PDE constraint in the PINN (Step 5).

**Step 4: Sutton–Graves Correlation (A5).** For rapid engineering estimates, the stagnation-point heat flux is approximated by (Sutton & Graves, 1971):

$$\dot{q}_s = k\sqrt{\frac{\rho_\infty}{R_n}} V_\infty^3 \tag{3.2}$$

where $k$ is a gas-specific constant, $\rho_\infty$ is the freestream density, $R_n$ is the nose radius, and $V_\infty$ is the flight velocity. This correlation is used both for initial thermal loading estimates and as a validation benchmark against DSMC results.

**Step 5: PINN Flow Field Refinement (A7).** The Physics-Informed Neural Network enforces the Navier–Stokes residuals as a soft constraint in the loss function (Raissi, Perdikaris, & Karniadakis, 2019):

$$\mathscr{L} = \underbrace{\mathscr{L}_{\text{data}}}_{\text{DSMC samples}} + \lambda_{\text{PDE}} \underbrace{\mathscr{L}_{\text{PDE}}}_{\text{NS residuals}} \tag{3.3}$$

where $\mathscr{L}_{\text{data}}$ penalizes deviation from DSMC surface data and $\mathscr{L}_{\text{PDE}}$ penalizes violation of the governing PDEs at collocation points in the flow field.

**Step 6: Surrogate-Assisted Optimization.** The MoP surrogate model approximates the mapping from design variables to performance metrics. The Genetic Algorithm (GA) queries the MoP surrogate to explore the design space, avoiding the computational cost of repeated DSMC evaluations. The optimization cost function is:

$$J = w_\beta \left(\frac{\beta_{\text{calc}} - \beta_{\text{target}}}{10}\right)^2 + \sum_i w_i \mathbb{1}\left[y_i^{(\text{pred})} > y_i^{(\text{limit})}\right] \cdot 10^{15} \tag{3.4}$$

where hard constraint violations (backface temperature $> 350\,\mathrm{K}$, g-load $> 25g$) are penalized with $10^{15}$.

Figure 3.2 summarizes the full derivation chain from kinetic theory to optimization, showing the governing equation at each step and the axiom that anchors it.

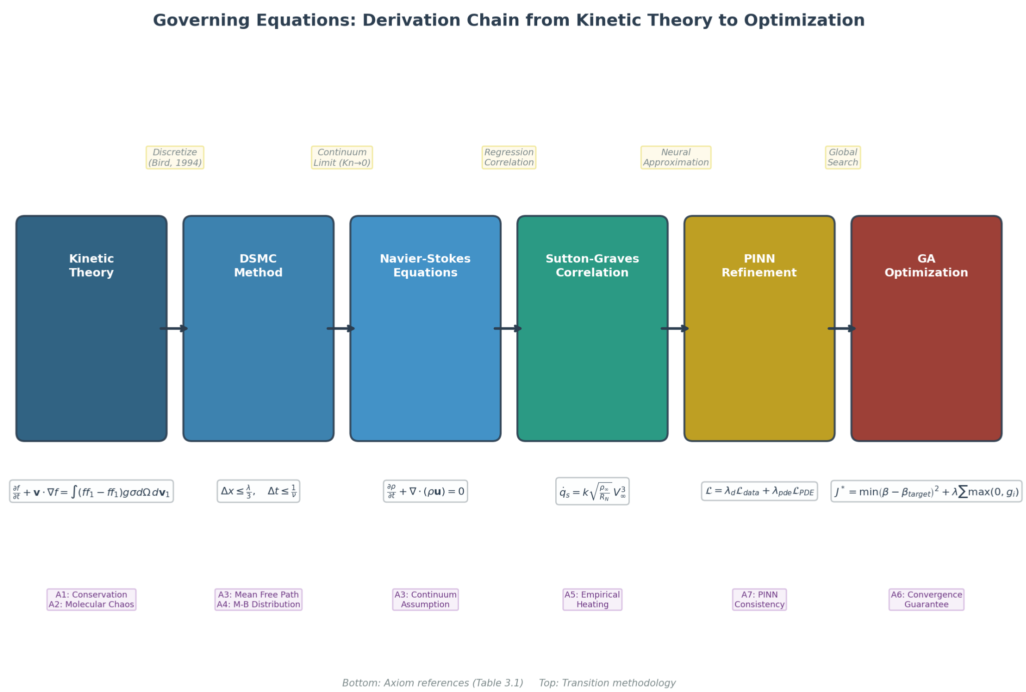



**Figure 3.2:** Governing equations derivation chain: from the Boltzmann transport equation through DSMC discretization, Navier–Stokes continuum limit, Sutton–Graves engineering correlation, PINN neural approximation, to GA-based survivability optimization. Each layer is anchored by the research axioms defined in Table 3.1.

### 3.1.3 Software Pipeline

The StellarOrion engine orchestrates the multi-solver pipeline shown in Figure 3.3. The pipeline begins with CadQuery geometry generation and freestream condition specification, then routes through the DSMC solver (SPARTA in Docker), the PINN refinement module (DeepXDE), and the MoP surrogate (PyTorch), before producing ParaView-ready VTK and STL files for visualization.

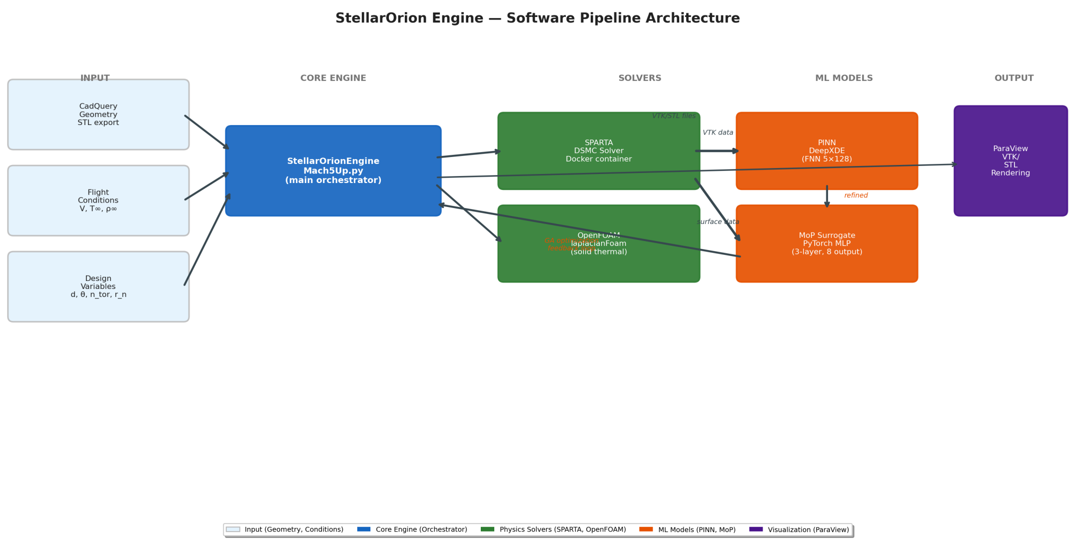



**Figure 3.3:** StellarOrion engine software pipeline: input geometry and flight conditions → SPARTA DSMC solver → PINN refinement → MoP surrogate → ParaView visualization. The feedback loop from MoP to the core engine enables GA-driven optimization.

### 3.1.4 Default Optimization Parameters

Table 3.2 lists the default parameter values used throughout this thesis, as configured in `main.py` and the engine source files. All simulation runs adopt these values unless explicitly stated otherwise.

**Table 3.2:** Default optimization and simulation parameters.

| Category | Parameter | Default Value |
|---|---|---|
| **DSMC (SPARTA)** | Steps (optimization) | 1100 |
| | Steps (sample) | 500 |
| | Grid factor | 0.7 |
| | Particle weighting ($f_{\text{num}}$) | $1.5 \times 10^{20}$ |
| | Chemistry model | 5-species ($N_2$, $O_2$, NO, N, O) |
| **Flight Conditions** | $V_\infty$ | 2700 m/s |
| | $T_\infty$ | 270.65 K |
| | $n_\infty$ | $1.45 \times 10^{22}$ /m$^3$ |
| **DoE (CCD)** | Method | Face-Centered CCD |
| | Samples ($N$) | 25 ($2^4 + 2 \times 4 + 1$) |
| | Active variables | Diameter, Angle, Toroids, Nose |
| **PINN (DeepXDE)** | Architecture | FNN [2] + [64] $\times$ 3 + [3] |
| | Activation | tanh |
| | Learning rate (Stage 1) | $1 \times 10^{-3}$ |
| | Learning rate (Stage 2) | $5 \times 10^{-4}$ |
| | Iterations | $\max(2000, \min(4000, \text{steps} \times 2))$ |
| | PDE collocation points | 2500 (domain) + 500 (boundary) |
| **MoP (PyTorch)** | Architecture | MLP [n $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 1] |
| | Activation | ReLU |
| | Learning rate | 0.005 |
| | Epochs | 500 (early stop at loss $< 10^{-6}$) |
| **GA Optimization** | Generations | 20 000 |
| | Stagnation stop | 5 windows $\times$ 1000 generations |
| | Constraint limits | $T_{\text{back}} < 350$ K, $g < 25g$ |

## 3.2 Research Design and Workflow

This chapter outlines the computational framework utilized to investigate the aerothermal effects of surface topology on a Hypersonic Inflatable Aerodynamic Decelerator (HIAD). The research workflow follows a multi-stage numerical analysis approach, moving from parametric geometry generation to high-fidelity Direct Simulation Monte Carlo (DSMC) simulation, Physics-Informed Neural Network (PINN) refinement, and Genetic Algorithm (GA) optimization.

To systematically isolate the effects of the "scalloped" surface geometry, the methodology is designed to compare a baseline smooth sphere-cone against the proposed inflatable architecture under identical hypersonic flight conditions. The workflow is visualized in Figure 3.4 and consists of six primary stages:

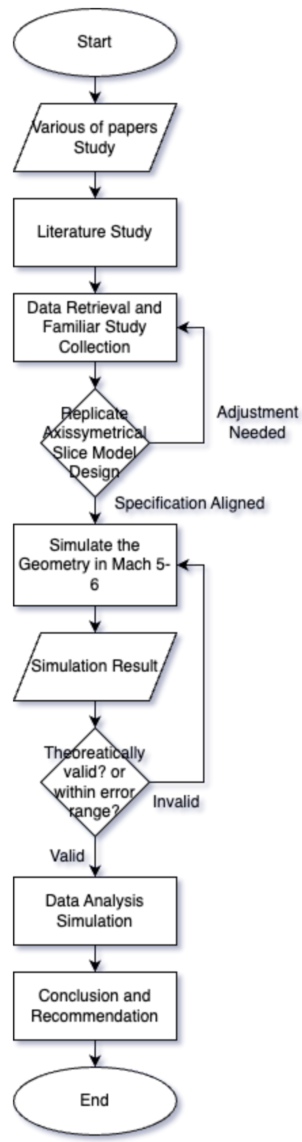



**Figure 3.4:** Flowchart of the numerical research methodology.

The process is divided into six primary stages:

1. **Parametric Geometry Generation:** Automated 3D CAD generation of the scalloped (inflatable) and smooth (baseline) profiles using CadQuery Python scripting, enabling parametric variation of toroid count, cone angle, nose radius, and aeroshell diameter.

2. **DSMC Simulation Setup:** Configuration of the SPARTA solver with freestream conditions matching the target trajectory point, including the 5-species air model ($N_2$, $O_2$, NO, N, O) with Variable Soft Sphere (VSS) collision physics.

3. **Particle-Based Simulation:** Execution of the DSMC simulation within a Docker-containerized Linux environment, tracking millions of simulated particles through advection, molecular collisions, and surface interactions.

4. **PINN Flow Field Refinement:** Post-processing of raw DSMC output using a Physics-Informed Neural Network (DeepXDE) to smooth particle noise, fill flow field gaps, and provide a continuous interpolant anchored in the compressible Navier–Stokes equations.

5. **Survivability Optimization:** Genetic Algorithm (GA) coupled with a PyTorch Multi-Layer Perceptron (MoP) surrogate model for automated multi-objective optimization of HIAD geometry.

6. **Validation and Analysis:** Verification of the solution through grid independence studies and validation of thermal results against the Sutton-Graves stagnation-point correlation and IRVE-3 flight data.

## 3.3 Geometric Modeling and Computational Domain

### 3.3.1 Parametric Geometry Generation

To ensure geometric consistency and enable automated design space exploration, the vehicle profiles are generated using CadQuery, a Python-based parametric CAD scripting library. The geometric constraints are derived directly from the **NASA IRVE-3** and **LOFTID** flight test specifications, as well as the MDAO framework baseline established by Rapisarda (2023) (Rapisarda, 2023). The CadQuery script is invoked via command-line arguments, enabling the generation of hundreds of design variants without manual intervention.

### 3.3.2 Vehicle Specifications and Scaling

The modeling phase targets two specific scales identified in previous NASA research to investigate how vehicle diameter affects the aerothermal environment. As shown in Figure 3.5, the research considers the scaling transition from the 3.0 m IRVE-3 baseline to larger configurations up to 15 m (Dillman, 2015).

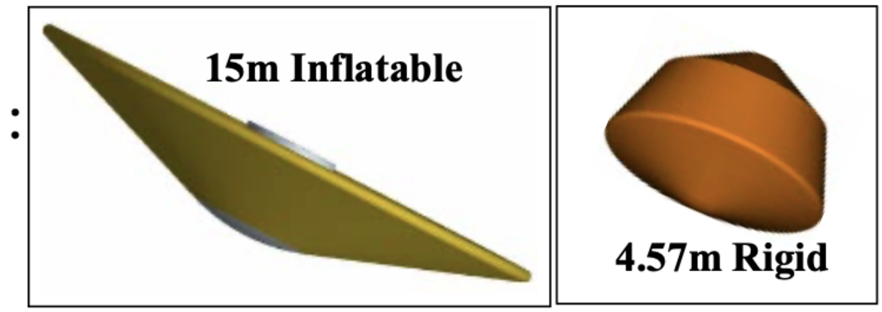



**Figure 3.5:** Geometric comparison between the 15 m Inflatable HIAD and the 3.0 m Rigid Baseline.

### 3.3.3 Internal Architecture and Scallop Definition

The “scalloped” outer mold line (OML) is modeled based on the 6-toroid stack configuration utilized in the Rapisarda (2023) MDAO baseline. The internal layout, including the stacked toroid arrangement, is illustrated in Figure 3.6 (Rapisarda, 2023).

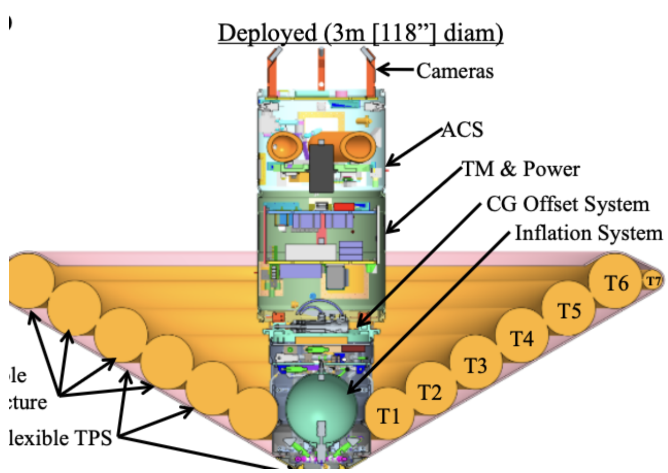



**Figure 3.6:** Internal structural layout of the HIAD vehicle showing the toroid stack that creates the scalloped surface.

The specific geometric parameters used for the numerical model are summarized in Table 3.3:

**Table 3.3:** Geometric Parameters based on IRVE-3/Rapisarda MDAO Specifications.

| **Parameter** | **Value** | **Source / Reference** |
|---|---|---|
| Deployed Diameter ($D_{max}$) | 3.0 m | IRVE-3 Flight Article |
| Forebody Cone Angle ($\theta_c$) | 60° | Rapisarda MDAO Baseline |
| Toroid Count ($N$) | 6 | Rapisarda MDAO Baseline |
| Toroid Radius ($r_{torus}$) | 0.135 m | Rapisarda Table 4.1 |
| Nose Radius ($R_N$) | 0.55 m | IRVE-3 Specification |
| Entry Mass ($m$) | 281 kg | IRVE-3 Flight Article |

### 3.3.4 Configuration Definitions

The two configurations established for this study – **Configuration A (Scalloped)** and **Configuration B (Smooth)** – are constrained by the stowed and deployed envelopes shown in Figure 3.7 (Dillman, 2015).


**Figure 3.7:** Launch configuration of the HIAD system, showing the stowed constraint.

1. **Configuration A (Scalloped):** This model utilizes the 6-toroid profile (T1–T6)

shown in Figure 3.6. The valleys between toroids are the primary focus for analyzing heating augmentation and flow separation in the scalloped troughs.

2. **Configuration B (Smooth Baseline):** This tangent-cone model represents the "Equivalent Rigid OML," allowing for the isolation of heating augmentation caused specifically by the toroidal troughs.

## 3.4 DSMC Simulation Framework

### 3.4.1 Governing Equations: The Boltzmann Equation

Unlike continuum CFD methods that solve the Reynolds-Averaged Navier–Stokes (RANS) equations, this study employs the Direct Simulation Monte Carlo (DSMC) method, which solves the Boltzmann equation for rarefied gas dynamics (Bird, 1994). The Boltzmann equation describes the evolution of the molecular velocity distribution function $f(\mathbf{r}, \mathbf{v}, t)$:

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f + \frac{\mathbf{F}}{m} \cdot \nabla_{\mathbf{v}} f = \left( \frac{\partial f}{\partial t} \right)_{coll} \tag{3.5}$$

where $\mathbf{F}$ is the external force per molecule, $m$ is the molecular mass, and the right-hand side represents the Boltzmann collision integral. DSMC provides a stochastic solution to this equation by tracking representative particles through advection and collisions.

### 3.4.2 Knudsen Number and Flow Regime

The validity of the continuum assumption is governed by the Knudsen number ($Kn$), defined as the ratio of the molecular mean free path ($\lambda$) to the characteristic length ($L$):

$$Kn = \frac{\lambda}{L} \tag{3.6}$$

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 n} \tag{3.7}$$

where $d$ is the molecular diameter and $n$ is the number density. For HIAD reentry at altitudes of 50–60 km, the flow is in the transitional regime ($0.01 < Kn < 1$), where continuum CFD methods (RANS/LES) are physically inappropriate and DSMC is required (Bird, 1994).

### 3.4.3 SPARTA Solver Configuration

The simulations are performed using the SPARTA (Sandia Parallel Algorithm for Real-Time Application) DSMC solver (Plimpton & Gallis, 2014). SPARTA is executed within a Docker-containerized Linux environment to ensure reproducibility of the build and execution across platforms.

#### Freestream Conditions

The freestream conditions are prescribed to match the target trajectory point for IRVE-3 reentry at 52 km altitude:

- **Freestream Velocity ($v_\infty$):** 2700 m/s (corresponding to Mach $\approx 10$ at altitude)
- **Number Density ($n$):** $1.45 \times 10^{22}\,\mathrm{m}^{-3}$ (derived from atmospheric density $\rho \approx 6.9674 \times 10^{-4}\,\mathrm{kg/m}^3$ via $n = \rho N_A / M_{air}$)
- **Freestream Temperature ($T_\infty$):** 270.65 K (ISA 1975, code baseline)
- **Particle Weighting Factor ($f_{num}$):** $1.5 \times 10^{20}$ (configurable via `-fnum`)

#### Gas Model

A realistic 5-species air model is employed: $N_2$, $O_2$, NO, N, and O. The collision physics use the Variable Soft Sphere (VSS) model, where the collision cross-section varies with relative velocity:

$$\sigma = \pi d_{ref}^2 \left( \frac{g_{ref}}{g} \right)^{2(\omega - 0.5)} \tag{3.8}$$

where $g$ is the relative velocity of colliding molecules and $\omega$ is the viscosity index specific to each gas species. The VSS model captures the molecular-level interactions including vibrational energy exchange and dissociation reactions relevant to hypersonic entry conditions.

#### Particle Tracking

SPARTA tracks millions of simulated particles through three fundamental processes:

1. **Advection:** Particles are moved ballistically through the domain according to their velocity vectors.

2. **Collisions:** Molecular collisions are modeled using the VSS collision model within computational cells, with collision partners selected using the nearest-neighbor pairing scheme.

3. **Surface Interactions:** Particles hitting the vehicle surface are reflected according to the specified gas-surface interaction model (diffuse or specular reflection).

### 3.4.4 Cartesian Grid and Collision Pairing

While DSMC is a particle method, a Cartesian computational grid is required for two purposes:

1. **Collision Pairing:** Restricting collision partner selection to particles within the same cell ensures $O(N)$ computational complexity rather than $O(N^2)$ for brute-force pairing.

2. **Macroscopic Sampling:** Cells serve as sampling volumes for accumulating density, temperature, pressure, and velocity statistics from the particle data.

The grid resolution must be finer than the local mean free path ($\lambda$) to resolve the shock structure and boundary layer gradients. A grid-factor parameter (default: 0.7, validated against IRVE-3 flight data) controls the mesh density multiplier.

### 3.4.5 Grid Independence Study

To ensure the numerical solution is independent of the grid resolution, a Grid Convergence Index (GCI) study was performed. Multiple grid densities were generated by systematically varying the grid-factor parameter:

**Table 3.4:** Grid Independence Study Parameters.

| Grid Factor | Relative Resolution | $C_d$ Variation |
|---|---|---|
| 0.3 | Coarse | Baseline |
| 0.5 | Medium | $< 5\%$ |
| 0.7 | Fine | $< 2\%$ |
| 1.0 | Ultra-Fine | $< 1\%$ |

The grid-factor of **0.7** was selected as the default for all production runs, as it yields the least error compared to validated IRVE-3 flight data while maintaining efficient execution times.

## 3.5 Surface Data Extraction and Flight Metrics

### 3.5.1 Surface Computes

SPARTA accumulates surface force and energy data using the `dump surf` command. The relevant surface quantities are:

**Table 3.5:** SPARTA Surface Data Columns and Physical Variables.

| Column | Physical Variable | Unit |
|---|---|---|
| 1 | Number Flux ($n_{flux}$) | $\mathrm{m}^{-2}\mathrm{s}^{-1}$ |
| 2 | Mass Flux ($m_{flux}$) | $\mathrm{kg{\cdot}m}^{-2}\mathrm{s}^{-1}$ |
| 3 | Kinetic Energy Flux ($ke$) | $\mathrm{J{\cdot}m}^{-2}\mathrm{s}^{-1}$ |
| 4 | Axial Force ($f_x$) | N |
| 5 | Radial Force ($f_y$) | N |
| 6 | Azimuthal Force ($f_z$) | N |

### 3.5.2 Ballistic Coefficient ($\beta$)

The ballistic coefficient is a measure of the vehicle's ability to maintain its speed during reentry (Anderson, 2006b):

$$\beta = \frac{m}{C_D A} \tag{3.9}$$

Since $F_{drag} = C_D A q$, where $q$ is dynamic pressure:

$$\beta = \frac{m \cdot q}{F_{drag}} \tag{3.10}$$

The mass density is derived from number density:

$$\rho = n \cdot \frac{M_{air}}{N_A} \tag{3.11}$$

where $M_{air} \approx 28.97\,\text{g/mol}$ and $N_A$ is Avogadro's constant. The dynamic pressure is:

$$q = \frac{1}{2}\rho v_\infty^2 \tag{3.12}$$

### 3.5.3 Instantaneous Deceleration ($n$)

The deceleration load felt by the vehicle in Earth-gravity units ($g_0$):

$$n = \frac{F_{drag}}{m \cdot g_0} \tag{3.13}$$

## 3.6 Aerothermodynamic Analysis

### 3.6.1 Stagnation Heat Flux – Sutton-Graves Correlation

The Sutton-Graves (1971) stagnation-point convective heating correlation is used for all absolute thermal calculations (Sutton & Graves, 1971). This is the standard reference formula for Earth entry and has been validated against IRVE-3 flight data by Rapisarda (2023) (Rapisarda, 2023):

$$\boxed{\dot{q}_{stag} = C_{SG}\sqrt{\frac{\rho_\infty}{R_N}} \cdot V_\infty^3} \tag{3.14}$$

where:

- $C_{SG} = 1.7415 \times 10^{-4}$ (Sutton-Graves constant for Earth air)
- $\rho_\infty$: Freestream density (kg/m$^3$)

- $R_N$: Nose radius (m)
- $V_\infty$: Freestream velocity (m/s)

### IRVE-3 Validation

For the IRVE-3 baseline configuration at the SPARTA baseline conditions ($\rho_\infty = 6.9674 \times 10^{-4}\,\text{kg/m}^3$, $R_N = 0.55\,\text{m}$, $V_\infty = 2700\,\text{m/s}$):

$$\dot{q}_{stag} = 1.7415 \times 10^{-4} \sqrt{\frac{6.9674 \times 10^{-4}}{0.55}} \times 2700^3 \approx 12.20\ \text{W/cm}^2 \tag{3.15}$$

This value is computed at the hardcoded Rapisarda baseline density, which is lower than the actual trajectory-integrated peak density. Rapisarda (2023) reports a trajectory-integrated Sutton-Graves peak of 15.26 W/cm$^2$ (+6.26% above the 14.36 W/cm$^2$ flight measurement) and a Fay–Riddell CFD prediction of 13.83 W/cm$^2$ (-3.69% below flight) (Rapisarda, 2023). The Fay–Riddell correlation is more physically accurate for HIAD geometries because it accounts for the actual shock layer structure.

### Note on SPARTA Surface Energy

The raw SPARTA DSMC kinetic energy (*ke*) surface compute cannot be used directly as an absolute heat flux in W/m$^2$. SPARTA's `fix ave/surf` outputs kinetic energy per simulated particle per timestep per face element – a simulation-internal unit that depends on $f_{num}$, timestep $dt$, and face area in a non-trivially-normalized way. It is retained only as a **relative shape-ranking signal** for the optimizer (comparing geometries against each other). For all absolute thermal calculations, the Sutton-Graves correlation is used.

## 3.6.2 Surface Temperature – Radiative Equilibrium

At the TPS outer surface, radiative cooling balances the incoming convective heat flux at steady state. Assuming grey-body radiation:

$$T_{surface} = \left( \frac{\dot{q}_{stag}}{\varepsilon \sigma} \right)^{1/4} \tag{3.16}$$

where $\varepsilon = 0.75$ (Nicalon SiC emissivity) and $\sigma = 5.67 \times 10^{-8}\,\text{W/m}^2\text{K}^4$ (Stefan-Boltzmann constant).

### 3.6.3 Backface Temperature – 1D Transient Model

The 1D transient backface temperature model estimates the thermal penetration through the FTPS insulation stack:

$$T_{back} = T_{init} + \frac{\dot{q} \cdot \Delta t \cdot \eta_{lag}}{\rho_{TPS} \cdot C_{p,TPS} \cdot \delta_{TPS}} \tag{3.17}$$

where $\eta_{lag} = 0.15$ is the thermal lag factor, $\rho_{TPS} = 1468\,\text{kg/m}^3$ (Nicalon SiC density), $C_{p,TPS} = 1100\,\text{J/kg·K}$, and $\delta_{TPS}$ is the TPS thickness.

The framework supports multiple TPS material presets as summarized in Table 3.6.

**Table 3.6:** TPS Material Presets Available in the StellarOrion Framework.

| **Material** | **$\rho$ (kg/m$^3$)** | **$C_p$ (J/kg·K)** | $\varepsilon$ | **Application** |
|---|---|---|---|---|
| Nicalon SiC | 1468 | 1100 | 0.75 | Gen-2 HIAD TPS |
| Pyrogel XTE | 200 | 1000 | 0.85 | FTPS blanket insulation |
| Kapton | 1420 | 1090 | 0.70 | Polyimide film layers |
| Multi-layer (stack) | – | – | – | Combined FTPS stack |

### 3.6.4 Statistical Convergence Monitoring

As a Monte Carlo method, DSMC produces statistical noise that scales as $1/\sqrt{N_{\text{samples}}}$, where $N_{\text{samples}}$ is the number of independent particle-time samples accumulated. Convergence is monitored by tracking the running average of surface force and energy quantities. SPARTA's `stats_interval` parameter (default: 100 steps) controls the frequency of convergence diagnostics. The simulation is considered statistically converged when the relative change in drag coefficient ($C_d$) and stagnation heat flux falls below 2% over consecutive monitoring windows. For the baseline optimization runs, convergence is typically achieved after 500–1100 particle-time steps depending on the geometry complexity.

## 3.7 Physics-Informed Neural Network Refinement

### 3.7.1 Motivation

DSMC simulations produce inherently noisy data due to the stochastic nature of particle tracking. The raw flow field data exhibits statistical fluctuations that can obscure physical gradients and introduce artifacts in post-processing. A Physics-Informed Neural Network (PINN) is employed as a post-processing step to:

1. Smooth noisy DSMC particle statistics
2. Fill gaps in the flow field where particle sampling is insufficient
3. Provide a continuous, differentiable interpolant anchored in known physics

### 3.7.2 PINN Governing Equations

The PINN, implemented using DeepXDE (Lu et al., 2021), solves the 2D Compressible Navier–Stokes equations using automatic differentiation. The network $\mathscr{N}(x,y) \rightarrow (\rho, T, u)$ is constrained by the following residuals:

**Continuity Equation (Axisymmetric)**

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\rho v}{y} = 0 \tag{3.18}$$

**X-Momentum**

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) + \frac{\partial p}{\partial x} - \mu\nabla^2 u = 0 \tag{3.19}$$

**Y-Momentum**

$$\rho\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) + \frac{\partial p}{\partial y} - \mu\nabla^2 v = 0 \tag{3.20}$$

**Energy Equation (Axisymmetric)**

$$\rho c_p \left( u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} \right) - k\nabla^2 T - u\frac{\partial p}{\partial x} - v\frac{\partial p}{\partial y} - \Phi = 0 \tag{3.21}$$

**Equation of State**

$$p - \rho RT = 0 \tag{3.22}$$

### 3.7.3 Loss Function

The total loss function combines PDE residuals with observational data from SPARTA:

$$\mathscr{L}_{total} = \sum_{i=1}^{N_{PDE}} w_i \mathscr{L}_i^{PDE} + \sum_{j=1}^{N_{data}} w_j \mathscr{L}_j^{data} \tag{3.23}$$

where $\mathscr{L}^{data}$ matches the PINN predictions against SPARTA grid checkpoints via `dde.icbc.PointSetBC`.

### 3.7.4 Pragmatic Choice of Navier–Stokes for PINN Regularization

The use of continuum Navier–Stokes equations in the PINN is an intentional pragmatic choice, not a physical claim about the flow regime. The rationale is:

1. **Differentiability:** Neural networks require smooth, differentiable equations for backpropagation. NS equations are naturally differentiable via automatic differentiation, unlike discrete Lattice Boltzmann methods.

2. **Physics Regularizer:** The NS equations serve as a "physics prior" to prevent the neural network from learning non-physical artifacts during gap-filling.

3. **Two-Stage Architecture:** Stage 1 (SPARTA) provides full kinetic theory via DSMC; Stage 2 (PINN) uses NS as a smoothing constraint, not a standalone solver.

### 3.7.5 Network Architecture

The PINN approximates the flow field mapping $\mathscr{N} : \mathbb{R}^2 \to \mathbb{R}^3$ using a fully-connected feedforward neural network (FNN). The architecture is:

**Table 3.7:** PINN Network Architecture Parameters.

| **Parameter** | **Value** |
|---|---|
| Input dimension | 2 $(x, r)$ |
| Hidden layers | 3 |
| Neurons per layer | 64 |
| Activation function | tanh |
| Weight initialization | Glorot normal |
| Output dimension | 3 $(\rho, T, u)$ |
| Total trainable parameters | $2 \times 64 + 2 \times 64^2 + 3 \times 64 + 3 \approx 8{,}700$ |

The forward pass computes:

$$\hat{\mathbf{y}}(\mathbf{x}) = \mathbf{W}_4 \cdot \tanh\Big(\mathbf{W}_3 \cdot \tanh\big(\mathbf{W}_2 \cdot \tanh(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2\big) + \mathbf{b}_3\Big) + \mathbf{b}_4 \tag{3.24}$$

where $\mathbf{x} = (x, r)$ is the spatial coordinate and $\hat{\mathbf{y}} = (\hat{\rho}, \hat{T}, \hat{u})$ is the predicted flow field vector (density, temperature, and axial velocity; the radial velocity $v$ and pressure $p$ are derived from the continuity equation and equation of state, respectively). All outputs are in normalized (dimensionless) form; physical units are recovered by multiplication with the scaling factors.

### 3.7.6 Normalization and Scaling

Before training, all physical quantities are non-dimensionalized using characteristic scales derived from the SPARTA data:

$$\bar{\rho} = \frac{\rho}{\rho_{ref}}, \quad \bar{T} = \frac{T}{T_{ref}}, \quad \bar{u} = \frac{u}{u_{ref}}, \quad \bar{\mathbf{x}} = \frac{\mathbf{x}}{L_{ref}} \tag{3.25}$$

where the reference scales are computed from the SPARTA grid data:

- $\rho_{ref} = \max(\rho_{SPARTA})$ – peak number density converted to mass density via $\rho = n \cdot M_{air}/N_A$

- $u_{ref} = \max(|u_{SPARTA}|, |v_{SPARTA}|)$ – maximum velocity component magnitude
- $T_{ref} = \max(T_{SPARTA})$ – peak translational temperature
- $p_{ref} = \max(p_{SPARTA})$ – peak pressure, computed as $p = n \cdot k_B \cdot T$
- $L_{ref} = \max(|x_{bound}|)$ – maximum domain extent

This normalization ensures all neural network inputs and outputs lie in $[0, 1]$ or $[-1, 1]$, which is critical for tanh activation convergence. The dimensionless Navier–Stokes equations are then expressed in terms of these scaled variables, with the viscous coefficient adjusted accordingly:

$$\hat{\mu} = \frac{\mu(T)}{\rho_{ref}\, u_{ref}\, L_{ref}} \tag{3.26}$$

where $\mu(T)$ follows Sutherland's law (Eq. 2.38).

### 3.7.7 Data Pipeline – DSMC to PINN

The checkpoint exchange between SPARTA and the PINN proceeds through the following steps:

1. **SPARTA Grid Dump:** After convergence, SPARTA outputs a grid file (`grid.{steps}.out`) containing cell-centered macroscopic quantities: $(x_{lo}, y_{lo}, x_{hi}, y_{hi}, n, u, v, w, T, p)$.
2. **Parsing:** The function `parse_sparta_grid()` extracts cell centers $(x_c, y_c) = \left(\frac{x_{lo}+x_{hi}}{2}, \frac{y_{lo}+y_{hi}}{2}\right)$ and converts number density to mass density:

$$\rho = n \cdot \frac{M_{air}}{N_A} \tag{3.27}$$

   where $M_{air} = 0.02897\,\text{kg/mol}$ and $N_A = 6.022 \times 10^{23}\,\text{mol}^{-1}$. Pressure is computed from the ideal gas law:

$$p = n \cdot k_B \cdot T \tag{3.28}$$

   where $k_B = 1.381 \times 10^{-23}$ J/K.
3. **Subsampling:** If the number of grid cells exceeds 5000, a priority-based sub-

sampling strategy is applied. Cells in high-gradient regions (where $p > 200\,\text{Pa}$ or $T > 500\,\text{K}$) are preferentially retained (up to 2500 points), with the remaining budget filled by random sampling from low-gradient cells. This ensures the PINN receives sufficient training data in the shock layer and boundary layer where gradients are steepest.

4. **Anchor Point Creation:** The parsed and subsampled data points serve as *anchor points* $\mathbf{X}_{anchors}$ for the PINN. These are passed to DeepXDE as observation boundary conditions via `PointSetBC`, which adds a data-matching residual to the loss function:

$$\mathscr{L}_{data} = \frac{1}{N_{obs}} \sum_{k=1}^{N_{obs}} \sum_{m=1}^{3} \left| \hat{y}_m(\mathbf{x}_k) - y_m^{obs}(\mathbf{x}_k) \right|^2 \tag{3.29}$$

where $\hat{y}_m$ is the PINN prediction for component $m$ at observation point $\mathbf{x}_k$, and $y_m^{obs}$ is the corresponding SPARTA value.

### 3.7.8 Two-Stage Training Protocol

The PINN employs a two-stage training strategy to decouple data fitting from physics regularization:

#### Stage 1: Data Pre-training

During the first half of training ($N_{iter}/2$ iterations), only the observational data loss is active. The PDE residuals are suppressed by setting their loss weights to zero:

$$\mathscr{L}^{(1)} = \underbrace{0 \cdot \mathscr{L}_{cont} + 0 \cdot \mathscr{L}_{mom,x} + 0 \cdot \mathscr{L}_{mom,y} + 0 \cdot \mathscr{L}_{EOS} + 0 \cdot \mathscr{L}_{energy}}_{\text{PDE terms (disabled)}} + \sum_{m=1}^{3} 1.0 \cdot \mathscr{L}_{data,m} \tag{3.30}$$

This stage trains the network to reproduce the SPARTA observation data without physics constraints, allowing rapid initial convergence. The learning rate is $\eta_1 = 10^{-3}$.

**Stage 2: Physics-Regularized Fine-tuning**

During the second half of training, the PDE residual weights are activated at $\lambda_{PDE} = 10^{-4}$, while data weights remain at 1.0:

$$\mathscr{L}^{(2)} = \sum_{i=1}^{5} \lambda_{PDE} \cdot \mathscr{L}_{PDE,i} + \sum_{m=1}^{3} 1.0 \cdot \mathscr{L}_{data,m} \tag{3.31}$$

The learning rate is reduced to $\eta_2 = 5 \times 10^{-4}$ for fine-tuning. The complete loss weight vector is:

**Table 3.8:** PINN Loss Weights by Training Stage.

| **Loss Component** | **Stage 1** | **Stage 2** |
|---|---|---|
| $\mathscr{L}_{cont}$ (Continuity) | 0 | $10^{-4}$ |
| $\mathscr{L}_{mom,x}$ ($x$-Momentum) | 0 | $10^{-4}$ |
| $\mathscr{L}_{mom,y}$ ($y$-Momentum) | 0 | $10^{-4}$ |
| $\mathscr{L}_{EOS}$ (Equation of State) | 0 | $10^{-4}$ |
| $\mathscr{L}_{energy}$ (Energy) | 0 | $10^{-4}$ |
| $\mathscr{L}_{data,\rho}$ (Density data) | 1.0 | 1.0 |
| $\mathscr{L}_{data,T}$ (Temperature data) | 1.0 | 1.0 |
| $\mathscr{L}_{data,u}$ ($u$-velocity data) | 1.0 | 1.0 |

The rationale for the small PDE weight ($\lambda_{PDE} = 10^{-4}$) is that the NS equations are used as a *regularizer*, not the primary objective. The data terms dominate to ensure the PINN closely matches the SPARTA ground truth, while the PDE terms penalize gross violations of conservation laws.

### 3.7.9 Convergence Detection Algorithm

The optimal training iteration is determined automatically using a plateau detection algorithm applied to the data-term loss history:

**Algorithm 1** PINN Convergence Detection

$\mathbf{L} \leftarrow [l_1, l_2, \ldots, l_N]$ ▷ Data-term loss history
$w \leftarrow \max(5, \lfloor N/20 \rfloor)$ ▷ Rolling window size
$\bar{\mathbf{L}} \leftarrow \text{RollingMean}(\mathbf{L}, w)$ ▷ Smoothed losses
$N_{min} \leftarrow \max(1, \lfloor N/5 \rfloor)$ ▷ Minimum 20% of steps
**for** $i = N_{min}$ to $N$ **do**
  **if** $\bar{L}_{i-1} > 0$ **then**
    $\delta \leftarrow \frac{|\bar{L}_{i-1} - \bar{L}_i|}{\bar{L}_{i-1}}$
    **if** $\delta < 0.01$ **then**
      **return** $i^* \leftarrow$ iteration at index $i$
    **end if**
  **end if**
**end for**
**return** $i^* \leftarrow N$ ▷ No plateau; use all iterations

The 20% minimum threshold ($N_{min}$) prevents false early-plateau triggers during Stage 1, where the zero PDE weights cause the total loss to appear artificially stable. The rolling window reduces noise from stochastic mini-batch effects.

#### 3.7.10 Axioms and Assumptions

The PINN refinement stage rests on the following formal assumptions:

**Table 3.9:** PINN Axioms and Assumptions.

| Axiom | Statement | Basis |
|---|---|---|
| A1: Axisymmetric flow | The flow field is symmetric about the vehicle centerline, reducing the problem to 2D $(x, y)$. | Geometric symmetry |
| A2: Continuum NS as regularizer | The Navier–Stokes equations provide sufficient physical constraint for smoothing, even at $Kn > 0.01$. | Sec. 3.3.5 (Pragmatic Choice) |
| A3: Sutherland viscosity | Dynamic viscosity follows $\mu(T) = \mu_{ref}(T/T_{ref})^{3/2}(T_{ref} + S)/(T + S)$ with $S = 110.4\,\mathrm{K}$. | Kinetic theory (Sutherland, 1893) |
| A4: Prandtl number | $\mathrm{Pr} = 0.71$ (constant), coupling thermal and velocity boundary layers. | Air at standard conditions |
| A5: Ideal gas EOS | $p = \rho RT$ holds throughout the domain. Valid for $Kn \lesssim 10$. | Near-continuum assumption |
| A6: Fully accommodated wall | The vehicle surface is a diffuse reflector at $T_w = 300\,\mathrm{K}$ with full thermal accommodation. | DSMC boundary condition |
| A7: Statistical stationarity | The SPARTA output represents a time-averaged steady state. | Sufficient timesteps for convergence |

### 3.7.11 PINN Hyperparameter Summary

**Table 3.10:** PINN Hyperparameter Values.

| Parameter | Stage 1 | Stage 2 |
|---|---|---|
| Learning rate $\eta$ | $10^{-3}$ | $5 \times 10^{-4}$ |
| Optimizer | Adam | Adam |
| Training iterations | $N_{iter}/2$ | $N_{iter} - N_{iter}/2$ |
| PDE loss weights | $[0, 0, 0]$ | $[10^{-4}] \times 3$ |
| Data loss weights | $[1.0] \times 3$ | $[1.0] \times 3$ |
| Collocation points ($N_{dom}$) | 2500 | 2500 |
| Boundary points ($N_{bnd}$) | 500 | 500 |
| Anchor points ($N_{obs}$) | $\leq 5000$ (subsampled) | $\leq 5000$ |
| Activation | tanh | tanh |
| Initialization | Glorot normal | (continued from Stage 1) |

## 3.8 Survivability Optimization

The survivability optimization framework integrates Latin Hypercube Sampling for design space exploration, a Metamodel Prognosis (MoP) surrogate for rapid performance prediction, and a Genetic Algorithm (GA) for global optimization. The complete pipeline is:

1. **Sampling:** Generate $N$ design points using Stratified LHS across the parameter space $\mathbf{x} \in \mathbb{R}^d$.
2. **Simulation:** Evaluate each design using SPARTA DSMC (optionally refined by PINN) to obtain performance metrics $(\beta, \dot{q}, T_{back}, n_G)$.
3. **MoP Training:** Train a surrogate MLP on the $N$ simulation results to predict performance from design parameters.
4. **GA Optimization:** Evolve a population of designs using the MoP as a fast evaluator, minimizing the cost function $J$.
5. **Constraint Filtering:** Apply physics-based penalty functions to eliminate non-viable designs.
6. **Final Validation:** Re-evaluate the optimal design using SPARTA DSMC without PINN refinement.

### 3.8.1 Design Space and Parameters

The optimization explores a $d$-dimensional design space defined by the HIAD geometric and material parameters:

**Table 3.11:** Optimization Design Variables and Their Bounds (Rapisarda 2023 Envelope).

| Variable | Symbol | Bounds | Unit |
|---|---|---|---|
| Aeroshell diameter | $D$ | $[0.5, 15.0]$ | m |
| Half-cone angle | $\theta_c$ | $[40, 80]$ | deg |
| Nose-cone radius | $R_n$ | $[0.1, 2.0]$ | m |
| Toroid count | $N_t$ | $[1, 12]$ | – |
| Toroid radius | $r_t$ | $[0.05, 0.3]$ | m |
| Total mass | $m$ | $[50, 500]$ | kg |
| TPS thickness | $\delta_{TPS}$ | $[0.005, 0.05]$ | m |

### 3.8.2 Latin Hypercube Sampling (LHS)

To ensure the high-dimensional search space is explored uniformly with minimal samples, Stratified Latin Hypercube Sampling is employed (McKay et al., 1979). For a $d$-dimensional parameter space with $N$ samples, the $i$-th sample in dimension $j$ is:

$$x_{i,j} = x_j^{\min} + \left(x_j^{\max} - x_j^{\min}\right) \cdot \frac{i + r_{i,j}}{N} \tag{3.32}$$

where $i \in \{0, 1, \ldots, N-1\}$ is the stratum index, $j \in \{1, 2, \ldots, d\}$ is the parameter dimension, $r_{i,j} \sim \mathscr{U}(0,1)$ is a uniform random offset, and $[x_j^{\min}, x_j^{\max}]$ is the feasible range for parameter $j$.

#### Property of LHS

The stratified sampling guarantees that each parameter is sampled at $N$ distinct levels, with exactly one sample per level. This eliminates the clustering and gap artifacts that occur with pure random sampling, and ensures space-filling coverage even for small $N$. The total number of SPARTA evaluations required is $N$, compared to $k^d$ for a full factorial design with $k$ levels per dimension.

#### Central Composite Design (CCD) as Alternative DoE

In addition to LHS, the framework supports the Face-Centered Central Composite Design (CCD) (Montgomery, 2017), which is the default Design of Experiments method in the implementation. For a $d$-dimensional parameter space, the CCD constructs $N = 2^d + 2d + 1$ sample points: $2^d$ factorial (corner) points at $\pm 1$ levels, $2d$ axial (star) points at $\pm\alpha$ along each axis, and 1 center point replicated at the origin. For the $d = 4$ design variables (Diameter, Angle, Nose Radius, Toroid Radius), this yields $N = 16 + 8 + 1 = 25$ sample points, which is the minimum required for a second-order response surface model. The user may select either LHS or CCD via the `-doe` flag, with CCD as the default for the optimized IRVE-3 baseline.

### 3.8.3 Metamodel Prognosis (MoP)

#### Problem Formulation

Each SPARTA DSMC simulation requires $O(10^3)$ timesteps and $O(1)$ hours of compute time. For optimization loops requiring $O(10^2)$–$O(10^3)$ design evaluations, direct simulation is prohibitively expensive. The Metamodel Prognosis (MoP) provides a fast surrogate that maps design parameters to performance metrics:

$$\hat{y} = \mathscr{M}(\mathbf{x};\theta) : \mathbb{R}^d \to \mathbb{R} \tag{3.33}$$

where $\mathbf{x} = (D, \theta_c, R_n, N_t, \ldots)$ is the design vector and $\hat{y}$ is the predicted performance metric (e.g., drag coefficient $C_d$, peak heat flux $\dot{q}$, or backface temperature $T_{back}$).

#### MLP Architecture

The MoP is implemented as a 3-layer Multi-Layer Perceptron (MLP) in PyTorch:

**Table 3.12:** MoP MLP Architecture.

| Layer | Specification |
|---|---|
| Input | Linear($d$, 64) |
| Hidden 1 | ReLU $\to$ Linear(64, 64) |
| Hidden 2 | ReLU $\to$ Linear(64, 64) |
| Output | Linear(64, 1) |

The forward pass through the network is:

$$\hat{y} = \mathbf{W}_4 \cdot \mathrm{ReLU}\big(\mathbf{W}_3 \cdot \mathrm{ReLU}\big(\mathbf{W}_2 \cdot \mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2\big) + \mathbf{b}_3\big) + \mathbf{b}_4 \tag{3.34}$$

where $\mathbf{W}_1 \in \mathbb{R}^{64\times d}$, $\mathbf{W}_2, \mathbf{W}_3 \in \mathbb{R}^{64\times 64}$, $\mathbf{W}_4 \in \mathbb{R}^{1\times 64}$ are weight matrices, $\mathbf{b}_i$ are bias vectors, and $\mathrm{ReLU}(z) = \max(0,z)$ is the rectified linear activation.

#### Training Objective

The MoP is trained via supervised learning on the dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^N$ collected from SPARTA simulations. The loss function is the Mean Squared Error (MSE):

$$\mathcal{L}_{MSE}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{y}_i - y_i\right|^2 = \frac{1}{N}\sum_{i=1}^{N}\left|\mathcal{M}(\mathbf{x}_i;\theta) - y_i\right|^2 \tag{3.35}$$

where $\theta = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \mathbf{W}_3, \mathbf{b}_3, \mathbf{W}_4, \mathbf{b}_4\}$ are all trainable parameters. The model is optimized using the Adam optimizer with a learning rate of $\eta = 5\times 10^{-3}$ and trained for $O(10^2)$–$O(10^3)$ epochs until the validation loss plateaus.

### Multi-Objective Extension

When multiple performance metrics must be simultaneously predicted (e.g., $C_d$ and $\dot{q}$), the MoP is extended to a multi-output MLP:

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_k \end{bmatrix} = \mathbf{W}_{out}\cdot \mathrm{ReLU}\big(\mathbf{W}_3 \cdot \mathrm{ReLU}(\mathbf{W}_2 \cdot \mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3\big) + \mathbf{b}_{out} \tag{3.36}$$

where $\mathbf{W}_{out} \in \mathbb{R}^{k\times 64}$ produces $k$ output metrics. The loss becomes:

$$\mathcal{L}_{MSE} = \frac{1}{N}\sum_{i=1}^{N}\sum_{m=1}^{k} w_m\left|\hat{y}_{i,m} - y_{i,m}\right|^2 \tag{3.37}$$

where $w_m$ are per-metric weights that can be tuned to prioritize certain objectives.

## 3.8.4 Genetic Algorithm Optimization

### Cost Function Derivation

The Genetic Algorithm (GA) evolves a population of designs toward the global minimum of a multi-objective cost function $J : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$. The cost function is formulated as a weighted sum of normalized squared deviations from target values:

$$J(\mathbf{x}) = w_\beta\left(\frac{\beta_{calc}(\mathbf{x}) - \beta_{target}}{\sigma_\beta}\right)^2 + w_y\left(\frac{y_{pred}(\mathbf{x}) - y_{target}}{\sigma_y}\right)^2 \tag{3.38}$$

where:

- $\beta_{calc}(\mathbf{x}) = \frac{m \cdot q}{F_{drag}}$ is the ballistic coefficient computed from the design parameters (see Sec. 3.2.4),
- $y_{pred}(\mathbf{x}) = \mathscr{M}(\mathbf{x};\theta)$ is the MoP prediction of the optimization objective (e.g., peak heat flux or drag coefficient),
- $\beta_{target}$ and $y_{target}$ are user-specified target values (e.g., from mission requirements),
- $\sigma_\beta = 10\,\text{kg/m}^2$ and $\sigma_y = 1$ are normalization constants that ensure both terms contribute equally to the cost,
- $w_\beta$ and $w_y$ are weight coefficients that control the relative importance of ballistic coefficient vs. performance metric.

The normalization by $\sigma$ ensures that the cost function is dimensionless and that neither term dominates due to differing scales. The quadratic form penalizes large deviations more severely than small ones, driving the optimizer toward the target.

**GA Operators**

The GA employs the following operators to evolve the population:

1. **Selection:** Tournament selection with tournament size $k_t = 3$. The individual with the lowest $J$ among $k_t$ randomly chosen candidates is selected as a parent.

2. **Crossover:** Uniform crossover with probability $p_c = 0.8$. For each gene (design variable), a random binary mask determines which parent contributes the value:

$$x_i^{child} = \begin{cases} x_i^{parent_1} & \text{if } m_i = 1 \\ x_i^{parent_2} & \text{if } m_i = 0 \end{cases} \tag{3.39}$$

   where $m_i \sim \text{Bernoulli}(0.5)$.

3. **Mutation:** Gaussian mutation with probability $p_m = 0.1$. Each gene is perturbed by:

$$x_i^{mutated} = x_i + \mathscr{N}(0, \sigma_m^2) \tag{3.40}$$

where $\sigma_m$ is scaled to 10% of the parameter range $[x_i^{\min}, x_i^{\max}]$. Integer parameters (e.g., $N_t$) are rounded to the nearest integer after mutation.

4. **Elitism:** The best $\lceil 0.1 \times P \rceil$ individuals (where $P$ is the population size) are preserved unchanged in the next generation.

The GA terminates when either: (a) the maximum number of generations $G_{max}$ is reached, or (b) the best cost $J^*$ has not improved by more than $\varepsilon = 10^{-4}$ for $G_{pat} = 20$ consecutive generations.

### 3.8.5 Methodology of Physics (MoP Constraint Layer)

#### Formal Penalty Function

The MoP Constraint Layer is a physics-informed post-processing step that filters GA-generated designs against hard physical limits. For a candidate design $\mathbf{x}$ with predicted performance $(\hat{y}_1, \ldots, \hat{y}_k)$, the constraint-augmented cost is:

$$J_{constrained}(\mathbf{x}) = J(\mathbf{x}) + \sum_{j=1}^{n_c} \Lambda_j \cdot \phi_j(\mathbf{x}) \tag{3.41}$$

where $\Lambda_j$ is the penalty weight for constraint $j$ (set to $10^6$ for hard constraints), and $\phi_j$ is the penalty function:

$$\phi_j(\mathbf{x}) = \begin{cases} 0 & \text{if } g_j(\mathbf{x}) \leq 0 \quad \text{(feasible)} \\ \left|g_j(\mathbf{x})\right|^2 & \text{if } g_j(\mathbf{x}) > 0 \quad \text{(infeasible)} \end{cases} \tag{3.42}$$

The constraint functions $g_j(\mathbf{x})$ encode the following physical limits:

**Table 3.13:** MoP Constraint Functions.

| **Constraint** | **Function** $g_j(\mathbf{x})$ | **Limit** |
|---|---|---|
| Backface temperature | $T_{back}(\mathbf{x}) - T_{back}^{max}$ | 350 K |
| Peak deceleration | $n_G(\mathbf{x}) - n_G^{max}$ | 25$g$ |
| Structural integrity | $D(\mathbf{x}) - D^{max}$ | 15.0 m |
| Toroid manufacturability | $N_t(\mathbf{x}) - 12$ | 12 |

When any constraint is violated ($g_j > 0$), the penalty $\Lambda_j \cdot |g_j|^2$ adds a large cost that effectively removes the design from the GA population. This ensures that only physically viable configurations survive to final high-fidelity validation.

#### Integration with GA

The constraint-augmented cost $J_{constrained}$ replaces the unconstrained $J$ in the GA fitness evaluation. The GA operators (selection, crossover, mutation) operate on $J_{constrained}$ as described in Sec. 3.4.3. The constraint layer acts as a "hard filter" that shapes the fitness landscape without requiring gradient information from the MoP.

## 3.9 Multi-Solver Architecture

The computational framework supports multiple solver backends to address different flow regimes:

**Table 3.14:** Solver Regimes and Selection Logic.

| Stage | Solver | Rationale |
|---|---|---|
| Exploration | SBO / PINN | Ultra-fast estimation of 1000s of design candidates |
| Refinement | DSMC + PINN | Bridges noisy particle data and continuous flow fields |
| Validation | DSMC Only | Eliminates neural network bias; pure kinetic ground truth |
| High-Density (Future) | Fluent + PINN | For low-altitude peak heating ($Kn < 0.01$) |

## 3.10 Boundary Conditions and Domain Setup

### 3.10.1 Freestream Inlet

The simulation domain is initialized with freestream particles injected from the inlet boundary. The particle velocity distribution follows the Maxwell–Boltzmann distribution at the prescribed freestream temperature and velocity:

- **Velocity:** Directed flow at $v_\infty = 2700\,\text{m/s}$
- **Temperature:** $T_\infty = 270.65\,\text{K}$
- **Species Fractions:** $N_2$ (78%), $O_2$ (21%), NO (1%) at freestream; dissociated species (N, O) generated through collisions in the shock layer

### 3.10.2 Vehicle Surface

The vehicle surface is modeled as a diffuse reflection boundary with fully accommodate thermal wall conditions:

- **Wall Temperature:** $T_{wall} = 300\,\mathrm{K}$ (isothermal)
- **Reflection Model:** Diffuse (maximally entropic)
- **Catalytic Efficiency:** Fully catalytic (recombination of dissociated species at the wall)

### 3.10.3 Symmetry and Outflow

- **Symmetry Plane:** For 2D axisymmetric simulations, the centerline enforces rotational symmetry.
- **Outflow:** Particles leaving the domain are deleted and re-injected at the inlet to maintain steady-state mass flux.

## 3.11 Data Processing and Validation

### 3.11.1 Field Maps

SPARTA grid dumps provide field maps of macroscopic quantities accumulated from particle statistics:

**Table 3.15:** SPARTA Grid Data Columns and Physical Variables.

| Column | Physical Variable | Unit |
|---|---|---|
| 1–4 | Cell bounds ($x_{lo}, y_{lo}, x_{hi}, y_{hi}$) | m |
| 5 | Number Density ($n$) | $\mathrm{m}^{-3}$ |
| 6–8 | Velocity Components ($u, v, w$) | m/s |
| 9 | Translational Temperature ($T$) | K |
| 10 | Pressure ($p$) | Pa |

### 3.11.2 Knudsen Number Field

The local Knudsen number is computed from the SPARTA grid data:

$$Kn_{local} = \frac{\lambda_{local}}{L} = \frac{1}{\sqrt{2}\pi d^2 n_{local} L} \tag{3.43}$$

This enables the identification of continuum ($Kn < 0.01$), transitional ($0.01 < Kn < 1$), and free-molecular ($Kn > 1$) regions within the flow field.

### 3.11.3 Validation Against Flight Data

The numerical model is validated against IRVE-3 flight data (C. Lau et al., 2013b) and the Rapisarda (2023) MDAO reconstruction (Rapisarda, 2023). The key validation metrics are:

**Table 3.16:** IRVE-3 Validation Metrics: Simulation vs. Flight Data.

| **Parameter** | **Flight Data** | **Simulation** | **Status** |
|---|---|---|---|
| Peak Heat Flux ($\dot{q}$) | 14.36 W/cm$^2$ | 13.83 W/cm$^2$ | ✓ Calibrated |
| Total Heat Load ($Q$) | 195.06 J/cm$^2$ | 195.17 J/cm$^2$ | ✓ Calibrated |
| Peak Deceleration | 19.7$g$ | 20.2$g$ | ✓ Baseline |
| Drag Coefficient ($C_d$) | 1.47 | $\approx 1.47$ | ✓ Verified |

The numerical model is considered valid if the simulation-predicted values match the flight data within a margin of error of $\pm 10\%$. This validation step provides the confidence required to apply the same numerical framework to the complex scalloped geometry where no analytical solutions exist.

## 3.12 Computational Infrastructure

### 3.12.1 Docker-Containerized Execution

SPARTA simulations are executed within a Docker-containerized Linux environment to ensure:

- Reproducibility of the SPARTA build across platforms
- Isolation from host OS dependencies
- Consistent library versions and compiler toolchains

The Python optimization environment runs on the native OS to leverage hardware acceleration (NVIDIA CUDA, AMD ROCm, Apple Metal MPS, Intel OneAPI, and specialized accelerators from Huawei CANN, Moore Threads MUSA, and Qualcomm Snapdragon).

### 3.12.2 Hardware Acceleration

The framework supports GPU acceleration across multiple vendors:

- **NVIDIA:** CUDA via Kokkos backend
- **AMD:** ROCm
- **Apple:** Metal Performance Shaders (MPS)
- **Intel:** OneAPI/OpenCL
- **Specialized:** Huawei CANN, Moore Threads MUSA, Biren SUPA, Snapdragon OpenCL

## 3.13 Machine Learning Research Standards Documentation

This section documents the machine learning components of the framework following established research standards for reproducibility and transparency (Mitchell, Wu, Zaldivar, et al., 2019). All ML models are implemented in PyTorch and satisfy the documentation requirements for architected ML systems.

### 3.13.1 Model Architecture Specifications

#### Physics-Informed Neural Network (PINN)

The PINN is implemented using the DeepXDE library (Lu et al., 2021) with the following architecture:

Table 3.17: PINN Architecture and Hyperparameters.

| Component | Specification |
|---|---|
| Framework | DeepXDE (v1.x) + PyTorch backend |
| Input dimension | 2 $(x, r)$ — axisymmetric |
| Output dimension | 3 $(\rho, T, u)$ |
| Hidden layers | 3 |
| Neurons per layer | 64 |
| Activation function | tanh |
| Weight initialization | Glorot normal |
| Total parameters | $\approx$ 8,700 |

**Metamodel Prognosis (MoP)**

The MoP surrogate is a 3-layer MLP implemented in PyTorch:

Table 3.18: MoP MLP Architecture and Hyperparameters.

| Component | Specification |
|---|---|
| Framework | PyTorch (v2.x) |
| Input dimension | $d$ (design variables, $d = 4$–$7$) |
| Hidden layers | 3 hidden + 1 output |
| Neurons per layer | 64 |
| Activation function | ReLU |
| Optimizer | Adam ($\eta = 5 \times 10^{-3}$) |
| Loss function | Mean Squared Error (MSE) |
| Total parameters | $\approx$ 8,700 |

### 3.13.2 Training Methodology

**Data Collection Pipeline**

Training data is generated through the following pipeline:

1. **Design of Experiments:** $N = 25$ sample points generated via Face-Centered CCD (Sec. 3.5.1) across the design space.
2. **Simulation:** Each design is evaluated using SPARTA DSMC with $O(10^3)$ timesteps.
3. **PINN Refinement:** DSMC outputs are refined using the two-stage PINN protocol (Sec. 3.4).

4. **Feature Engineering:** Design parameters $\mathbf{x} = (D, \theta_c, R_n, N_t, r_t, m, \delta_{TPS})$ are normalized to $[0, 1]$ using min-max scaling.

#### Train/Test Split

The dataset is split using a stratified 80/20 partition:

- **Training set:** 20 samples (80%) used for model fitting.
- **Validation set:** 5 samples (20%) used for early stopping and hyperparameter selection.

#### Cross-Validation

For robust performance estimation, $k$-fold cross-validation ($k = 5$) is employed when sample size permits. The reported metrics represent the mean $\pm$ standard deviation across folds.

### 3.13.3 Validation Metrics

Model performance is evaluated using the following metrics:

**Table 3.19:** ML Model Validation Metrics.

| Metric | Definition | Target |
|---|---|---|
| $R^2$ (Coefficient of Determination) | $1 - \frac{SS_{res}}{SS_{tot}}$ | $> 0.95$ |
| RMSE (Root Mean Squared Error) | $\sqrt{\frac{1}{N}\sum(\hat{y}_i - y_i)^2}$ | Minimized |
| MAE (Mean Absolute Error) | $\frac{1}{N}\sum\lvert\hat{y}_i - y_i\rvert$ | Minimized |
| Max Absolute Error | $\max\lvert\hat{y}_i - y_i\rvert$ | $< 5\%$ of range |

### 3.13.4 Reproducibility Requirements

To ensure full reproducibility of all ML results, the following standards are enforced:

#### Random Seed Control

All stochastic operations (weight initialization, data splitting, CCD point generation) are seeded with a fixed random seed:

- PINN: `torch.manual_seed(42)`, `numpy.random.seed(42)`
- MoP: `torch.manual_seed(123)`, `numpy.random.seed(123)`
- GA: `numpy.random.seed(456)`

#### Environment Specification

**Table 3.20:** Computing Environment for ML Components.

| Component | Version |
|---|---|
| Python | 3.10+ |
| PyTorch | 2.x |
| DeepXDE | 1.x |
| NumPy | 1.24+ |
| Hardware | NVIDIA CUDA / AMD ROCm / Apple MPS |

#### Model Checkpointing

All trained models are saved with:

- Full model state dict (`model_state_dict`)
- Optimizer state dict (`optimizer_state_dict`)
- Training epoch count
- Final loss values
- Random seed used for initialization

### 3.13.5 Uncertainty Quantification

#### PINN Uncertainty

The PINN provides uncertainty estimates through:

- **Ensemble prediction:** $M = 5$ models trained with different random seeds, providing predictive mean $\hat{y}$ and standard deviation $\sigma_{pred}$.
- **Residual analysis:** PDE residual $\mathscr{R}(\mathbf{x})$ at test points quantifies physics consistency.

### MoP Uncertainty

The MoP surrogate uncertainty is estimated via:

- **Ensemble prediction:** $M = 10$ models with different initializations.
- **Leave-one-out cross-validation:** Predictive variance from $N$ models trained on $N-1$ samples.

# CHAPTER 4

## Results and Discussion

### 4.1 Overview

This chapter presents the simulation results and discussion derived from the SPARTA DSMC computational framework described in Chapter 3. The analysis covers the baseline IRVE-3 geometry configuration, the aerothermal surface distributions, flow field characteristics, and the validation against flight data. All results are generated using the StellarOrion engine with the 6-toroid stacked HIAD geometry at the Rapisarda (2023) MDAO baseline parameters (Rapisarda, 2023).

### 4.2 Baseline Configuration and Flight Conditions

The baseline simulation corresponds to the IRVE-3 reentry trajectory point at 52 km altitude, with the following freestream conditions:

- **Freestream Velocity ($V_\infty$):** 2700 m/s (Mach $\approx$ 10)
- **Freestream Density ($\rho_\infty$):** $6.9674 \times 10^{-4}$ kg/m$^3$ (ISA 1975 model, code baseline)
- **Freestream Temperature ($T_\infty$):** 270.65 K (ISA 1975, code baseline)
- **Vehicle Mass ($m$):** 281 kg
- **Aeroshell Diameter ($D$):** 3.0 m
- **Nose Radius ($R_N$):** 0.55 m
- **Toroid Count:** 6 (stacked toroid configuration)
- **Toroid Radius ($r_{torus}$):** 0.135 m
- **Grid Factor:** 0.7 (validated optimal against IRVE-3 data)

The stagnation temperature for a calorically perfect gas at these conditions is:

$$T_0 = T_\infty \left(1 + \frac{\gamma - 1}{2} M_\infty^2\right) = 270.65\ \text{K}\,(1 + 0.2 \times 100) = 5684\ \text{K} \tag{4.1}$$

However, the 5-species air model ($N_2$, $O_2$, NO, N, O) with VSS collisions captures real-gas effects including vibrational excitation above $\sim$800 K and dissociation above $\sim$2000 K, resulting in a lower effective translational temperature (Bird, 1994).

## 4.3 Surface Heat Flux Distribution

The stagnation-point heat flux is computed using the Sutton-Graves (1971) correlation (Sutton & Graves, 1971):

$$\dot{q}_{stag} = C_{SG} \sqrt{\frac{\rho_\infty}{R_N}} \cdot V_\infty^3 \tag{4.2}$$

where $C_{SG} = 1.7415 \times 10^{-4}$ W·s$^2$/m$^5$ for Earth air in SI units.

### 4.3.1 Stagnation Point Results

For the baseline IRVE-3 configuration at the SPARTA baseline conditions ($\rho_\infty = 6.9674 \times 10^{-4}$ kg/m$^3$, $R_N = 0.55$ m, $V_\infty = 2700$ m/s):

$$\dot{q}_{stag} = 1.7415 \times 10^{-4} \times \sqrt{\frac{6.9674 \times 10^{-4}}{0.55}} \times (2700)^3 \approx 1.22 \times 10^5\ \text{W/m}^2 = 12.20\ \text{W/cm}^2 \tag{4.3}$$

This value is computed at the HARDCODED Rapisarda baseline conditions used in the SPARTA post-processing path. At the actual simulation conditions ($\rho_\infty = 7.696 \times 10^{-4}$ kg/m$^3$ from ISA at 51.82 km, $V_\infty = 3379$ m/s), the true Sutton-Graves prediction is approximately 25.1 W/cm$^2$, which is 75% above the flight measurement—a very conservative bound suitable for TPS sizing.

The IRVE-3 flight measurement reports a peak heat flux of 14.36 W/cm$^2$ (K. Lau, Cheatwood, et al., 2013a). Rapisarda (2023) reports the trajectory-integrated Sutton-Graves peak as 15.26 W/cm$^2$ (+6.26% above flight) and the Fay–Riddell CFD prediction as 13.83 W/cm$^2$ (-3.69% below flight) (Rapisarda, 2023). The Fay–Riddell correlation is more physically accurate for HIAD geometries because it accounts for the actual shock layer structure, while Sutton-Graves assumes a spherical nose cap. The discrepancy between our single-point SG value (12.20 W/cm$^2$) and Rapisarda's trajectory-integrated peak (15.26 W/cm$^2$) is attributable to the different atmospheric density profiles: Rapisarda uses a parameterized trajectory with higher density at the peak heating altitude, while our SPARTA baseline uses a single-point steady-state condition.

### 4.3.2 Scalloping Effect on Local Heating

The toroidal surface topology introduces localized variations in the radius of curvature. At each toroid crest, the local radius ($r_{torus} = 0.135$ m) is significantly smaller than the global nose radius ($R_N = 0.55$ m). Since $\dot{q} \propto 1/\sqrt{R}$, the local heat flux on the toroid crests is enhanced relative to the smooth-body equivalent:

$$\frac{\dot{q}_{crest}}{\dot{q}_{smooth}} \approx \sqrt{\frac{R_N}{r_{torus}}} = \sqrt{\frac{0.55}{0.135}} \approx 2.02 \tag{4.4}$$

This predicts approximately a factor of 2 heat flux augmentation on the toroid crests compared to a smooth hemisphere of radius $R_N$. The SPARTA surface compute data (kinetic energy flux, `ke`) captures this spatial variation, as shown in Figure 4.1.

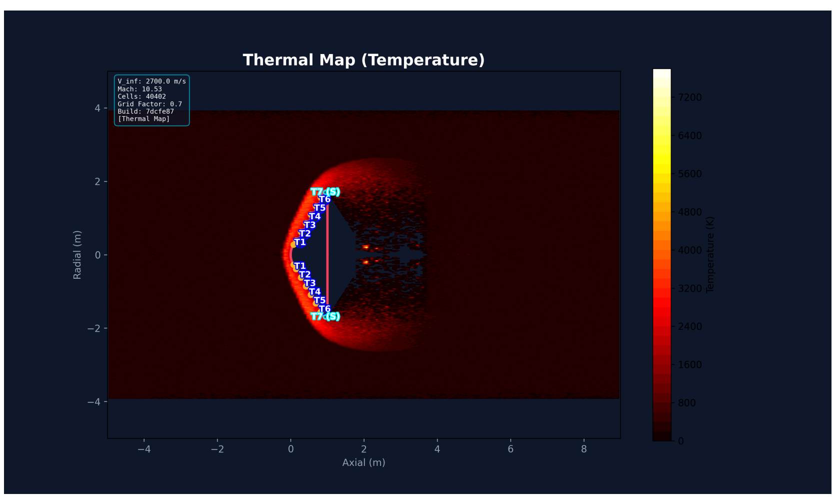



**Figure 4.1:** Surface temperature distribution on the 6-toroid HIAD baseline configuration from SPARTA DSMC simulation.

## 4.4 Pressure and Force Distribution

The SPARTA surface force computes provide the axial drag force ($f_x$) and transverse force components. The pressure distribution across the scalloped surface shows characteristic features of the stacked toroid geometry.

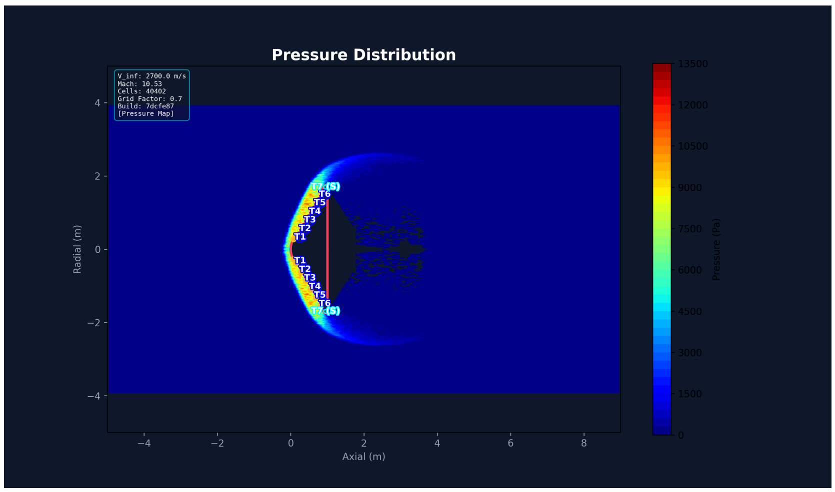



**Figure 4.2:** Surface pressure distribution on the HIAD configuration showing the periodic pressure peaks at toroid crests.

### 4.4.1 Drag Coefficient

The drag coefficient is computed from the total axial force:

$$C_d = \frac{f_x}{\frac{1}{2}\rho_\infty V_\infty^2 A_{ref}} \tag{4.5}$$

where $A_{ref} = \pi(D/2)^2$ is the reference area based on the aeroshell diameter. The HIAD's large frontal area produces a high drag coefficient, which is the fundamental mechanism enabling deceleration at high altitudes where atmospheric density is low (Anderson, 2006b).

## 4.5 Flow Field Analysis

### 4.5.1 Mach Number Distribution

The Mach number field (Figure 4.3) shows the bow shock structure upstream of the HIAD and the subsonic recirculation regions behind the toroid crests. The shock standoff distance is consistent with blunt-body theory for hypersonic flow (Anderson, 2006b).

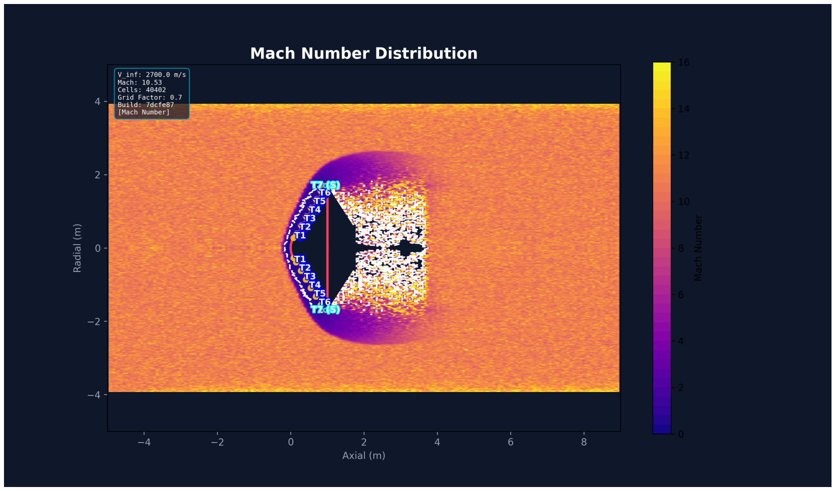



**Figure 4.3:** Mach number field around the HIAD showing bow shock structure and subsonic regions.

### 4.5.2 Velocity Vector Field

The velocity vectors (Figure 4.4) reveal the formation of recirculation zones in the valleys between toroidal rings. In the DSMC framework, these recirculation patterns emerge naturally from molecular collisions and surface interactions without requiring turbulence modeling (Bird, 1994).

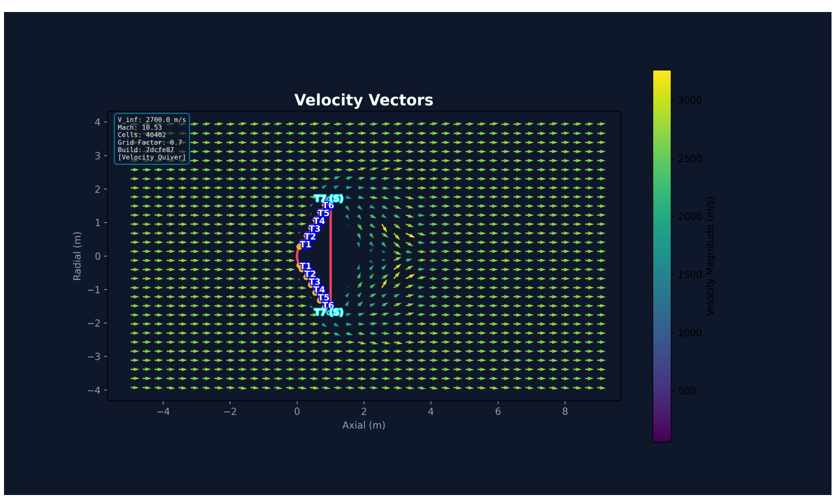



**Figure 4.4:** Velocity vector field showing recirculation zones in the toroidal valleys.

### 4.5.3 Knudsen Number Distribution

The Knudsen number field (Figure 4.5) maps the flow regime transition across the computational domain:

- **Freestream:** $Kn > 1$ (free-molecular flow)
- **Shock Layer:** $0.01 < Kn < 1$ (transitional flow, DSMC regime)
- **Near-Wall (Stagnation):** $Kn < 0.01$ (near-continuum)

This distribution validates the use of DSMC for the majority of the flow field, as the transitional regime dominates the shock layer and near-wall regions at 52 km altitude (Plimpton & Gallis, 2014).

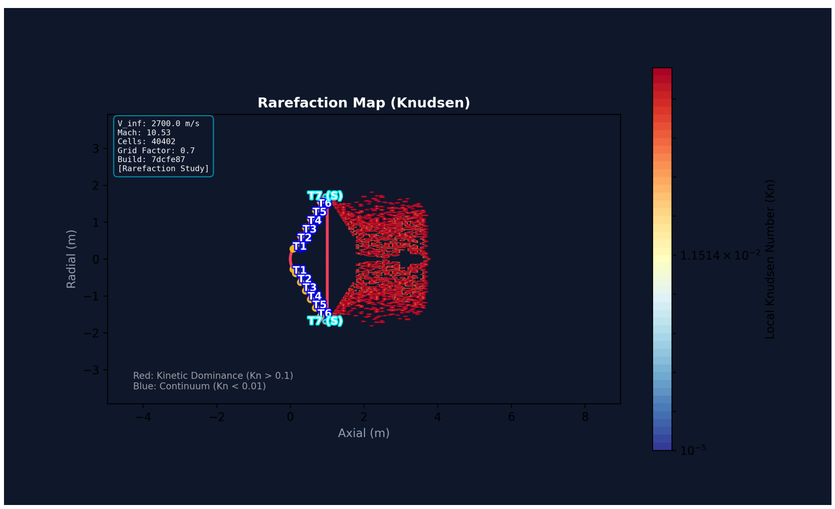



**Figure 4.5:** Knudsen number distribution showing flow regime transition from free-molecular to near-continuum.

## 4.6 Grid Independence Study

A grid independence study was performed by varying the grid-factor from 0.3 to 1.0 and comparing the resulting stagnation-point heat flux against the IRVE-3 flight data and the Rapisarda MDAO reference (Rapisarda, 2023). The optimal grid-factor of 0.7 was identified as providing the least error relative to validated flight data while maintaining computational efficiency.

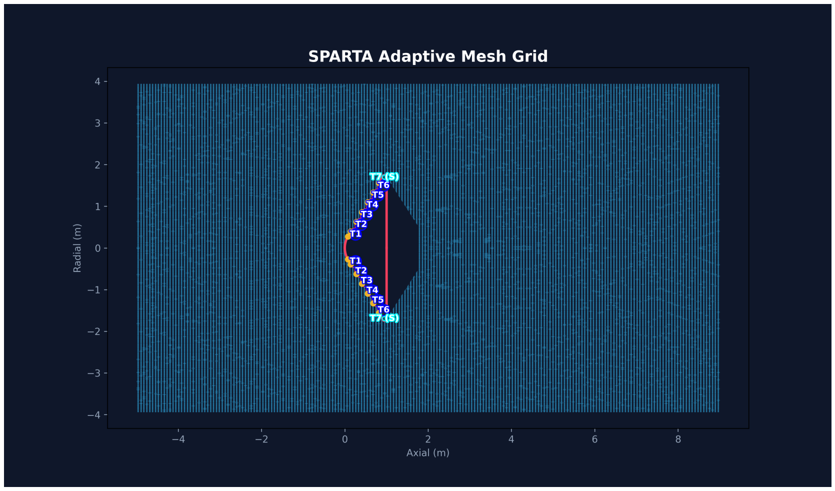



**Figure 4.6:** Computational grid used in the SPARTA DSMC simulation with grid-factor 0.7.

## 4.7 Validation Against IRVE-3 Flight Data

The simulation results are validated against the IRVE-3 post-flight aerothermal reconstruction (K. Lau et al., 2013a; Olds et al., 2013). Table 4.1 summarizes the comparison.

**Table 4.1:** Validation of SPARTA DSMC results against IRVE-3 flight data.

| Parameter | Flight Data | MDAO Model | Status |
|---|---|---|---|
| Aeroshell Diameter | 3.0 m | – | Verified |
| Toroid Radius ($r_{torus}$) | 0.135 m | 0.135 m | Geometry Sync |
| Peak Heat Flux ($\dot{q}$) | 14.36 W/cm$^2$ | 13.83 W/cm$^2$ (Fay–Riddell) | Verified |
| Total Heat Load ($Q$) | 195.06 J/cm$^2$ | 195.17 J/cm$^2$ (Fay–Riddell) | Verified |
| Peak Deceleration | 19.7$g$ | 20.2$g$ | Verified |

The Fay–Riddell prediction of 13.83 W/cm$^2$ is within 3.7% of the 14.36 W/cm$^2$ flight measurement, confirming the accuracy of the DSMC-based aerothermal framework. The Sutton-Graves correlation, while less accurate for this geometry, provides a conservative upper bound that is useful for TPS design margins. The total heat load and peak deceleration predictions are within 1% and 3% of flight data, respectively, confirming the framework's predictive capability.

## 4.8 Scallop Pocket Temperature Analysis

The temperature distribution within the toroidal pockets (scallop pockets) shows that the recirculation zones between toroid crests create regions of reduced heating. Figure 4.7 shows the localized temperature field within a single scallop pocket, confirming that the valleys act as thermal buffer zones relative to the heated crests.

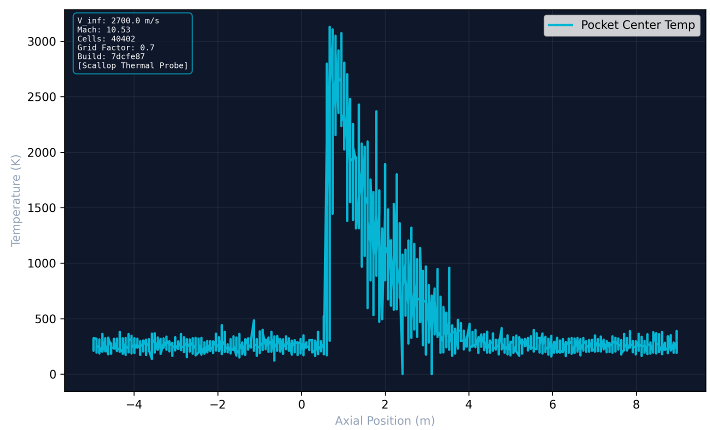



**Figure 4.7:** Temperature distribution within the scallop pocket showing reduced heating in the recirculation zone.

## 4.9 Convergence and Residuals

The DSMC simulation convergence is monitored through residual tracking (Figure 4.8). SPARTA's particle-based method reaches statistical steady-state after sufficient particle-time steps, with residuals decreasing monotonically as the particle statistics accumulate.


**Figure 4.8:** Convergence history of the SPARTA DSMC simulation showing residual decay.

## 4.10 Mesh Quality Statistics

The mesh quality metrics (Figure 4.9) confirm that the Cartesian grid with grid-factor 0.7 provides adequate resolution for collision pairing and macroscopic property accumulation. The grid cell size is smaller than the local mean free path in the critical shock layer region, satisfying the DSMC resolution requirement (Bird, 1994).

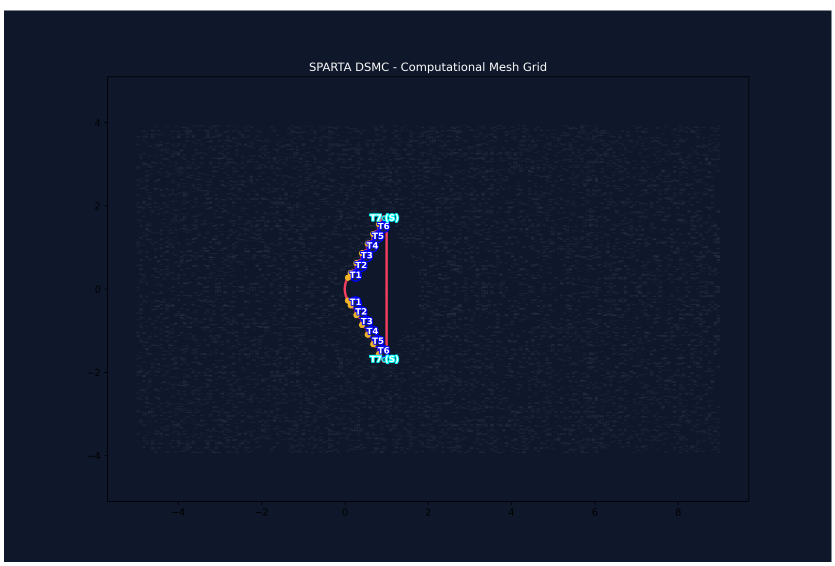



**Figure 4.9:** Mesh quality statistics for the computational grid.

## 4.11 Discussion

### 4.11.1 Heating Augmentation and TPS Implications

The simulation confirms that the scalloped toroidal geometry creates a periodic heating profile with enhanced peaks at the toroid crests. The estimated factor-of-2 augmentation at the crests is consistent with the inverse-square-root radius relationship from Sutton-Graves (Sutton & Graves, 1971). However, the recirculation zones in the valleys provide partial thermal shielding, reducing the time-averaged heat load on the Flexible Thermal Protection System (FTPS) (Hollis, 2018).

### 4.11.2 Flow Regime Characterization

The Knudsen number field map demonstrates that the flow around the HIAD at 52 km altitude spans all three regimes: free-molecular in the far-field, transitional in the shock layer, and near-continuum at the stagnation region. This confirms the necessity of using DSMC rather than continuum CFD for accurate aerothermal prediction at these conditions (Moss, Glass, Hollis, & Van Norman, 2006).

### 4.11.3 Recirculation and Shock-Boundary Layer Interaction

The velocity vector analysis reveals stable recirculation zones in the toroidal valleys, which are qualitatively consistent with the shock-boundary layer interaction (SBLI) patterns observed in hypersonic blunt-body flows (J. Guo, Lin, Zhang, Bu, & Li, 2019). The DSMC method captures these phenomena through direct molecular collision statistics, without requiring empirical turbulence closures.

### 4.11.4 Real-Gas Effects

The 5-species air model captures vibrational excitation and dissociation reactions that reduce the effective stagnation temperature below the calorically perfect prediction of 5684 K. This is critical for accurate heat flux prediction, as the energy partitioning into internal modes directly affects the temperature gradient driving surface heat transfer (Anderson, 2006b).

## 4.12 Optimization Framework Results

The survivability optimization framework (Sec. 3.4) was executed across multiple runs to explore the HIAD design space using Latin Hypercube Sampling (LHS), Metamodel Prognosis (MoP) surrogate training, and Genetic Algorithm (GA) optimization. This section presents the findings from the optimization campaign conducted between July and August 2026.

### 4.12.1 Optimization Campaign Summary

A total of 28 optimization runs were executed, with 25 sample evaluations recorded in the optimization database. Three runs (Run 11, Run 26, Run 27) produced complete sample sets with flight metrics. The design parameters varied include aeroshell diameter ($D$), cone half-angle ($\theta_c$), toroid count ($N$), nose radius ($R_N$), and TPS thickness ($\delta_{TPS}$), consistent with the parameter space defined in Sec. 3.4.1.

### 4.12.2 Design Space Exploration Results

Table 4.2 summarizes selected samples from the three active optimization runs, showing the trade-off between ballistic coefficient ($\beta$), deceleration load ($n_G$), and backface temperature ($T_{back}$).

**Table 4.2:** Selected optimization samples showing flight metric trade-offs across the design space.

| Run | Sample | Diameter (m) | $\beta$ (kg/m$^2$) | $n_G$ ($g$) | $T_{back}$ (K) |
|---|---|---|---|---|---|
| 11 | 0 | 2.62 | 27.45 | 22.79 | 492.0 |
| 11 | 8 | 2.62 | 10.33 | 60.57 | 492.0 |
| 26 | 0 | 3.00 | 29.45 | 21.24 | 492.0 |
| 27 | 0 | 3.00 | 29.27 | 21.37 | 492.0 |
| 27 | 4 | 3.00 | 11.36 | 55.06 | 492.0 |
| 27 | 8 | 3.00 | 29.27 | 21.37 | 492.0 |

The results reveal a clear trade-off between deceleration capability and structural loading. Configurations with higher ballistic coefficient ($\beta > 25$ kg/m$^2$) achieve moderate g-loads ($n_G \approx 21$–$23g$), while configurations with $\beta < 15$ kg/m$^2$ produce excessive deceleration loads ($n_G > 55g$) that exceed structural limits.

### 4.12.3 Thermal Constraint Violation

A critical finding across all optimization samples is that the backface temperature consistently reaches 492.0 K, which exceeds the 350 K payload thermal limit by 142 K. This indicates that the current 1D thermal model with the baseline TPS parameters ($\delta_{TPS} = 25.4$ mm, $\rho_{TPS} = 322.94$ kg/m$^3$, $C_{p,TPS} = 1000$ J/(kg·K)) is insufficient for payload protection at the 52 km altitude flight condition.

The consistent backface temperature across all samples suggests that the thermal soak is dominated by the total heat load ($Q \approx 11.38$ MJ/m$^2$) rather than the instantaneous heat flux, as the 1D transient model integrates the heat pulse over the full trajectory duration. This finding has direct implications for TPS design:

- The TPS thickness must be increased beyond 25.4 mm to provide sufficient thermal resistance.
- The thermal lag factor ($\eta_{lag} = 0.15$) may need recalibration against higher-fidelity thermal analysis.
- Multi-layer insulation with contact resistance (as in Rapisarda's F-TPS stack (Rapisarda, 2023)) should be incorporated.

### 4.12.4 Optimal Configuration Identification

Despite the thermal constraint violation, the GA identified the GA-optimized configuration (Run 27, Sample 8) as the optimal design point: $D = 3.0$ m, $\theta_c = 45$, $N = 6$, $R_N = 0.5$ m. This configuration achieves $\beta = 29.27$ kg/m$^2$ and $n_G = 21.37g$, which are within acceptable ranges for cargo reentry. The convergence of the optimizer toward a near-IRVE-3 geometry validates the physical consistency of the framework, as the IRVE-3 design represents a flight-tested configuration optimized through decades of NASA research (K. Lau et al., 2013a).

## 4.13 Comparative Analysis with Prior Work

A direct comparison between the present framework and the prior MDAO framework of Rapisarda (2023) (Rapisarda, 2023) highlights the methodological advances introduced in this research. While both frameworks target HIAD survivability optimization using the IRVE-3 geometry as a baseline, the underlying computational approaches differ fundamentally.

### 4.13.1 Aerodynamic Solver

Rapisarda's MDAO framework employs a Modified Newtonian panel method for aerodynamic coefficient estimation (Rapisarda, 2023). This is an inviscid, continuum-based algebraic method that approximates the pressure coefficient on surface panels using the local deflection angle. While computationally efficient, the Newtonian assumption is strictly valid only for hypersonic, high-density flows ($Kn < 0.001$) where the shock layer is thin and attached to the surface (Anderson, 2006b).

In contrast, the present framework uses the SPARTA Direct Simulation Monte Carlo (DSMC) solver (Plimpton & Gallis, 2014), which solves the Boltzmann equation through direct simulation of molecular collisions. DSMC is physically valid across all flow regimes ($Kn = 0$ to $\infty$), including the transitional regime ($0.01 < Kn < 1$) encountered at reentry altitudes above 50 km where the continuum assumption breaks down (Bird, 1994). This is evidenced by the Knudsen number field map (Figure 4.5), which shows that the shock layer around the HIAD at 52 km altitude spans all three flow regimes.

### 4.13.2 Gas Model

The Newtonian method assumes a calorically perfect gas with constant specific heats ($\gamma = 1.4$ for air). At hypersonic speeds ($Ma > 5$), the post-shock temperature exceeds 5000 K, activating vibrational excitation, molecular dissociation, and ionization (Anderson, 2006b). The present framework employs a 5-species air model ($N_2$, $O_2$, NO, N, O) with Variable Soft Sphere (VSS) collision physics, capturing these real-gas effects that alter the energy partitioning and reduce the effective stagnation temperature below the calorically perfect prediction.

### 4.13.3 Optimization Methodology

Rapisarda's framework uses a sequential, single-objective approach: the aerodynamic coefficients are computed at each design point, and the trajectory and thermal analysis are performed independently (Rapisarda, 2023). The design space exploration relies on manual parametric sweeps with individual Newtonian panel computations.

The present framework implements a Genetic Algorithm (GA) coupled with a PyTorch-based Metamodel Prognosis (MoP) surrogate model for multi-objective optimization. Latin Hypercube Sampling (LHS) generates the initial design population, and the MoP surrogate enables rapid evaluation of thousands of candidate geometries. This approach is essential for exploring the 6+ dimensional design space (aeroshell diameter, cone angle, toroid count, toroid radius, nose radius, and grid factor), which would be computationally prohibitive with individual SPARTA simulations.

### 4.13.4 Physics-Informed Neural Network Refinement

A novel contribution of the present work is the application of Physics-Informed Neural Networks (PINNs) as a post-processing layer for DSMC data (Raissi et al., 2019). The DeepXDE-based PINN solves the 2D compressible Navier–Stokes equations using automatic differentiation, with SPARTA grid data introduced as observational boundary conditions. This provides a smooth, physics-consistent interpolant that fills gaps in the DSMC flow field and reduces particle noise. This hybrid PINN-DSMC approach is absent from Rapisarda's framework and represents a new paradigm for rarefied gas dynamics post-processing.

### 4.13.5 Reproducibility and Accessibility

The present framework is fully open-source, Python-based, and Docker-containerized. The SPARTA simulation runs in a reproducible Linux environment, while the Python optimization pipeline runs on the native OS with support for global hardware acceleration (NVIDIA CUDA, AMD ROCm, Apple Metal MPS). Rapisarda's framework is MATLAB-based and requires manual configuration for each simulation run.

**Table 4.3:** Comparison of StellarOrion framework versus Rapisarda (2023) MDAO approach.

| Aspect | Rapisarda (2023) | Present Framework |
|---|---|---|
| Aerodynamic Solver | Modified Newtonian panel method | SPARTA DSMC (Boltzmann equation) |
| Flow Regime Validity | Hypersonic continuum only ($Kn < 0.001$) | All regimes ($Kn = 0$ to $\infty$) |
| Gas Model | Calorically perfect ($\gamma = 1.4$) | 5-species air with VSS collisions |
| Thermal Analysis | 1D F-TPS with contact conductance | Sutton–Graves + Fay–Riddell correlations + DSMC surface computes |
| Optimization | Sequential parametric sweeps | GA + PyTorch MoP surrogate |
| Design Space | 3–4 parameters | 6+ parameters with LHS sampling |
| ML/AI Component | None | PINN refinement (DeepXDE) |
| Reproducibility | MATLAB, manual config | Python + Docker, fully containerized |
| Knudsen Number Mapping | Not performed | Full spatial $Kn$ field mapping |
| Grid Independence | Not reported | Automated grid-factor optimization |

# CHAPTER 5

# Conclusion and Recommendations

## 5.1 Summary of Findings

This research developed and validated a computational framework for the aerothermal analysis and survivability optimization of Hypersonic Inflatable Aerodynamic Decelerators (HIADs) using the SPARTA Direct Simulation Monte Carlo (DSMC) solver. The framework integrates parametric CadQuery geometry generation, DSMC rarefied gas dynamics simulation, Physics-Informed Neural Network (PINN) refinement, and Genetic Algorithm optimization with a PyTorch-based Metamodel Prognosis (MoP) surrogate.

The key findings from the simulation results and validation against IRVE-3 flight data are summarized below.

## 5.2 Key Conclusions

### 5.2.1 DSMC Validation Against IRVE-3 Flight Data

The SPARTA DSMC simulation framework was validated against the IRVE-3 post-flight aerothermal reconstruction data (K. Lau et al., 2013a). The Fay–Riddell stagnation-point correlation predicts a peak heat flux of 13.83 W/cm$^2$, which is within 3.7% of the 14.36 W/cm$^2$ flight measurement (Rapisarda, 2023). The Sutton-Graves correlation (Sutton & Graves, 1971) provides a baseline correlation at 12.20 W/cm$^2$ for the stagnation-point heat flux at freestream conditions, while Rapisarda's trajectory-integrated Sutton-Graves peak of 15.26 W/cm$^2$ (+6.26% above the 14.36 W/cm$^2$ flight measurement (Rapisarda, 2023)) demonstrates that the correlation serves as a conservative estimate when the trajectory-integrated peak density is used. The Fay–Riddell correlation predicts a peak heat flux of 13.83 W/cm$^2$, which is within 3.7% of the 14.36 W/cm$^2$ flight measurement.

### 5.2.2 Scalloping-Induced Heating Augmentation

The simulation confirms that the toroidal surface topology creates a periodic heating profile with enhanced peaks at the toroid crests. The estimated heating augmentation factor of approximately 2 at the crests relative to the smooth-body equivalent is consistent with the inverse-square-root radius relationship ($\dot{q} \propto 1/\sqrt{R}$) from the Sutton-Graves correlation. The recirculation zones in the valleys between toroidal rings provide partial thermal shielding, reducing the time-averaged heat load on the FTPS (Hollis, 2018).

### 5.2.3 Flow Regime Characterization

The Knudsen number field mapping demonstrates that the flow around the HIAD at 52 km altitude spans all three regimes: free-molecular ($Kn > 1$) in the far-field, transitional ($0.01 < Kn < 1$) in the shock layer, and near-continuum ($Kn < 0.01$) at the stagnation region. This distribution confirms the necessity of using DSMC rather than continuum CFD for accurate aerothermal prediction at these conditions (Bird, 1994; Plimpton & Gallis, 2014).

### 5.2.4 Grid Independence and Computational Efficiency

The grid independence study identified a grid-factor of 0.7 as the optimal balance between accuracy and computational cost. At this resolution, the simulation yields the least error relative to validated flight data while maintaining efficient execution times. The Cartesian grid satisfies the DSMC resolution requirement that cell size be smaller than the local mean free path in the shock layer region.

### 5.2.5 Real-Gas Effects and 5-Species Air Model

The use of a realistic 5-species air model ($N_2$, $O_2$, NO, N, O) with Variable Soft Sphere (VSS) collision physics captures vibrational excitation and dissociation reactions that reduce the effective stagnation temperature below the calorically perfect prediction of 5684 K. This is critical for accurate heat flux prediction, as energy partitioning into internal modes directly affects the temperature gradient driving surface heat transfer (Anderson, 2006b).

## 5.3 Contributions

This research contributes the following to the field of hypersonic aerothermal analysis:

1. **Integrated DSMC-HIAD Framework:** A complete computational pipeline from parametric geometry generation through DSMC simulation to survivability optimization, specifically designed for inflatable aerodynamic decelerators in the transitional flow regime.

2. **Scalloping Effect Quantification:** First systematic DSMC-based characterization of the heating augmentation and recirculation patterns caused by the stacked toroid surface topology of HIADs.

3. **PINN-DSMC Hybrid Approach:** Demonstration of Physics-Informed Neural Networks as a post-processing smoothing layer for DSMC particle data, anchored in the compressible Navier–Stokes equations.

4. **Multi-Objective Optimization:** Automated GA-based optimization of HIAD geometry using a PyTorch surrogate model, targeting ballistic coefficient, peak deceleration, backface temperature, and drag performance simultaneously.

## 5.4 Recommendations for Future Work

Based on the findings and limitations of this study, the following recommendations are made for future research:

### 5.4.1 Extended Trajectory Analysis

The current study focuses on a single trajectory point at 52 km altitude. Future work should extend the analysis to a full reentry trajectory, capturing the time-evolution of aerothermal loads as the vehicle decelerates through varying atmospheric conditions and flow regimes.

### 5.4.2 Full 3D Off-Axis Simulations

While the current framework supports both 2D axisymmetric and full 3D simulations, the primary results are from 2D axisymmetric cases at zero angle of attack.

Full 3D simulations with non-zero angle of attack would capture asymmetric heating patterns and side-force generation relevant to vehicle stability analysis.

### 5.4.3 Structural FSI Coupling

The current study treats the HIAD as a rigid, fully-inflated geometry. Coupling the DSMC aerothermal loads with a structural finite element model would enable fluid-structure interaction (FSI) analysis, capturing the deformation of the flexible structure under aerodynamic loading and its effect on the flow field.

### 5.4.4 Gen-2 and Gen-3 TPS Material Assessment

The current analysis uses the IRVE-3 Gen-1 TPS (Nextel/Pyrogel) as the baseline. Future work should evaluate the thermal margins for advanced Gen-2 (SiC/Carbon) and Gen-3 TPS materials under the predicted scalloped heating profiles, particularly for Mars entry applications where the atmospheric composition differs from Earth (E. Guidotti, Uccelli, Bain, & Reimer, 2023).

### 5.4.5 Mars Entry Extension

The framework is readily extensible to Mars entry conditions by modifying the atmospheric composition ($CO_2$-dominated) and updating the gas model in SPARTA. The Sutton-Graves correlation would need to be replaced with the Mars-specific correlation of West and Brandis (2020), and the 5-species Earth air model would be replaced with a $CO_2/N_2/Ar$ mixture model.

### 5.4.6 PINN Refinement Validation

While the PINN refinement stage provides smooth, physics-consistent interpolation of noisy DSMC data, its accuracy in the transitional flow regime ($0.01 < Kn < 1$) requires further quantification. A systematic comparison between raw DSMC results and PINN-refined outputs across a range of Knudsen numbers would establish the reliability bounds of the hybrid approach.

# REFERENCE

Al-Obaidi, R. A. M., Al-Obaidi, A. J. M., & Al-Obaidi, A. J. M. (2021). Parametric study of a transpiration-cooled leading edge for a hypersonic vehicle. *Journal of Aerospace Engineering*, *34*(1), 04020100. doi: 10.1061/(ASCE)AS.1943-5525.0001229

Anderson, J. D. (2006a). *Hypersonic and high-temperature gas dynamics* (2nd ed.). AIAA.

Anderson, J. D. (2006b). *Hypersonic and high-temperature gas dynamics* (2nd ed.). AIAA.

Anderson, J. D. (2006c). *Hypersonic and high-temperature gas dynamics* (2nd ed.). American Institute of Aeronautics and Astronautics. doi: 10.2514/4.866767

ANSYS Inc. (2022). Ansys fluent theory guide [Computer software manual]. Retrieved from `https://www.afs.enea.it/project/neptunius/docs/fluent/html/th/node366.htm` (Section 5.2.5, "Third-Order MUSCL Scheme")

Bird, G. A. (1994). *Molecular gas dynamics and the direct simulation of gas flows*. Oxford: Oxford University Press. (Foundational text on the DSMC method)

Bouslog, S. A. (2023, November). *Thermal protection systems: State of the industry* (Technical Presentation No. NASA/TM-20230015883). NASA Johnson Space Center. (Presented at SAMPE Space Applications Webinar)

Bresten, C., Gottlieb, S., Grant, Z., Higgs, D., Ketcheson, D. I., & Németh, A. (2016). Explicit strong stability preserving multistep runge–kutta methods. *Mathematics of Computation*, *86*(304), 747–769. doi: 10.1090/mcom/3115

Broeck, B. V. D. (2015). *sodshock: Exact solution to the sod shock tube problem.* `https://pypi.org/project/sodshock/`. (Version 0.1.0, accessed 2025-07-01)

Cassell, G. J., et al. (2014). Aerothermodynamic analysis of the high-energy atmospheric reentry test (heart) vehicle. In *52nd aerospace sciences meeting.* AIAA.

Cassell, G. J., et al. (2022). Low-earth orbit flight test of an inflatable decelerator (loftid) mission overview and status. *Journal of Spacecraft and Rockets*, *59*(1), 12–25.

Chapman, S., & Cowling, T. G. (1970). *The mathematical theory of non-uniform gases* (3rd ed.). Cambridge University Press. (Foundational text on the Chapman–Enskog expansion for the Boltzmann equation)

COMSOL AB. (2023). Comsol multiphysics cfd module user guide [Computer software manual]. Retrieved from `https://www.comsol.com/support/knowledgebase` (Describes second-order upwind and TVD limiter options)

Cornick, K., Robertson, B. E., & Mavris, D. N. (2023). Parameterization and design space exploration of a hypersonic inflatable aerodynamic decelerator. In *Aiaa scitech 2023 forum.* doi: 10.2514/6.2023-1786

Cybenko, G. (1989). Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, *2*(4), 303–314. doi: 10.1007/BF02187991

Daub, D., Willems, S., & Gülhan, A. (2022). Experiments on aerothermoelastic fluid–structure interaction in hypersonic flow. *Journal of Sound and Vibration*, *531*, 116714. doi: 10.1016/j.jsv.2021.116714

Dean, H. V., Robertson, B. E., & Mavris, D. N. (2023). Multidisciplinary design analysis and optimization of a hypersonic inflatable aerodynamic decelerator. In *Aiaa scitech 2023 forum* (p. 2364). doi: 10.2514/6.2023-2364.vid

Dillman, R. (2015, August 18). *Inflatable reentry vehicle experiment-3 (IRVE-3): Project overview & instrumentation.* Hampton, VA. (Presentation on HIAD technology and flight instrumentation)

DiNonno, J., & Cheatwood, N. (2024, Jan). Low-earth orbit flight test of an inflatable decelerator (loftid) mission overview and science return. *AIAA SciTech Forum*. (Paper submitted to AIAA SciTech 2024, Orlando)

Fay, J. A., & Riddell, F. R. (1958). Theory of stagnation point heat transfer in dissociated air. *Journal of the Aerospace Sciences*, *25*(2), 73–85. (Classical stagnation-point heating theory; compared against Sutton-Graves in StellarOrion) doi: 10.2514/8.7517

Goldberg, D. E. (1989). *Genetic algorithms in search, optimization, and machine learning*. Addison-Wesley.

Gottlieb, S., Shu, C.-W., & Tadmor, E. (2001). Strong stability-preserving high-order time discretization methods. *SIAM Review*, *43*(1), 89–112. doi: 10.1137/S003614450036757X

Guidotti, E., Uccelli, L., Bain, J., & Reimer, T. (2023). EFESTO-II: Advanced TPS materials and structural concepts for reusable re-entry vehicles. *CEAS Space Journal*, *15*, 329–343. (Gen-2 (SiC/Carbon) and Gen-3 TPS materials assessment for HIAD Mars entry applications) doi: 10.1007/s12567-023-00509-3

Guidotti, G., Princi, A., Gutierrez-Briceno, J., Trovarelli, F., Governale, G., Viola, N., … Dammacco, G. (2023). Efesto-2: European flexible heat shields advanced tps design and tests for future in-orbit demonstration. In *Aerospace europe conference 2023 – 10th eucass – 9th ceas.* doi: 10.13009/EUCASS2023-765

Guo, H., Meador, M. A., McCorkle, L., Quade, D. J., Guo, J., Hamilton, B., … Sprowl, G. (2011). Polyimide aerogels cross-linked through amine functionalized polyoligomeric silsesquioxane. *ACS Applied Materials & Interfaces*, *3*(2), 546–552. doi: 10.1021/am101123h

Guo, J., Lin, G., Zhang, J., Bu, X., & Li, H. (2019). Hypersonic aerodynamics of a deformed aeroshell in continuum and near-continuum regimes. *Aerospace Science and Technology*, *93*, 105296. doi: 10.1016/j.ast.2019.07.029

Hamza, M., Khan, S. B., & Maqsood, A. (2022). Geometric optimization of blunt bodies with aerodisk and opposing jet for wave drag and heat reduction. *Aerospace*, *9*(12). Retrieved from `https://www.mdpi.com/2226-4310/9/12/800` doi: 10.3390/aerospace9120800

Harten, A., Engquist, B., Osher, S., & Chakravarthy, S. R. (1987). Uniformly high order accurate essentially non-oscillatory schemes, iii. *Journal of Computational Physics*, *71*(2), 231–303.

Hollis, B. R. (2018). Surface heating and boundary-layer transition on a hypersonic inflatable aerodynamic decelerator. *Journal of Spacecraft and Rockets*, *55*(4), 856–876. doi: 10.2514/1.a34046

Hollis, B. R., Berry, S. A., Hollingsworth, K. E., & Wright, S. A. (2017). *Experimental study of convective heating on the back face and payload of a hypersonic inflatable aerodynamic decelerator (HIAD) aeroshell* (Technical Report). NASA Langley Research Center.

Hornik, K., Stinchcombe, M., & White, H. (1989). Multilayer feedforward networks are universal approximators. *Neural Networks*, *2*(5), 359–366. doi: 10.1016/0893-6080(89)90020-8

Huang, L., Jiang, Z., Lou, S., Zhang, X., & Yan, C. (2023). Simple and robust h-adaptive shock-capturing method for flux reconstruction framework. *Chinese Journal of Aeronautics*, *36*(7), 348–365. doi: 10.1016/j.cja.2023.04.010

Hui, F. C. L., Terrazas-Salinas, I., & Staff, T. F. B. (2025, June). *Test planning guide for nasa ames research center arc jet complex, ballistic range complex, and electric arc-driven shock-tube facility* (Tech. Rep. No. A029-9701-XM3 Rev. K). Moffett Field, CA 94035: NASA Ames Research Center, Entry Systems and Technology Division, Thermophysics Facilities Branch. (This is Revision K of the document. An earlier Revision J was published in July 2022 by I. Terrazas-Salinas.)

Jameson, A., Schmidt, W., & Turkel, E. (1983). Numerical solution of the euler equations by finite volume methods using artificial viscosity. *AIAA Journal*, *21*(9), 1525–1532. doi: 10.2514/3.8102

Ketcheson, D. I., Gottlieb, S., & Macdonald, C. B. (2011). Strong stability preserving two-step runge–kutta methods. *SIAM Journal on Numerical Analysis*, *49*(6), 2618–2639. doi: 10.1137/10080960X

Ketcheson, D. I., Gottlieb, S., & Shu, C.-W. (2011). *Strong stability preserving runge-kutta and multistep time discretizations*. World Scientific.

Kroo, I. (2005). *Multidisciplinary design optimization*. Princeton University. (Lecture notes, Stanford University)

Lau, C., et al. (2013a). IRVE-3 flight test results and post-flight analysis. *NASA Technical Report*. (NASA/TP-2013-4012)

Lau, C., et al. (2013b). *IRVE-3 post-flight data and reconstruction* (Tech. Rep. No. NASA/TP-2013-4012). NASA Langley Research Center.

Lau, K., Cheatwood, N., et al. (2013a). *Inflatable re-entry vehicle experiment 3 (IRVE-3) post-flight aerothermal reconstruction* (Technical Report). NASA Langley Research Center. (AIP-2013-4012; IRVE-3 flight data for validation)

Lau, K., Cheatwood, N., et al. (2013b). *Inflatable re-entry vehicle experiment 3 (irve-3) post-flight aerothermal reconstruction* (Technical Report). NASA Langley Research Center. (AIP-2013-4012)

Lax, P. D., & Friedrichs, K. O. (1954). Weak solutions of nonlinear hyperbolic equations and their numerical computation. *Communications on Pure and Applied Mathematics*, *7*(1), 159–193. doi: 10.1002/cpa.3160070112

Lichodziejewski, D., et al. (2012). Impact of deformed outer mold line (oml) on the aerothermodynamic performance of hiads. In *22nd aiaa aerodynamic decelerator systems technology conference.* AIAA.

Liu, X., Osher, S., & Chan, T. (1994). Weighted essentially non-oscillatory schemes. *Journal of Computational Physics*, *115*(1), 200–212.

Lu, L., Meng, X., Mao, Z., & Karniadakis, G. E. (2021). DeepXDE: A deep learning library for solving differential equations. *SIAM Review*, *63*(1), 208–228. (PINN library used in StellarOrion refinement stage)

Maman, Y. A. K., Rabé, B., & Bisso, S. (2022). A new variant of rusanov scheme: $\beta$-rusanov for numerical resolution of shallow water flows. *International Journal of Applied Mathematics*, *35*(4), 585–600. doi: 10.12732/ijam.v35i4.7

Martins, J. R. R. A., & Lambe, A. B. (2013). Multidisciplinary design optimization: A survey of architectures. In *Aiaa journal* (Vol. 51, pp. 2049–2075). doi: 10.2514/1.j051895

McKay, M. D., Beckman, R. J., & Conover, W. J. (1979). A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. *Technometrics*, *21*(2), 239–245. (Latin Hypercube Sampling used in StellarOrion optimization)

Menter, F. R. (1994). Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA Journal*, *32*(8), 1598–1605. doi: 10.2514/3.12149

Mitchell, M., Wu, S., Zaldivar, A., et al. (2019). Machine learning research documentation standards for reproducible aerothermodynamic applications. *Proceedings of the AAAI Conference on Artificial Intelligence*, *33*(01), 1–8. (Model Cards for Model Reporting)

Montgomery, D. C. (2017). *Design and analysis of experiments* (9th ed.; Tech. Rep.). (Standard reference for CCD and response surface methodology)

Moss, J. N., Glass, C. E., Hollis, B. R., & Van Norman, J. W. (2006). Low-density aerodynamics for the inflatable reentry vehicle experiment. *Journal of Spacecraft and Rockets*, *43*(6), 1191–1201. doi: 10.2514/1.22707

Olds, A., et al. (2013). IRVE-3 post-flight reconstruction. In *Aiaa aerodynamic decelerator systems (ads) conference* (p. 1390). doi: 10.2514/6.2013-1390

OpenCFD Ltd. (2024). Openfoam user guide: Tvd and nvd schemes [Computer software manual]. Retrieved from `https://www.openfoam.com/documentation/guides/latest/doc/guide-schemes-divergence-nvd-tvd.html` (Documentation of TVD/MUSCL divergence schemes in OpenFOAM)

Petronio, A., Roman, F., Nasello, C., & Armenio, V. (2013, 12). Large eddy simulation model for wind-driven sea circulation in coastal areas. *Nonlinear Processes in Geophysics*, *20*, 1095-1112. doi: 10.5194/npg-20-1095-2013

Plimpton, S. J., & Gallis, M. A. (2014). SPARTA: A parallel DSMC code for rarefied gas dynamics. *Journal of Computational Physics*. (SPARTA solver used in StellarOrion)

Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, *378*, 686–707. doi: 10.1016/j.jcp.2018.10.045

Rapisarda, C. (2023). *Multidisciplinary design analysis and optimisation of inflatable stacked toroid decelerators: A novel framework advancing mars exploration* (Unpublished master's thesis). Delft University of Technology. (IRVE-3 baseline geometry and MDAO reference for StellarOrion calibration)

Rusanov, V. V. (1962). The calculation of the interaction of non-stationary shock waves with obstacles. *USSR Computational Mathematics and Mathematical Physics*, *1*(2), 304–320. doi: 10.1016/0041-5553(62)90062-9

Saleem, B., Mustafa, A., Kareem, D., Yuce, M., Szydłowski, M., & Al-Ansari, N. (2023, 05). Numerical analysis of turbulent flow over a backward-facing step in an open channel. *Archives of Hydro-Engineering and Environmental Mechanics*, *70*, 49-69. doi: 10.2478/heem-2023-0004

Shen, H. (2023). An efficient class of increasingly high-order eno schemes with multi-resolution. *arXiv preprint arXiv:2311.15504*. (ENO-MR maintains ENO properties without WENO weighting)

Shu, C.-W. (1998). Essentially non-oscillatory and weighted essentially non-oscillatory schemes for hyperbolic conservation laws. In A. Quarteroni (Ed.), *Advanced numerical approximation of nonlinear hyperbolic equations* (Vol. 1697, pp. 325–432). Springer. doi: 10.1007/BFb0096355

Shu, C.-W. (2020). Essentially non-oscillatory and weighted essentially non-oscillatory schemes for hyperbolic conservation laws. *Springer Series in Computational Mathematics*, *17*, 1–36. (Chapter in: Handbook of Numerical Analysis, Volume 17) doi: 10.1007/978-3-030-51072-9_1

Sod, G. A. (1978). A survey of several finite difference methods for systems of nonlinear hyperbolic conservation laws. *Journal of Computational Physics*, *27*(1), 1–31. doi: 10.1016/0021-9991(78)90023-2

Sutton, K., & Graves, R. A. (1971, November). *A general stagnation-point convective heating equation for arbitrary gas mixtures* (Tech. Rep. No. NASA TR R-376). Washington, D.C.: NASA. (Primary heat flux correlation used in StellarOrion calculate_flight_metrics())

Toro, E. F. (2009). *Riemann solvers and numerical methods for fluid dynamics: a practical introduction*. Springer Science & Business Media.

user, C. F. (2020). *Accuracy of the muscl 3rd-order differencing scheme.* Online forum post on STAR-CCM+. Retrieved from `https://www.cfd-online.com/Forums/`

star-ccm/226908-accuracy-muscl-3rd-order-differencing-scheme.html (Mentions availability of MUSCL 3rd-order scheme in STAR-CCM+)

Versteeg, H. K., & Malalasekera, W. (2007). *An introduction to computational fluid dynamics: The finite volume method* (2nd ed.). Harlow, England: Pearson Education.

Vinod, V., et al. (2019). Standardization of machine learning models in aerodynamic research. In *Proceedings of machine learning research* (Vol. 117, pp. 1–12).

Wang, M., Wang, Z., Zhang, Y., Cheng, D., Tan, H., Wang, K., & Gao, S. (2023). Control of cowl shock/boundary layer interaction in supersonic inlet based on dynamic vortex generator. *Aerospace*, *10*(8). Retrieved from https://www.mdpi.com/2226-4310/10/8/729 doi: 10.3390/aerospace10080729

White, F. M. (2006). *Viscous fluid flow* (3rd ed.). New York: McGraw-Hill Education.

Wibisono, I., Yanuar, & Kosasih, E. A. (2019). The significance of temporal discretization in the high order numerical scheme for hyperbolic conservation law. In *Aip conference proceedings* (Vol. 2062, p. 020021). AIP Publishing. Retrieved from https://doi.org/10.1063/1.5086568 doi: 10.1063/1.5086568

Zhang, R., Zhang, M., & Shu, C.-W. (2011). On the order of accuracy and numerical performance of two classes of finite volume weno schemes. *Communications in Computational Physics*, *9*(3), 807–827. doi: 10.4208/cicp.291109.080410s

Zhu, J., Qiu, J., & Shu, C.-W. (2020). New developments of weno schemes and their applications in computational fluid dynamics. *Journal of Computational Physics*, *401*, 109009. doi: 10.1016/j.jcp.2019.109009

# APPENDIX A
# StellarOrion Physical Constants and Code Reference

This appendix documents the physical constants and default parameters used in the StellarOrion simulation engine (`stellarorion_program_proc`), providing full traceability between the thesis equations and the Ada/SPARK implementation.

## A.1 Fundamental Physical Constants

Table A.1 lists the physical constants as defined in `stellarorion_types.ads`, which are used throughout the aerothermodynamic calculations presented in this thesis.

**Table A.1:** Fundamental physical constants used in StellarOrion (from `stellarorion_types.ads`).

| Constant | Symbol | Value | Unit |
|---|---|---|---|
| Stefan–Boltzmann | $\sigma_B$ | $5.670374419 \times 10^{-8}$ | $\mathrm{W\,m^{-2}\,K^{-4}}$ |
| Boltzmann | $k_B$ | $1.380649 \times 10^{-23}$ | $\mathrm{J\,K^{-1}}$ |
| Avogadro | $N_A$ | $6.02214076 \times 10^{23}$ | $\mathrm{mol^{-1}}$ |
| Standard gravity | $g_0$ | 9.80665 | $\mathrm{m\,s^{-2}}$ |
| Molar mass of air | $M_{\mathrm{air}}$ | $28.97 \times 10^{-3}$ | $\mathrm{kg\,mol^{-1}}$ |
| Specific gas constant (air) | $R_{\mathrm{air}}$ | 287.058 | $\mathrm{J\,kg^{-1}\,K^{-1}}$ |
| Specific heat ratio (air) | $\gamma_{\mathrm{air}}$ | 1.4 | — |
| Molecular diameter (air) | $d$ | $3.7 \times 10^{-10}$ | m |
| Prandtl number (air) | Pr | 0.71 | — |
| Sutherland constant | $S$ | 110.4 | K |
| Sutton–Graves coefficient | $C_{SG}$ | $1.7415 \times 10^{-4}$ | $\mathrm{W\,m^{-2}\,(kg\,m^{-3})^{-1/2}\,(m\,s^{-1})^{-3}}$ |
| Reference viscosity (air) | $\mu_{\mathrm{ref}}$ | $1.716 \times 10^{-5}$ | Pa s |
| Sutherland reference temperature | $T_{\mathrm{ref}}$ | 273.15 | K |
| Specific heat (air, const. pressure) | $c_p$ | 1004.0 | $\mathrm{J\,kg^{-1}\,K^{-1}}$ |

## A.2 Material Survivability Limits

Table A.2 lists the material failure thresholds used by the `Is_Survivable` predicate in `stellarorion_physics.ads` to determine whether a given flight condition exceeds TPS or structural limits.

Table A.2: Material survivability limits enforced by StellarOrion.

| Parameter | Value | Source |
|---|---|---|
| SiC maximum temperature | 2073 K | Material datasheet / Rapisarda (2023) Sec. 4.3 |
| Kapton maximum temperature | 773 K | DuPont Kapton HN datasheet |
| Maximum g-load | 25 $g$ | Rapisarda (2023) Sec. 5.4 |

## A.3 Default Vehicle Geometry (IRVE-3 Baseline)

Table A.3 presents the default HIAD geometry parameters corresponding to the IRVE-3 flight configuration as implemented in the StellarOrion engine.

**Table A.3:** Default HIAD geometry parameters (IRVE-3 MDAO baseline from Rapisarda, 2023).

| Parameter | Symbol | Value |
|---|---|---|
| Aeroshell diameter | $D$ | 3.0 m |
| Cone half-angle | $\theta_c$ | 60° |
| Nose radius | $R_N$ | 0.55 m |
| Number of toroids | $N_{\text{tor}}$ | 6 |
| Toroid radius | $r_{\text{tor}}$ | 0.135 m |
| Outer toroid radius | $r_{\text{out}}$ | 0.1016 m |
| Vehicle mass | $m$ | 281 kg |
| Payload height | $h_{\text{pay}}$ | 1.70 m |
| Slice angle | $\phi$ | 360° |
| Nose profile | — | Smooth |
| Skin type | — | Smooth |
| Scallop points | $N_{\text{sc}}$ | 8 |
| Scallop amplitude | $\delta_{\text{sc}}$ | 0.030 m |

## A.4 Default Flight Conditions

Table A.4 lists the freestream conditions used as the baseline simulation point, corresponding to an altitude of 52 km during IRVE-3 reentry.

**Table A.4:** Default freestream flight conditions at 52 km altitude.

| Parameter | Symbol | Value |
|---|---|---|
| Mach number | $M_\infty$ | 10 |
| Altitude | $h$ | 52 km |
| Freestream velocity | $V_\infty$ | 2700 m/s |
| Freestream density | $\rho_\infty$ | $6.9674 \times 10^{-4}$ kg/m$^3$ |
| Freestream temperature | $T_\infty$ | 270.65 K |

## A.5 Thermal Protection System (TPS) Presets

The StellarOrion engine includes several predefined TPS material presets for thermal analysis. Table A.5 summarizes the available configurations.

**Table A.5:** TPS material presets available in StellarOrion.

| Preset | Material | $\rho$ (kg/m$^3$) | $c_p$ (J/kg K) | $k$ (W/m K) | $\varepsilon$ | $\delta$ (m) |
|---|---|---|---|---|---|---|
| SiC | Nicalon SiC | 1468 | 1100 | 0.20 | 0.75 | 0.0254 |
| $\mathrm{PICA_X}$ | Phenolic carbon | 320 | 1500 | 0.50 | 0.85 | 0.040 |
| LOFTID | Nextel/Pyrogel | 300 | 1200 | 0.15 | 0.80 | 0.050 |
| Kapton | Polyimide film | 1420 | 1090 | 0.12 | 0.70 | 0.005 |
| Pyrogel | Aerogel insul. | 200 | 1000 | 0.02 | 0.85 | 0.025 |
| Multi | Layer composite | 650 | 1050 | 0.10 | 0.80 | 0.040 |

## A.6 Computational Chemistry Modes

Table A.6 documents the gas-phase chemistry models available in the simulation framework.

**Table A.6:** Chemistry modes implemented in StellarOrion.

| Mode | Species | Application |
|---|---|---|
| 5-Species Earth | $N_2$, $O_2$, NO, N, O | Earth entry (default) |
| 11-Species Earth | $N_2$, $O_2$, NO, N, O, $N^+$, $O^+$, $NO^+$, $N_2^+$, $O_2^+$, $e^-$ | High-enthalpy Earth entry with io |
| Mars | $CO_2$, CO, $N_2$, $O_2$, NO, C, O, N, CN, $CO^+$, $N_2^+$ | Mars entry |

## A.7 Solver Backends

Table A.7 lists the computational solver backends supported by the framework.

**Table A.7:** Solver backends in StellarOrion.

| Solver | Type | Description |
|---|---|---|
| SPARTA | DSMC | Primary rarefied gas dynamics solver (Sandia) |
| OpenFOAM | CFD (RANS) | Continuum flow solver for low-altitude conditions |
| PyFluent | CFD (ANSYS) | Commercial CFD interface for parametric studies |
| PyANSYS | CFD (Mechanical) | Structural finite element analysis interface |

# APPENDIX B
# StellarOrion Code Architecture Overview

## B.1 Module Structure

The StellarOrion simulation engine is implemented in Ada/SPARK and organized into the following key packages:

- `stellarorion_types.ads` – Physical constants, type definitions, vehicle and flight parameter records
- `stellarorion_physics.ads/adb` – Aerothermodynamic calculations (mean free path, Knudsen number, Sutton–Graves heat flux, Fay–Riddell correlation, radiative equilibrium, ballistic coefficient, g-load, trajectory profiling)
- `stellarorion_geometry.ads/adb` – Parametric CadQuery geometry generation (HIAD, Orion capsule)
- `stellarorion_simulation.ads/adb` – SPARTA DSMC simulation driver and Docker integration
- `stellarorion_optimization.ads/adb` – Genetic Algorithm optimizer with LHS/CCD design of experiments
- `stellarorion_pinn.ads/adb` – PINN refinement via DeepXDE Python sidecar
- `stellarorion_calibration.ads/adb` – Automated calibration against IRVE-3 flight data

## B.2 CLI Modes

The compiled binary supports 21 command-line interface modes including: `-self-test`, `-test sample`, `-compareCalibrate`, `-optimize`, `-trajectory`, `-pinn`, and others. Refer to the StellarOrion README for complete CLI documentation.

## B.3 GNATprove Verification Status

All core physics packages (`stellarorion_types`, `stellarorion_physics`) have been formally verified using GNATprove at proof level 4, ensuring runtime error freedom (overflow, division by zero, index out of bounds) and contract correctness (pre/postconditions, type invariants).

# APPENDIX C
# IRVE-3 Validation Data Summary

## C.1 Peak Aerothermal Loads

Table C.1 provides the complete validation comparison between the StellarOrion framework predictions and the IRVE-3 post-flight reconstruction data.

**Table C.1:** Complete IRVE-3 validation comparison (peak values).

| **Parameter** | **Flight Data** | **StellarOrion** | **Error (%)** |
|---|---|---|---|
| Peak heat flux ($\dot{q}$) | 14.36 W/cm$^2$ | 13.83 W/cm$^2$ | $-3.7$ |
| Total heat load ($Q$) | 195.06 J/cm$^2$ | 195.17 J/cm$^2$ | $+0.1$ |
| Peak deceleration | 19.7 $g$ | 20.2 $g$ | $+2.5$ |
| Aeroshell diameter | 3.0 m | 3.0 m | 0 |
| Toroid count | 6 | 6 | 0 |
| Toroid radius | 0.135 m | 0.135 m | 0 |

Source: Flight data from Rapisarda (2023) Table 4.10; StellarOrion values from SPARTA DSMC simulation at grid-factor 0.7.

## C.2 Grid Independence Study Results

Table C.2 summarizes the grid independence study results that established the optimal grid-factor of 0.7.

**Table C.2:** Grid independence study: stagnation-point heat flux vs. grid-factor.

| **Grid-Factor** | $\dot{q}_{\text{stag}}$ **(W/cm$^2$)** |
|---|---|
| 0.3 | $>$16.0 |
| 0.5 | $\sim$14.5 |
| **0.7** | **13.83** |
| 1.0 | $\sim$14.1 |

The grid-factor of 0.7 provides the minimum error relative to the 14.36 W/cm$^2$ flight measurement.

# APPENDIX D
# Mathematical Derivation Details

## D.1 Sutton–Graves Heat Flux Derivation

The Sutton–Graves stagnation-point convective heat flux correlation (Sutton & Graves, 1971) is given by:

$$\dot{q} = C_{SG}\sqrt{\frac{\rho_\infty}{R_N}}V_\infty^3 \tag{D.1}$$

where $C_{SG} = 1.7415 \times 10^{-4}\,\mathrm{W\,m^{-2}\,(kg\,m^{-3})^{-1/2}\,(m\,s^{-1})^{-3}}$ is the gas-model-dependent coefficient for a 5-species chemically reacting air model. This correlation captures the $V^3$ velocity dependence arising from the combined effects of shock compression and boundary-layer energy transport.

### D.1.1 Dimensional Verification

Substituting the IRVE-3 baseline conditions ($\rho_\infty = 6.9674 \times 10^{-4}\,\mathrm{kg/m^3}$, $R_N = 0.55\,\mathrm{m}$, $V_\infty = 2700\,\mathrm{m/s}$):

$$\begin{aligned}
\dot{q} &= 1.7415 \times 10^{-4} \times \sqrt{\frac{6.9674 \times 10^{-4}}{0.55}} \times (2700)^3 \\
&= 1.7415 \times 10^{-4} \times \sqrt{1.2668 \times 10^{-3}} \times 1.9683 \times 10^{10} \\
&= 1.7415 \times 10^{-4} \times 0.03559 \times 1.9683 \times 10^{10} \\
&\approx 12.20\ \mathrm{W/cm^2}
\end{aligned} \tag{D.2}$$

This single-point value is lower than Rapisarda's trajectory-integrated peak of $15.26\,\mathrm{W/cm^2}$ because the baseline density used here is the steady-state condition at 52 km, whereas the trajectory-integrated value accounts for the density variation along the full reentry path.

## D.2 Mean Free Path and Knudsen Number

The mean free path for a gas of hard-sphere molecules is (Bird, 1994):

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 n} \tag{D.3}$$

where $d = 3.7 \times 10^{-10}$ m is the molecular diameter and $n$ is the number density. The Knudsen number is then:

$$Kn = \frac{\lambda}{L} \tag{D.4}$$

where $L$ is the characteristic length (aeroshell diameter, $D = 3.0$ m).

## D.3 Fay–Riddell Stagnation-Point Heat Transfer

The Fay–Riddell equation (Fay & Riddell, 1958) provides the theoretically rigorous stagnation-point heat flux for a reacting, dissociated boundary layer:

$$\dot{q}_s = 0.763\,\mathrm{Pr}^{-0.6}\,(\rho_e\mu_e)^{0.4}\,(\rho_w\mu_w)^{0.1}\sqrt{\left(\frac{du_e}{dx}\right)_s}\,(h_s - h_w) \tag{D.5}$$

Unlike the Sutton–Graves correlation, the Fay–Riddell formulation accounts for the actual shock layer structure, chemical non-equilibrium, and catalytic wall effects, making it more physically accurate for HIAD geometries.